

# SUPPLEMENTARY MATERIAL

## STATISTIC PRESERVATION DOES NOT IDENTIFY ATTACK-SUPPRESSION PRESERVATION IN COMPOSED FEDERATED-LEARNING DEFENSES

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This supplement holds per-cell measurement detail for the composability screen, together with nine self-contained blocks relocated from the main paper’s appendix: the composition audit’s rubric, retrieval frame and coded rows (§S16); one pre-registered question about whether randomization is itself costly (§S17); the equivalence testing at the existing seed counts (§S18); the composition and adaptive-attack construction details (§S19); and the scale and architecture checks (§S20); the one use we claim for the screen, evaluation prioritization, and the designs an evaluator after discovery should run instead (§S21); that use priced over the nine variants an evaluator could run, precision against recall against evaluation cost (§S22); and the closed census of the region Prop. 11(a) settles algebraically, which holds the one falsifiable prospective positive (§S23); and the subordinate boundary study that prices the screen itself, with the 42-pair development and held-out menus, the prospective suite’s 13/20, the metric-swap ablation, the CIFAR-100 composition menu and the out-of-sample PASS battery (§S24). Every claim any of it supports is stated and qualified in the main paper, and section, table and theorem numbers without an **S** prefix refer to it, as do citations, whose full entries are in its reference list. What decided what was moved is that no claim in the main paper is stated only here. Five of the nine are then skippable; **the equivalence testing, the boundary study, the frontier and the census are not**: the first is where each arm’s binding  $m^*$  and the one-sided bound behind every non-increase are computed, the second is what the screen buys and the two boundary conditions on it, and the other two carry what the screen costs and the one falsifiable prospective positive.

### S1 THE PROSPECTIVE SUITE, ARM BY ARM

(i) **FLTrust as  $d_1$  (7 pairs): out of scope, 2/7.** The five predicted-HIGH misses are  $\text{fltrust} \rightarrow \{\text{rep, nc, krum, mkrum, tm}\}$ . Their C2/C3 failures are real (FLTrust does destroy Reputation’s consensus-distance signal), and are costless because  $d_2$  was never load-bearing: the C0 mechanism of §S24.1.

(ii) **FLTrust as  $d_2$  (6 pairs): in scope, 6/6, and the near-identity of §S24.1 is visible per seed rather than only in the aggregate.**  $\text{Krum} \rightarrow \text{FLTrust}$  and  $\text{MKrum} \rightarrow \text{FLTrust}$  are **bit-identical** to FLTrust alone on all six runs (0.0519/0.0459/0.0453 scaling, 0.0492/0.0466/0.0426 pixel), as expected for a pass-through  $d_1$ .  $\text{NC} \rightarrow \text{FLTrust}$  is *bit-identical on the pixel attack*, where no client exceeds  $\tau=5$  so  $c_i \equiv 1$ , and differs only within seed noise on model-scaling, where clipping actually fires.  $\text{Rep} \rightarrow \text{FLTrust}$  and  $\text{RFA} \rightarrow \text{FLTrust}$  likewise differ only within noise. The single genuine deviation is  $\text{FG} \rightarrow \text{FLTrust}$  (0.064, per-seed up to 0.084): the one  $d_1$  that produces exact *zeros* rather than positive rescalings, changing FLTrust’s effective client set instead of only its magnitudes, which is the Lemma 2 interaction. Every  $d_1$  in the menu is a non-negative per-client rescaling whose positive part FLTrust cancels exactly, so only the *zero pattern* is free to matter, which is where the one deviation appears.

(iii) **Neither defense is FLTrust (7 pairs): 5/7.** Per-pair values are in Table S15; both misses are diagnosed in §S24.1 and neither is a mechanism error. This arm is the one §S24.1 calls the only one that could have surprised us, and §S2 reports that 5 of its 7 pairs reduce to a single defense, leaving 2 that compose two real mechanisms.

### S2 MULTIKRUM AT $K=5$ , AND THE MATCHED CONTRASTS WE WITHDREW

**The mechanism.** `multi.krum` scores each participant by its summed distance to nearest neighbours, keeps the  $k$  lowest and averages them, with  $k$  defaulting to 5. Every experiment in this project runs `clients_per_round = 5` and none passes  $k$ , so the kept set is *all five* participants and the call reduces to plain averaging. Measured rather than argued (`experiments/verify_wave3_invariance.py`, test

D; `experiments/analyze_mkrum_degeneracy.py`): at  $K=5$  the largest absolute difference between `multi_krum` and `fedavg` on identical inputs is  $1.9 \times 10^{-6}$ , which is float summation order alone; at  $K=10$  it is 25.2, a genuine selector again. MultiKrum is therefore a distinct rule in this codebase only when more clients participate than it selects, which no run reported here does.

**Two degeneracies compose, and we had recognized only one.** (A) *Pass-through  $d_1$ .* A rank aggregator emits one vector rather than per-client updates, so `krum`, `multi_krum`, `trimmed_mean` and `coord_median` supply no upstream transform and  $d_1 \rightarrow d_2$  measures  $d_2$  alone. This was known and is what the table’s DEGEN label records; it makes `Krum`  $\rightarrow$  `FLTrust` and `MKrum`  $\rightarrow$  `FLTrust`, and `Krum`  $\rightarrow$  `RFA` and `MKrum`  $\rightarrow$  `RFA`, the same measurement twice (per-seed  $|\Delta \text{ASR}| = 0.000$  on both attacks). (B) *MultiKrum as  $d_2$ .* Because MultiKrum is FedAvg here,  $d_1 \rightarrow$  `MKrum` applies  $d_1$ ’s per-client transform and then averages, which for every weighting or clipping  $d_1$  in this menu is that defense standalone. This is the new fact, and the affected rows carry frozen labels of C1-fail or C2-fail rather than DEGEN because it was not known when the labels were committed. Counting both, 8 of the suite’s 20 rows reduce to a single defense, not the 5 marked DEGEN: the five already marked, plus `RFA`  $\rightarrow$  `MKrum`, `NC`  $\rightarrow$  `MKrum` and `FLTrust`  $\rightarrow$  `MKrum`.

**The matched contrasts do not survive, and substituting Krum is what shows it.** §S24.1 previously drew three contrasts holding  $d_1$  fixed while  $d_2$  moved from `FLTrust` to a Euclidean rule, reading the gap as evidence for Proposition 11. Two used MultiKrum and so had no downstream defense on one side. Replacing it with the genuine selector Krum gives a repaired-looking  $12.6\text{--}19.1\times$  gap, and the repair is illusory: for the gap to be about invariance, C1 would have to hold across it, and it does not. Every Krum arm takes its max-committed ASR from the pixel backdoor, on which Krum’s own standalone ASR is 0.583, above threshold.

| $d_1$ | $\rightarrow$ FLTrust | $\rightarrow$ Krum | attack | Krum alone | excess over Krum alone          |
|-------|-----------------------|--------------------|--------|------------|---------------------------------|
| Rep   | 0.046                 | 0.880              | pixel  | 0.583      | +0.297                          |
| NC    | 0.046                 | 0.583              | pixel  | 0.583      | +0.000, bit-identical 3/3 seeds |
| RFA   | 0.047                 | 0.725              | pixel  | 0.583      | +0.142                          |

The NormClip row is the sharpest form of the point: `NC`  $\rightarrow$  `Krum` is bit-identical to Krum alone on all three seeds (0.779/0.039/0.932, accuracy 0.396/0.592/0.467), because under the pixel backdoor the maximum client update norm over 45 client-rounds is  $3.04 < \tau=5$ , so NormClip is the exact identity in 9/9 measured rounds (test F; under model-scaling norms reach 24.5 and it binds in 11/45). Nothing about the invariance class of  $d_2$  is being tested there. In 0 of the 3 contrasts does the non-invariant  $d_2$  suppress its own max-committed attack standalone, which is the condition that would make the contrast informative. The Krum arms for NC and RFA come from the metric-swap suite (§S24.2) rather than the prospective suite, at the identical protocol and the same seeds 42–44; we note the change of source rather than smooth it over.

**What arm (iii) is left with.** Of its 7 pairs, `Krum`  $\rightarrow$  `CM` is `CoordMedian` alone, `Krum`  $\rightarrow$  `RFA` and `MKrum`  $\rightarrow$  `RFA` and `RFA`  $\rightarrow$  `MKrum` are `RFA` alone, and `NC`  $\rightarrow$  `MKrum` is NormClip alone. Two compose two real mechanisms: `FG`  $\rightarrow$  `Krum`, which is a miss at 0.078 and 9.9% clean accuracy (the collapsed-model artifact), and `Rep`  $\rightarrow$  `Krum`, which is a hit at 0.880. So the arm’s 5/7 is 1/2 on the pairs that put two mechanisms in series, and the surviving hit is a predicted HIGH, so it still bears on specificity only.

**A noise floor this degeneracy hands us, and a second use for the accuracy gate.**  $d_1 \rightarrow$  `MKrum` and  $d_1 \rightarrow$  `FedAvg` are the same function computed through two code paths that differ by  $1.9 \times 10^{-6}$  per aggregation, as are MultiKrum and FedAvg standalone. Over 50 rounds that difference is amplified chaotically through the training trajectory, so whatever such a cell pair disagrees by is a measured floor on how small an ASR difference can be called real at this protocol. At matched seeds, across the five such cells with clean accuracy above the 0.35 gate, the per-seed  $|\Delta \text{ASR}|$  reaches 0.046; that is what licenses the paper’s repeated “differ only within seed noise”. In the one cell below the gate, MultiKrum against FedAvg under model-scaling at 0.10 accuracy, the same function disagrees with itself by 0.270 (per-seed ASR 0.000/0.995/0.730 against 0.000/1.000/1.000). So the accuracy gate is a precondition for reading small ASR differences at all, not only for reading suppression claims: on a collapsed model ASR is not a stable quantity. Neither figure is to be confused with the exactly 0.000 of degeneracy (A), where the two rows traverse an identical code path.

**What we did not change.** The suite’s frozen categories, predictions and 13/20 score stand exactly as committed at 262cf35; relabelling them after seeing the data is what the pre-registration forbids. What is retracted is the body’s interpretation of three rows and of the three contrasts, and Table S15 carries a marker and a footnote rather than an edited label.

### S3 METRIC-SWAP ABLATION: VERIFICATION AND PER-CELL DETAIL

**The mechanism is verified in code before it is claimed.** An outcome alone would leave open whether the mirrors are invariant at all, so we check the statistic itself rather than only the ASR (`experiments/verify_cos_invariance.py`, `results/cos_invariance_check.json`), com-

paring each  $d_2$ 's internal state on  $\{u_i\}$  against  $\{c_i u_i\}$  for the  $c_i$  that NormClip and RFA actually produce on live rounds. Over 18 rounds (3 seeds  $\times$  2 upstream defenses  $\times$  3 rounds), `cos_reputation`'s weight vector agrees to  $\leq 2.5 \times 10^{-7}$  and its weight *ordering* never changes (0/18), and `cos_krum` selects the same client in 18/18. The Euclidean originals do not: Reputation's weights move by up to 0.222 and its ordering changes in 12/18 rounds; Krum's selection changes in 6/18. The size of the disturbance tracks the realized weight ratio  $\rho$ , as Proposition 11 predicts: under NormClip ( $\rho \in [1.00, 4.90]$ ) Reputation's ordering changes in 3/9 rounds and Krum's selection in 0/9; under RFA ( $\rho \in [2.11, 17.38]$ ) they change in 9/9 and 6/9. On 1000 synthetic draws with larger rescalings the separation is cleaner: ordering changes in 0/1000 trials for `cos_reputation` against 992/1000 for Reputation, and 0/1000 against 618/1000 for the two Krums. The mirrors are additionally checked against the shipped implementations they stand in for (selected index identical, aggregate relative error  $\approx 10^{-8}$ , below the  $\approx 7 \times 10^{-5}$  MPS-versus-CPU nondeterminism of the same computation), so the invariance is a property of the statistic and not of a divergent reimplement.

One prediction in the suite was genuinely two-sided, and it is the only prospective positive result either screening suite produced. `cos_krum`'s C1 verdict on model-scaling turns entirely on whether the accuracy gate is applied: 0.078 passes the numeric threshold, 0.146 accuracy fails the gate. Rather than choose a reading after the fact we froze *both*, with a resolution rule fixed in advance. Primary (accuracy-gated)  $\Rightarrow$  C1 fails  $\Rightarrow$  HIGH; secondary (numeric C1 only)  $\Rightarrow$  the pair is certified  $\Rightarrow$  LOW; and if the outcome had been LOW at accuracy  $\geq 0.35$ , the gate would have been wrong and our two primary labels for those cells would have been recorded as misses. Both pairs came out **HIGH** (NormClip  $\rightarrow$  `cos_krum` 0.508, RFA  $\rightarrow$  `cos_krum` 0.566), refuting the secondary reading and vindicating the accuracy gate against a pre-registered alternative that would have overturned it. This is a small result but a real one: the gate is usually defended by argument, and here it beat a competing specification of itself on data neither of us had seen. We flag the weaker half honestly: NormClip  $\rightarrow$  `cos_krum`'s 0.508 sits barely above threshold on per-seed values 0.084/0.676/0.766, so that cell is threshold-noise even though its label is correct, and the RFA arm at 0.566 (per-seed 1.000/0.121/0.576) is what carries the conclusion.

The pattern holds across the suite once cells without power are excluded.  $\Delta$  is only observable where the standalone  $d_2$  genuinely suppresses the attack, since a baseline already above 0.5 has no suppression left to lose and will show  $\Delta \approx 0$  whatever the transform did: a ceiling effect, not preservation. Restricting to standalone ASR  $< 0.5$  at accuracy  $\geq 0.35$  leaves six cells, and measured disturbance splits them perfectly at both seed counts. At the frozen  $n=3$ : undisturbed (0/9)  $\Delta \in [-0.027, +0.001]$ , disturbed ( $> 0/9$ )  $\Delta \in [+0.630, +0.865]$ , a separation of +0.629. At  $n=5$  after the post-hoc top-up, with each standalone recomputed at the same five seeds from the dose ladder's  $\kappa=0$  rung (`results/dose_response/summary.json`, that  $d_2$  alone under the identical configuration, bit-identical to the published standalone baselines on every seed they share, largest deviation 0.00e+00:  $[-0.044, +0.001]$  versus  $[+0.717, +0.878]$ , a separation of +0.716. Neither has any overlap, and the extra seeds widen the split rather than narrow it. At both seed counts the two extremes are the two Krum cells, which share a standalone, so the width of the split is algebraically the Krum contrast's own gap and not independent evidence beyond it; the other four cells extend the halves outward, not the distance between them. Two of the three undisturbed cells are `cos_krum` on the pixel backdoor, where the mirror suppresses standalone (0.300 at 0.598 accuracy on the frozen three seeds, 0.490 on all five) and carries that suppression through under *both* upstreams: at  $n=3$  intact (0.300,  $\Delta = +0.000$  under NormClip; 0.273,  $\Delta = -0.027$  under RFA), at  $n=5$  still intact (0.490,  $\Delta = +0.000$ ; 0.447,  $\Delta = -0.044$ ; both at 0.61 accuracy, on new per-seed values 0.864/0.686 and 0.849/0.565 against a baseline that moves to 0.490 on those same seeds). The NormClip arm is intact *exactly*: NormClip never binds on this attack (0/9 rounds; max client norm 3.04 against  $\tau = 5$ ), so its composition is the same five runs as the `cos_krum` baseline, bit-identical per seed, and  $\Delta = +0.000$  is an identity rather than a measurement. Had the frozen three-seed baseline been kept under the five-seed compositions, the same two rows would have read  $\Delta = +0.190$  and  $+0.147$ , and the natural reading of that, the magnitude channel of Lemma 2 moving  $\Delta$  where the selection is exactly invariant, would have credited an effect to a transform that provably did nothing: the movement is entirely the baseline's seed set, which is why the subtrahend has to move with the composition. So the confirming half of the split is contributed by the invariant construction behaving as C2 predicts (which is the one thing the metric-swap arm does deliver, even though C1's failure on model-scaling denied it a frozen LOW label), and this contrast has no power on the magnitude channel in either direction: its only exactly-invariant downstream sits under an upstream that never rescales. That is a limitation of this post-hoc contrast and a direct motivation for arm 3 of the ladder, which moves that dial on purpose.

**Excluded cells.** All four `cos_*` model-scaling cells are excluded as uninterpretable (hollow standalone baseline), and the six Reputation / `cos_reputation` / Krum pixel cells for lack of power (standalone  $\geq 0.5$ ).

## S4 OUT-OF-SAMPLE PASS VALIDATION: PER-CELL DETAIL

**(A) DBA attack transfer, at two adversarial fractions.** The Distributed Backdoor Attack (Xie et al., 2020) uses spatially-distributed sub-triggers at four image corners, a categorically different attack family, and is not in the development menu ( $f=0.4$  is required for viability at  $K=5$ ). We ran three PASS-predicted compositions

and three FAIL controls under DBA at  $N=10$ ,  $n=5$  seeds, first at  $f=0.4$  and then, to address the power limitation the first wave exposed, again at  $f=0.6$  ( $K \cdot f=3.0$ ):

| Pair                | Pred.  | DBA $f=0.4$       | DBA $f=0.6$       |
|---------------------|--------|-------------------|-------------------|
| fg→rfa              | PASS   | $0.109 \pm 0.065$ | $0.125 \pm 0.050$ |
| fg→cm               | PASS   | $0.050 \pm 0.027$ | $0.045 \pm 0.021$ |
| rep→cm              | PASS   | $0.041 \pm 0.014$ | $0.043 \pm 0.015$ |
| nc→rep              | FAIL   | $0.297 \pm 0.224$ | $0.378 \pm 0.185$ |
| rfa→rep             | FAIL   | $0.297 \pm 0.159$ | $0.384 \pm 0.217$ |
| fg→tm               | FAIL   | $0.049 \pm 0.031$ | $0.068 \pm 0.018$ |
| fedavg (no defense) | sanity | $0.236 \pm 0.190$ | $0.386 \pm 0.192$ |

**The FAIL side is uninformative at both fractions, and the mechanism is known rather than inferred.** The sanity baseline never clears 0.5, so a FAIL control that also sits below 0.5 carries no information: at  $f=0.4$  it is  $0.236 \pm 0.190$  and bimodal (4/5 seeds  $< 0.22$ , one  $> 0.6$ ), because all four adversarial sub-triggers must be sampled in the same round, probability  $\binom{6}{1}/\binom{10}{5} = 2.4\%$  at  $K=5$ ; at  $f=0.6$  it rises only to 0.386 (§S19.3 shows the residual bimodality is a per-seed admissibility latent in the distributed trigger, not purely a co-participation effect). The distributions still separate at the tails, max per-seed ASR 0.202 across the PASS pairs against 0.761 across the FAIL pairs, which is what shows DBA embeds when co-sampling succeeds; and the PASS-FAIL gap is not merely DBA’s own weakness, since fg→rfa reaches 0.125 where the undefended baseline reaches 0.386, a  $3\times$  suppression. The symmetric verdict: PASS predictions are confirmed robustly at both fractions, and FAIL-side falsifiability requires a DBA variant that embeds reliably before those controls mean anything.

**(B) Scale and architecture transfer, and (C) the independent survivor.** All predictions were pre-registered at commit `ca627e5` before any of these ran,  $n=5$  seeds throughout the pre-registered conditions (the reference policy below is  $n=10$ ). (B) tests fg→rfa and fg→cm at  $N=100$ ,  $K=10$ ,  $f=0.05$  on ResNet18, a  $10\times$  client-pool expansion with a different architecture, both pre-registered PASS under committed scaling and committed pixel. (C) tests fg→cm, selected *solely* by criterion PASS status and independently of the 189-config payoff-matrix scan that selected rep+tm, against temporal mixing of fg/cm (50/50 at random per round) at  $N=10$ ,  $f=0.2$ :

| Condition                                                                                       | Pred.     | committed_scaling | committed_pixel   |
|-------------------------------------------------------------------------------------------------|-----------|-------------------|-------------------|
| <i>(B) <math>N=100</math>, <math>K=10</math>, <math>f=0.05</math>, ResNet18</i>                 |           |                   |                   |
| fg→rfa                                                                                          | PASS      | $0.063 \pm 0.031$ | $0.069 \pm 0.029$ |
| fg→cm                                                                                           | PASS      | $0.152 \pm 0.118$ | $0.141 \pm 0.099$ |
| fedavg/nc 50/50 ref. ( $n=10$ )                                                                 | reference | $0.410 \pm 0.386$ |                   |
| <i>(C) <math>N=10</math>, <math>f=0.2</math>; oracle equals the scaling column in both rows</i> |           |                   |                   |
| fg→cm composed                                                                                  | PASS      | $0.107 \pm 0.031$ | $0.099 \pm 0.020$ |
| fg/cm mixed 50/50                                                                               | —         | $0.327 \pm 0.417$ | $0.679 \pm 0.159$ |

**(B): all four conditions PASS**, at a maximum of 0.152. Two caveats travel with that. These runs reach only  $\approx 0.29$  clean accuracy at 50 rounds, ResNet18 being far from converged at this budget, so they establish that *suppression transfers* across scale and architecture and **not** that the composition is deployable at  $N=100$  as configured; and the two FAIL-predicted pairs at the same scale (nc→rep C2-FAIL 0.279, fg→tm C3-FAIL 0.167) again land below 0.5, so FAIL-side discrimination is unavailable at this scale and fraction for the same reason as in (A). **The reference is not a single-defense contrast, and we do not read a suppression ratio off it:** RFA standalone was never run at this scale, so the only reference policy available is the FedAvg/NormClip 50/50 mix at  $n=10$ , which reaches  $0.410 \pm 0.386$  on committed scaling and is bimodal across seeds. What that licenses is the weaker but necessary point that the attack is *viable* at  $N=100$ , so fg→rfa’s  $0.063 \pm 0.031$  is suppression rather than an attack that simply fails at scale.

**(C): composition dominates mixing, on the committed-pixel policy only.** We anchor there because it is the only policy under which both conditions retain a usable model (clean accuracy 0.66 composed, 0.77 mixed), giving  $0.099 \pm 0.020$  against  $0.679 \pm 0.159$ , a  $6.9\times$  suppression. On committed scaling the mixed model is *destroyed* (0.11 clean accuracy, CIFAR-10 chance), so its ASR is not a meaningful backdoor comparison and we exclude it from the headline. Two limits on the claim: on that excluded scaling column the mixed condition’s standard deviation exceeds its mean ( $0.327 \pm 0.417$ ), which is the per-seed instability the pixel anchor avoids, and this is independence from the survivor scan rather than full data-independence, since fg→cm also appears in the development set at 0.107 (Table S13).

## S5 3-WAY COMPOSITIONS: THE CRITERION APPLIED WHERE EXHAUSTIVE SEARCH CANNOT RUN

Table S1: 3-way composition predictions, pre-registered and then scored (base rates below the table, not in it). Max ASR = max(mean\_scaling, mean\_pixel) across seeds.

| Triple ( $d_1 \rightarrow d_2 \rightarrow d_3$ ) | Predicted | Max ASR           | Verdict        |
|--------------------------------------------------|-----------|-------------------|----------------|
| fg→rep→rfa                                       | LOW       | 0.048             | <b>CORRECT</b> |
| fg→rep→cm                                        | LOW       | 0.087             | <b>CORRECT</b> |
| rep→fg→rfa                                       | LOW       | 0.045             | <b>CORRECT</b> |
| rep→fg→cm                                        | LOW       | 0.136             | <b>CORRECT</b> |
| fg→nc→cm                                         | LOW       | 0.109             | <b>CORRECT</b> |
| fg→rfa→cm                                        | LOW       | 0.041             | <b>CORRECT</b> |
| rfa→nc→rep                                       | HIGH      | 0.894             | <b>CORRECT</b> |
| fedavg→nc→rep                                    | HIGH      | 0.842             | <b>CORRECT</b> |
| nc→rep→tm                                        | HIGH      | 0.636             | <b>CORRECT</b> |
| nc→fg→tm                                         | HIGH      | 0.267             | <b>MISS</b>    |
| <b>Overall accuracy</b>                          |           | <b>9/10 = 90%</b> |                |

**Selection-bias note.** All 6 LOW triples in Table S1 use FoolsGold or reputation as  $d_1$ : the same successful weighting defenses from the 2-way analysis. This reflects the structural fact that these are the only defenses satisfying C1 for both attacks in our menu. The criterion’s 3-way value is not discovering new  $d_1$  choices but correctly *filtering* from the  $2 \times 6 \times 5 = 60$  triples starting with FG/rep: it predicts LOW for only 6–8 of these 60 (those where the  $d_2 \rightarrow d_3$  chain preserves signals), correctly rejecting the majority. A random 6-of-10 draw from our measured set matches the criterion’s mean ASR (0.078) only 0.3% of the time ( $p = 0.003$ , resampling test; §S24).

**And the 90% against what a constant predictor scores.** Six of these ten triples are LOW, so always-LOW already scores 6/10 and always-HIGH 4/10: unlike the 2-way suite, whose 38/42 sits one pair above a constant-HIGH 37/42, this figure is above both trivial policies. It is still ten triples, still ordered rather than validated, and it certifies nothing about security, which is what limitation (L2) of the main text states in full.

## S6 DOSE-RESPONSE DESIGN: THE FULL PRE-REGISTERED PROTOCOL

Four properties make this a dial on the theory rather than an arbitrary perturbation. (i) It is exactly Proposition 11’s form  $T(u_i) = c_i u_i$  with  $c_i > 0$ , so every rung lies inside the theory’s scope. (ii) **The dose is  $\rho$ :**  $\max_j c_j / \min_j c_j = e^{2\kappa}$  exactly, the same upstream weight ratio Theorem 9 and both mechanism-preserving regimes are stated in, verified by reading  $\rho$  back off the *transformed update norms* on live rounds, where realized  $\rho$  matches nominal to three decimals at every rung and  $\text{mean}(c)$  stays in  $[0.99999, 1.00002]$ . (iii)  $\kappa=0$  gives  $c \equiv 1$ : the implementation short-circuits, so the identity rung is bit-identical to  $d_2$  standalone by construction rather than by rounding. (iv) Normalizing the *mean* to 1 holds the aggregate’s scale fixed, separating the quantity under test (disturbance of  $d_2$ ’s *statistic*, i.e. C2) from the magnitude-contraction channel of Lemma 2 that costs accuracy; without it the ladder would confound C2 with attenuation. The permutation is keyed on (seed, round) and **not** on client identity: assigning by id would place the adversaries at a fixed end of the ladder and systematically attenuate them, manufacturing a success, whereas this keying gives every arm at a given seed and round the *identical* coefficient vector, so the arms differ in  $d_2$  and nothing else. **dose\_kappa( $\kappa$ ) is an instrument, not a defense**, and is never evaluated or recommended as one.

**Frozen grid and arms.**  $\kappa \in \{0, 0.5, 1.0, 2.0\}$ , i.e.  $\rho \in \{1.00, 2.72, 7.39, 54.60\}$ : four rungs spanning a  $\approx 55 \times$  ratio. Arms are selected by the paper’s own power rule (§S24.2), fixed before the freeze: the ladder is informative only where standalone  $d_2$  genuinely suppresses the attack ( $\text{ASR} < 0.5$  at clean accuracy  $\geq 0.35$ ), since a baseline already above threshold has no suppression left to lose and would give a flat curve by ceiling effect. Applying that rule to the measured baselines leaves exactly four ( $d_2$ , attack) cells, and they happen to cover three invariance classes of Proposition 11 (Table A5). The  $\kappa=0$  rung is run *fresh* at all five seeds for every arm rather than reused from the published baselines, so every rung of every curve comes from one code path at one seed set; the published baselines then serve as an independent per-seed cross-check of that rung (tolerance  $10^{-6}$ ) instead of as data in it.

**The abscissa is measured, not nominal.** What matters to C2 is not  $\rho$  but what the rescaling does to the specific statistic  $d_2$  reads, so we measure that separately and *before* freezing any prediction, pushing live FL updates through the shipped transform and recording each arm’s decision at every  $\kappa$  on the *same* raw updates (measure\_dose\_disturbance.py, results/dose\_disturbance.json; 5 seeds  $\times$  3 rounds per

attack). No ASR is computed there, which is why it can legitimately precede the freeze. It establishes three things before any outcome exists (Table A5): *there is a dose*: disturbance is exactly 0 at the identity rung and rises monotonically with  $\kappa$  in every non-invariant arm; *the class-(a) invariance claim holds in the shipped code*: `cos_krum`'s selection is unchanged in 0 of 120 cells at  $\rho$  up to 54.6, its cosine scores moving by at most  $1.1 \times 10^{-4}$  relative, the float32 floor for a  $\approx 1.1\text{M}$ -dimensional vector, which we record rather than assert on; and *class (b) is graded rather than binary*, at roughly half the class-(c) disturbance (0.487 vs. 0.867/0.933 at  $\kappa=2.0$ ), which is what arm 4's prediction turns on.

**Frozen predictions.** The *primary* test is pooled: across all 16 (arm,  $\kappa$ ) cells, the rise in ASR over that arm's own  $\kappa=0$  rung is predicted to be positively rank-correlated with the measured disturbance rate (Spearman, one-sided,  $\alpha = 0.05$ ). This is the dose-response claim the design exists for, and its predictor is a *measured property of the statistic*, not the identity of the defense and not its standalone effectiveness, both held fixed within every arm. The *secondary* predictions are the per-arm shapes of Table A5, tested by a one-sided Jonckheere–Terpstra trend test across the four ordered rungs at  $n=5$ . For arm 3, “flat” is scored as an *equivalence* claim rather than a failed rejection: confirmed if  $|\overline{\text{ASR}}(\kappa=2.0) - \overline{\text{ASR}}(\kappa=0)| < 0.15$  and every rung stays below 0.5. That margin is fixed and justified in advance: it is under a quarter of the  $+0.629$  within-defense gap for Krum on record at the freeze (the post-hoc top-up later put that gap at  $+0.716$  at  $n=5$ , which only makes the margin more conservative; it is not revised), and  $0.300 + 0.15 = 0.450 < 0.5$ , so “within the margin” and “still suppressing” coincide by design for this arm. Arm 4's class-(b) signature is the *shallower slope*, so its criterion includes its  $\kappa=2.0$  rise being smaller than both class-(c) arms'.

## S7 DOSE-RESPONSE CAVEATS RECORDED BEFORE THE FREEZE

**Caveats recorded in advance.** (i) Arms 1–2 are on model-scaling and arms 3–4 on the pixel backdoor. This is not a free choice: on model-scaling *no* invariant  $d_2$  suppresses at usable accuracy, that is the finding of §S24.2 (`cos_krum` 0.078 at 0.146 accuracy), which is also why `cos_krum/scaling` is not an arm. “Same attack” is required *within* each ladder, which is where the causal claim lives; across arms we compare curve *shapes and slopes*, never levels. (ii) Arm 4 has thin headroom: `coord_median/pixel` sits only 0.057 below threshold, so the binary above/below-0.5 reading is weak evidence either way and we report the continuous curve with intervals. (iii) The dose is uncorrelated with adversary status by construction, but it is not adversary-adversarial: a dose assignment that deliberately up-weights adversaries would be a strictly harder test and is not run here. (iv) A synthetic upstream transform is not a real defense, so we test ecological validity by overlaying the two *real* upstream transforms already measured on these same cells (NormClip and RFA, §S24.2) as anchor points at their own measured disturbance; if they fall far off the synthetic curve at matched disturbance, that is a finding about the instrument and is reported as one. (v) Disturbance is measured on FedAvg trajectories on one shared set of raw updates per round, which is what makes it a clean abscissa, but means the rates in Table A5 are not re-measured along each arm's own 50-round trajectory.

## S8 DOSE-RESPONSE LADDER: PER-SEED DETAIL, INTERVALS, AND THE ANCHOR RESIDUALS

Every value below is recomputed per seed by `analyze_dose_response.py` from the ladder and disturbance JSON files; nothing is transcribed. Seeds are 42–46 in order; Table S2 carries the per-rung detail.

**The primary test, and one post-hoc sensitivity we report but do not lean on.** Pooled over all 16 cells, Spearman  $\rho_s = +0.351$ , one-sided  $p = 0.091$  (the frozen test; refuted). Dropping the four  $\kappa=0$  cells, which sit at (0, 0) by construction rather than by measurement, raises this to  $\rho_s = +0.780$ ,  $p = 0.001$  over 12 cells. We were not entitled to that analysis in advance and it does not change the verdict, and its sign is in any case driven by the *invariant* arm's ASR falling at zero disturbance being paired with the reputation arm's rising at high disturbance, not by the fragility pattern the framework claims: arms 1 and 4 contradict it. Against the *nominal* dose  $\kappa$  (equivalently  $\log \rho$ ) instead of measured disturbance, the pooled correlation is  $\rho_s = -0.354$ . Neither predictor works pooled.

**Per-arm trend tests.** Jonckheere–Terpstra across the four ordered rungs  $\times$  5 seeds, one-sided for increasing trend (normal approximation; a 20,000-draw permutation p-value with a fixed seed is reported alongside by the analysis script and agrees to the third decimal in every arm): arm 1  $J = 70.0$ ,  $z = -0.34$ ,  $p_{\uparrow} = 0.63$ ; arm 2  $J = 144.0$ ,  $z = +4.64$ ,  $p_{\uparrow} = 1.7 \times 10^{-6}$ ; arm 3  $J = 29.0$ ,  $z = -3.10$ ,  $p_{\uparrow} = 0.999$  ( $p_{\downarrow} = 9.8 \times 10^{-4}$ ); dropping the gate-failing  $\kappa=2.0$  rung leaves  $z = -2.06$ ,  $p_{\downarrow} = 0.020$ ; arm 4  $J = 31.0$ ,  $z = -2.96$ ,  $p_{\uparrow} = 0.998$  ( $p_{\downarrow} = 1.5 \times 10^{-3}$ ). No frozen rule tests a decreasing trend, so those  $p_{\downarrow}$  values are descriptive.

**Clean accuracy against dose (descriptive, no frozen rule).** Arm 1  $0.565 \rightarrow 0.652 \rightarrow 0.682 \rightarrow 0.619$  ( $z = +1.35$ ); arm 2  $0.777 \rightarrow 0.779 \rightarrow 0.765 \rightarrow 0.701$  ( $z = -3.30$ ); arm 3  $0.607 \rightarrow 0.569 \rightarrow 0.466 \rightarrow 0.220$  ( $z = -3.50$ ); arm 4  $0.767 \rightarrow 0.764 \rightarrow 0.749 \rightarrow 0.690$  ( $z = -4.11$ ). This is what separates a suppression

Table S2: Per-rung mean ASR with Student-t 95% intervals at  $n=5$ , the measured disturbance of that arm’s statistic, mean clean accuracy, and the per-seed values. <sup>†</sup> fails the 0.35 accuracy gate.

| arm                    | $\kappa(\rho)$ | disturb. | mean ASR [95% CI]     | acc                | per-seed ASR                      |
|------------------------|----------------|----------|-----------------------|--------------------|-----------------------------------|
| 1 krum / scaling       | 0 (1.00)       | 0.000    | 0.062 [0.019, 0.105]  | 0.565              | 0.115, 0.021, 0.047, 0.057, 0.069 |
|                        | 0.5 (2.72)     | 0.600    | 0.011 [−0.001, 0.023] | 0.652              | 0.000, 0.010, 0.011, 0.027, 0.007 |
|                        | 1.0 (7.39)     | 0.733    | 0.014 [0.003, 0.024]  | 0.682              | 0.003, 0.015, 0.021, 0.021, 0.008 |
|                        | 2.0 (54.6)     | 0.867    | 0.054 [0.002, 0.106]  | 0.619              | 0.005, 0.039, 0.077, 0.113, 0.037 |
| 2 reputation / scaling | 0 (1.00)       | 0.000    | 0.017 [0.012, 0.022]  | 0.777              | 0.014, 0.012, 0.017, 0.023, 0.019 |
|                        | 0.5 (2.72)     | 0.867    | 0.022 [0.014, 0.031]  | 0.779              | 0.016, 0.018, 0.024, 0.033, 0.020 |
|                        | 1.0 (7.39)     | 0.867    | 0.487 [0.349, 0.624]  | 0.765              | 0.439, 0.547, 0.578, 0.557, 0.313 |
|                        | 2.0 (54.6)     | 0.933    | 0.779 [0.673, 0.885]  | 0.701              | 0.710, 0.843, 0.730, 0.896, 0.714 |
| 3 cos_krum / pixel     | 0 (1.00)       | 0.000    | 0.490 [0.058, 0.922]  | 0.607              | 0.003, 0.269, 0.630, 0.864, 0.686 |
|                        | 0.5 (2.72)     | 0.000    | 0.450 [0.130, 0.769]  | 0.569              | 0.764, 0.236, 0.165, 0.648, 0.435 |
|                        | 1.0 (7.39)     | 0.000    | 0.153 [−0.127, 0.433] | 0.466              | 0.132, 0.090, 0.000, 0.543, 0.001 |
|                        | 2.0 (54.6)     | 0.000    | 0.086 [−0.100, 0.271] | 0.220 <sup>†</sup> | 0.083, 0.000, 0.000, 0.345, 0.000 |
| 4 coord_median / pixel | 0 (1.00)       | 0.000    | 0.443 [0.263, 0.622]  | 0.767              | 0.395, 0.287, 0.449, 0.677, 0.405 |
|                        | 0.5 (2.72)     | 0.251    | 0.428 [0.241, 0.615]  | 0.764              | 0.454, 0.215, 0.430, 0.638, 0.402 |
|                        | 1.0 (7.39)     | 0.378    | 0.327 [0.181, 0.472]  | 0.749              | 0.351, 0.139, 0.352, 0.463, 0.328 |
|                        | 2.0 (54.6)     | 0.487    | 0.170 [0.113, 0.228]  | 0.690              | 0.179, 0.090, 0.204, 0.192, 0.187 |

effect from a contraction artifact: arm 4 loses 0.077 of accuracy while its ASR more than halves, whereas arm 3 loses 0.39.

**Anchor residuals (pre-registered caveat (iv)).** Disturbance rates from `cos_invariance_check.json`, ASRs recomputed from `metric_swap/` at the frozen  $n=3$  and from `metric_swap_topup/` at  $n=5$ ; the residual is against the synthetic rung nearest in measured disturbance. NormClip→Krum 0/9 = 0.000, ASR 0.062 → 0.062, residual +0.001; RFA→Krum 6/9 = 0.667, 0.691 → 0.779, residual +**0.767**; NormClip→Reputation 3/9 = 0.333, 0.792 → 0.806, residual +**0.789**; RFA→Reputation 9/9 = 1.000, 0.882 → 0.895, residual +0.116; NormClip→cos\_krum 0/9, 0.300 → 0.490, residual +0.000; RFA→cos\_krum 0/9, 0.273 → 0.447, residual −0.044. Realized upstream  $\rho$  on the same rounds: Norm-Clip mean 3.61 (max 4.90), RFA mean 11.93 (max 17.38).

## S9 DECOMPOSING THE EMITTED DISPLACEMENT: RE-SELECTION AGAINST RESCALING

The  $\Delta$  agg. column of Table A8 mixes two things the causal claim of §D.2 needs apart. For a selector,  $\text{agg}(U) = u_{s_0}$  and  $\text{agg}(T(U)) = c_{s_1} u_{s_1}$ , with  $s_0$  the client selected on the raw stack and  $s_1$  the client selected on the transformed one, so exactly

$$c_{s_1} u_{s_1} - u_{s_0} = \underbrace{(u_{s_1} - u_{s_0})}_{\text{re-selection}} + \underbrace{(c_{s_1} - 1) u_{s_1}}_{\text{rescaling}},$$

both components reported relative to  $\|u_{s_0}\|$ . The re-selection component is what a magnitude-controlled dose would still move (it takes both updates *untransformed*) while the rescaling component is what such a control closes. They are **not orthogonal**: the measured angle runs from  $\cos=-0.99$  to  $-0.43$ , mean  $\approx-0.67$ , so their norms need not sum to the total, and the table prints the angle rather than leaving that discrepancy unexplained. What is asserted instead is the exact vector identity above: worst residual 2.66e−07 over all 240 (arm, family, rung, seed, round) cells, float32 noise. **This decomposes the published number, not a second similar one:** every recomputed total agrees with the corresponding  $\Delta$  agg. value in `results/admission_measurement.json` to 0.00e+00, on the same 5 seeds  $\times$  3 live rounds, the same rungs and the same configuration as that measurement. **Selectors only:** a weighted averager and a coordinate-wise order statistic emit no single selected client, so “the selected update” and hence this split are undefined for them, and `coord_median` and `reputation` do not appear. `experiments/measure_displacement_decomposition.py` trains no model per rung, computes no ASR, writes a new file (`results/displacement_decomposition.json`) and leaves every frozen artifact byte-identical.

| Design              | Selector | $\kappa$ | total | re-selection | rescaling | cos   | sel. chg. |
|---------------------|----------|----------|-------|--------------|-----------|-------|-----------|
| Confounded ladder   | krum     | 0        | 0.000 | <b>0.000</b> | 0.000     | —     | 0.000     |
|                     |          | 0.5      | 0.637 | <b>0.685</b> | 0.333     | −0.65 | 0.600     |
|                     |          | 1        | 0.731 | <b>0.851</b> | 0.685     | −0.66 | 0.733     |
|                     |          | 2        | 0.880 | <b>1.675</b> | 1.674     | −0.72 | 0.867     |
| Instrument (Mode S) | krum     | 0        | 0.000 | <b>0.000</b> | 0.000     | —     | 0.000     |
|                     |          | 0.5      | 0.663 | <b>0.646</b> | 0.308     | −0.64 | 0.533     |
|                     |          | 1        | 0.847 | <b>1.075</b> | 0.666     | −0.69 | 0.800     |
|                     |          | 2        | 0.892 | <b>0.986</b> | 0.973     | −0.66 | 0.733     |
| Confounded ladder   | cos_krum | 0        | 0.000 | <b>0.000</b> | 0.000     | —     | 0.000     |
|                     |          | 0.5      | 0.270 | <b>0.000</b> | 0.270     | —     | 0.000     |
|                     |          | 1        | 0.515 | <b>0.000</b> | 0.515     | —     | 0.000     |
|                     |          | 2        | 0.834 | <b>0.000</b> | 0.834     | —     | 0.000     |
| Instrument (Mode S) | cos_krum | 0        | 0.000 | <b>0.000</b> | 0.000     | —     | 0.000     |
|                     |          | 0.5      | 0.395 | <b>0.000</b> | 0.395     | —     | 0.000     |
|                     |          | 1        | 0.754 | <b>0.000</b> | 0.754     | —     | 0.000     |
|                     |          | 2        | 1.297 | <b>0.000</b> | 1.297     | —     | 0.000     |

**What it says, and what it does not.** Mode S into Krum at  $\kappa=2$ : the published 0.892 is 0.986 of re-selection and 0.973 of rescaling, so the dose moves *which* benign client is aggregated and *how large* it arrives, in comparable measure: the instrument is not quietly a magnitude-only perturbation, nor quietly a decision-only one. Of the 11/15 rounds whose selection changes there, **none** lands on an adversary; that is the quantitative form of the claim that Krum’s decision changes are benign-to-benign, and the reason the support of the admitted adversarial mass changes in **none of the 60 measured rounds** while the decision channel moves by 0.733: on this arm no adversary is admitted at *any* rung, so that count is a floor, and the arms carrying adversarial mass at baseline are reputation and coord\_median (48 of 60 rounds each; Table A8). cos\_krum is the informative contrast: its statistic is exactly invariant to positive rescaling, so 0/15 selections change at every rung of *both* designs and its entire displacement (1.297 under the instrument, 0.834 under the confounded ladder) is rescaling. Its fall therefore cannot travel through the decision channel at all, which corroborates §D.2’s attribution of that fall to the magnitude channel from a quantity measured independently of any ASR. **This is a decomposition of a displacement, not of an ASR effect:** it says what the dose moves in the update entering training, makes no claim about suppression, and is an analysis of runs already reported rather than a new experiment.

## S10 THE TWO MAGNITUDE-CHANNEL ARMS: PER-SEED DETAIL

Every value below is emitted by `experiments/emit_perseed_magnitude_arms.py`, which reads the two arms’ frozen artifacts through the same readers their analyzers use rather than re-implementing a leg, and refuses to print a short row if any seed is missing from any leg. Seeds are 42–61 in order,  $n=20$  on every column, and both arms are paired on the same seed set. Arm A is krum under committed scaling with score-only Krum as the control (§D.2 of the paper); Arm B is coord\_median under the committed pixel backdoor with the coordinate-wise control (§D.10 of the paper). The  $\kappa=0$  column of each arm is the imported identity leg, which is the downstream defense alone. The last column is the *uncontrolled* Mode S leg at  $\kappa=2$  on Arm B’s cell, the published +0.125 rise’s own numerator, shown so that the magnitude channel’s contribution is readable per seed rather than only as a mean.

**One thing Table S3 shows that neither mean does.** On Arm B every seed’s uncontrolled  $\kappa=2$  value exceeds its controlled one, so the magnitude channel raises ASR on 20 of 20 seeds on that cell, while the controlled contrast against the identity falls on 19 of 20. The two facts together are why the arm is reported as reversing the controlled leg’s own sign rather than as attenuating it: the control does not shrink a rise towards zero here, it crosses it.

## S11 ADVERSARY-CORRELATED DISPERSION $\Delta_c$ , MEASURED

$\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$  for every per-client transform in the paper, over the same 5 seeds  $\times$  3 rounds and, within a round, the *same* raw update stack for every transform, so the rows are comparable by construction (`measure_coefficient_targeting.py`, `results/coefficient_targeting.json`). **No ASR is computed and no model is trained per transform**, which is what lets this be added without rescoring anything; the figures are **post-hoc and exploratory** and no frozen prediction depends on them. Coefficients are read back from transformed norms,  $c_i = \|T(u_i)\|/\|u_i\|$ , so the table measures the shipped code path rather than the builder’s description of it. As a provenance check the synthetic families’ adversarial coeffi-

Table S3: Per-seed trigger ASR on the two arms that close the aggregate-magnitude channel.  $\Delta$  is that seed’s paired difference, the quantity each arm’s primary estimand averages; the *mean* row reproduces the two analyzers’ printed primaries ( $-0.0088$  and  $-0.1722$ ) and the two Arm B columns’ difference reproduces  $\Delta_{\text{mag}} = +0.2973$ . **Arm B’s sign is not carried by a few seeds:** its  $\Delta$  is negative on 19 of 20, the single exception being seed 57 at  $+0.012$ . Arm A’s is negative on 8 of 20, which is what an effect inside the margin looks like per seed and is why its interval does not exclude zero.

| seed | Arm A: krum / scaling |                 |           | Arm B: coord.median / pixel |                 |           | Arm B<br>$\kappa=2$ uncontr. |
|------|-----------------------|-----------------|-----------|-----------------------------|-----------------|-----------|------------------------------|
|      | $\kappa=0$            | $\kappa=2$ ctrl | $\Delta$  | $\kappa=0$                  | $\kappa=2$ ctrl | $\Delta$  |                              |
| 42   | 0.115                 | 0.000           | $-0.114$  | 0.395                       | 0.259           | $-0.136$  | 0.597                        |
| 43   | 0.021                 | 0.063           | $+0.042$  | 0.287                       | 0.221           | $-0.066$  | 0.319                        |
| 44   | 0.047                 | 0.087           | $+0.040$  | 0.449                       | 0.193           | $-0.256$  | 0.516                        |
| 45   | 0.057                 | 0.039           | $-0.017$  | 0.677                       | 0.462           | $-0.216$  | 0.756                        |
| 46   | 0.069                 | 0.003           | $-0.067$  | 0.405                       | 0.299           | $-0.106$  | 0.514                        |
| 47   | 0.062                 | 0.086           | $+0.024$  | 0.291                       | 0.224           | $-0.067$  | 0.484                        |
| 48   | 0.054                 | 0.024           | $-0.031$  | 0.708                       | 0.614           | $-0.094$  | 0.813                        |
| 49   | 0.065                 | 0.041           | $-0.024$  | 0.495                       | 0.332           | $-0.162$  | 0.744                        |
| 50   | 0.033                 | 0.033           | $-0.000$  | 0.500                       | 0.172           | $-0.328$  | 0.622                        |
| 51   | 0.009                 | 0.043           | $+0.034$  | 0.489                       | 0.243           | $-0.245$  | 0.667                        |
| 52   | 0.024                 | 0.047           | $+0.023$  | 0.285                       | 0.093           | $-0.193$  | 0.483                        |
| 53   | 0.024                 | 0.027           | $+0.003$  | 0.674                       | 0.518           | $-0.156$  | 0.763                        |
| 54   | 0.035                 | 0.004           | $-0.032$  | 0.398                       | 0.224           | $-0.174$  | 0.565                        |
| 55   | 0.188                 | 0.028           | $-0.159$  | 0.397                       | 0.219           | $-0.178$  | 0.500                        |
| 56   | 0.002                 | 0.043           | $+0.040$  | 0.458                       | 0.287           | $-0.170$  | 0.586                        |
| 57   | 0.010                 | 0.048           | $+0.038$  | 0.372                       | 0.385           | $+0.012$  | 0.478                        |
| 58   | 0.014                 | 0.025           | $+0.011$  | 0.708                       | 0.459           | $-0.249$  | 0.744                        |
| 59   | 0.044                 | 0.052           | $+0.008$  | 0.424                       | 0.269           | $-0.156$  | 0.577                        |
| 60   | 0.006                 | 0.009           | $+0.003$  | 0.422                       | 0.241           | $-0.181$  | 0.595                        |
| 61   | 0.000                 | 0.002           | $+0.002$  | 0.734                       | 0.409           | $-0.324$  | 0.746                        |
| mean | 0.044                 | 0.035           | $-0.0088$ | 0.478                       | 0.306           | $-0.1722$ | 0.603                        |

cient shares reproduce `results/admission_measurement.json` to 0.0 absolute on all 13 cells, so the rows of Table S4 may be read on the same axis as the Mode-S and Mode-A figures of §D.2.

Table S4:  $\rho$  says how much a transform spreads the coefficients;  $\Delta_c$  says who receives the spread. They are not ordered together: RFA and NormClip target the adversary  $2\text{--}3.6\times$  harder than the ladder’s most dispersed rung at  $\rho$  that is  $3.7\text{--}13\times$  smaller. Means over non-degenerate rounds; the synthetic rows are attack-pooled since their coefficients do not depend on the attack.

| transform                                               | attack        | $\rho$               | $\mathbb{E}[c \mid A]$ | $\mathbb{E}[c \mid \neg A]$ | $\Delta_c$                 |
|---------------------------------------------------------|---------------|----------------------|------------------------|-----------------------------|----------------------------|
| NormClip                                                | model-scaling | 4.19                 | 0.268                  | 0.989                       | $-0.721$                   |
| NormClip                                                | pixel         | 1.00                 | 1.000                  | 1.000                       | 0.000                      |
| Reputation                                              | model-scaling | $8.3 \times 10^5$    | 0.003                  | 1.387                       | $-1.383$                   |
| Reputation                                              | pixel         | 1.68                 | 0.915                  | 1.019                       | $-0.105$                   |
| FoolsGold                                               | model-scaling | $1.9 \times 10^{12}$ | 1.497                  | 0.834                       | <b><math>+0.663</math></b> |
| FoolsGold                                               | pixel         | $1.8 \times 10^{12}$ | 1.365                  | 0.986                       | <b><math>+0.378</math></b> |
| RFA                                                     | model-scaling | 14.8                 | 0.125                  | 1.332                       | $-1.208$                   |
| RFA                                                     | pixel         | 1.69                 | 0.923                  | 1.012                       | $-0.089$                   |
| <i>Synthetic families of §D.1–§D.2, same code path:</i> |               |                      |                        |                             |                            |
| ladder, $\kappa=0.5/1.0/2.0$                            | —             | 2.72/7.39/54.6       | —                      | —                           | $-0.140/-0.240/-0.336$     |
| Mode S, all rungs                                       | —             | 1.00–54.6            | —                      | —                           | <b>0.000</b>               |
| Mode A, $\nu=-2 \dots +2$                               | —             | 7.39–7.39            | —                      | —                           | $-1.138 \dots +2.469$      |

Three readings, in decreasing confidence. (1)  $\rho$  is not a proxy for  $\Delta_c$ : the two are anti-ordered on the anchors, and NormClip on the pixel backdoor has  $\rho = 1$  exactly because no client exceeds  $\tau=5$ , so  $\Delta_c = 0$  there for a reason the invariance classification cannot see. A clip is  $\Delta_c$ -neutral only where it never binds, so that zero is a property of the whole rescaling family rather than of this row, which App. D.6 measures on a second dataset. (2) Both transforms carrying a missed anchor are targeted far beyond any ladder rung, which puts the residuals of §D.1 on a measured axis rather than a range mismatch;  $\Delta_c$  does not sort the anchors themselves, since NormClip carries one that misses and one that lands. (3)  $\Delta_c$  is nonetheless not a predictor of suppression: RFA is strongly negative at  $-1.208$  and `rfa`→`krum` fails at 0.779, while Reputation is more negative still at  $-1.383$  and `rep`→`coord.median` suppresses. That is the two-regime structure of §C again: strong negative  $\Delta_c$  can mean either annihilation or mere dilution, and  $\Delta_c$ ’s magnitude alone does not say

which. We report it as the axis on which the ladder’s anchors were unmatched, not as the quantity that replaces the statistic.

## S12 ANTICIPATED OBJECTIONS

Eight objections we expect, answered against text that is already in the paper rather than by new argument. Nothing here is a new claim; each answer is a pointer plus the scope it turns on.

**1. This is small-scale federated learning:  $N \in \{10, 100\}$  clients with  $K \in \{5, 10\}$  sampled per round, three architectures, mostly one dataset.** Agreed that the scale is small, and every empirical magnitude is scoped accordingly, per arm, never pooled across arms, and enumerated in Table A2 with the scale and architecture evidence in §S20. What does *not* depend on scale is the identification argument: any criterion whose eligibility gate is defined on the same outcome it predicts conditions on a descendant of both mechanism and effectiveness (Proposition 7), and that holds at any client count, model size or sample size. The boundary results forbid an inference; they do not measure a system.

**2. The interventions are oracles: no deployed system can pin an adversary’s coefficient or read adversary identity.** Correct, and that is why Mode S is an **instrument, not a screen**. Mode S measures level (P4) only by reading adversary identity, which no deployable defense can, so these experiments **identify the intervention’s effect at a fixed adversarial coefficient share and are not a (P4) screen**. Pinning  $c=1.0$  has one purpose, to cut the confounded path of Figure A6(a) by construction, exactly as an oracle-labelled control closes one path in an ablation: one path, not every path, the magnitude of the emitted aggregate still moving with the dose. The one control that *removes* an oracle component (score-only, which scores the transformed stack but aggregates the untransformed selected update) does not change the verdict (§D.2, Appendix S9).

**3.  $n=5$  seeds is small.** It is:  $n=5$  per arm,  $n=8$  for `cos_krum`, with Student- $t$  95% intervals throughout and no claim that a small effect is zero. Where suppression does not move we report **no evidence of a practically meaningful change under the pre-specified  $\pm 0.15$  equivalence margin**, a margin frozen before the runs (Table A2), never a claim that suppression is unchanged. On the arms where we report it, that margin is one-sided in a way the phrase does not say on its own: identity-rung mean ASR is below 0.15 there, so  $\Delta > -0.15$  holds by arithmetic before any seed runs and the only leg that tests anything is the upper one, which bounds a *rise*. Section S18 scores every arm on both legs and prints the one-sided bound beside each two-sided interval. Two design choices do more for precision than seeds would: the identity rungs are imported **bit-identical per seed** (largest deviation 0.00e+00), so the contrast is within-seed rather than across noise, and the central claim is a **sign reversal** ( $-0.273$  against  $+0.125$  at  $n=20$  on the identical cell, Table A3), which is not a null result. Finally, Proposition 7 says the cell that would validate the criterion is *empty by design*: more seeds cannot repair it (Table S12).

**4. If ASR does not move, why does a change of decision matter?** Because the change of decision is what a preservation criterion *measures*, and the gap between the two is the finding. An invariance argument certifies the top three levels of Definition 2 while the security content sits at level (P4), the adversarial contribution admitted. A criterion reading the Krum arm would call that composition broken (the decision flips in up to 80% of rounds) for a composition whose suppression shows no measurable change: a false alarm in the direction that costs coverage (Proposition 8, recall bound  $4/5$ ). The converse fails too: `cos_krum`’s selection is bit-identical at every rung yet it never suppresses (0.584 at the identity, Mode S at  $n=8$ ) and its ASR *falls* 0.425 across that ladder, so exact invariance of the statistic neither implies suppression nor holds it fixed. As in the body, the  $\kappa=2$  endpoint carrying that 0.425 sits below our own clean-accuracy floor, so **the witness we rest on is the first gate-passing rung** (0.173); the full-ladder figure is quoted here only for its sign (§D.2).

**5. The theory covers only positive per-client rescalings.** Yes, and we say so wherever it is used (Proposition 11, Theorem 9, §C.3). The class is not arbitrary (it is what norm clipping, contribution weighting and reputation weighting do to a client’s update) but it excludes coordinate-wise filters and any transform that changes an update’s direction, and the theorem’s own margin closes under heterogeneity ( $m=3.24 \rightarrow 1.01$  at  $\alpha=0.1$ ), so it **certifies nothing where heterogeneity is high**. The negative does not rest on the theory: it is a *measured* counterexample, and a narrower theory makes the counterexample sharper rather than weaker, since **the theorem certifies mechanism preservation, not security**; it bounds no ASR.

**6. The identification failure is a textbook fact, conditioning on a descendant of the outcome. What is the paper for?** The textbook fact is the smallest of the three layers, and we score it as such: §2 states the violation in one paragraph and Table 1 labels it *general*, claiming no novelty for it. What is not textbook is what the fact is attached to. First the estimand: for a composed FL defense the quantity a preservation check should recover is the effect of an intervention on the upstream transform, and Definition 2 is what says which *level* of that effect a check verifies (the statistic) against which level the security content

sits at (the admitted adversarial contribution). Second the instantiation: the screening move examined here, which admits a pair for analysis only once its composition already suppresses the attack, gates eligibility on a descendant of the very thing it is then read as evidence about, and Proposition 7 formalizes that for this move rather than in general. Third the measurement: the two levels are shown to come apart in *sign* on one cell ( $-0.273$  against  $+0.125$ , Table A3), which no causal argument supplies. A reader who already knows the collider fact does not know from it that the statistic and suppression disagree in sign on a defense pair anyone would compose.

**7. The adversary is held fixed while the defense changes. Would an attacker adapting to the composition not erase the effect?** It might, and nothing here bounds that. Every number in this paper is scoped to a *committed* attack, fixed before the defense and held across the rungs, which is what leaves the upstream intervention as the only thing that varies, and it is why §6 and the abstract deny any claim about adaptive robustness. One arm moves in this direction: regime A of §D.4 runs an adversary that solves for the downstream defense, on both ladders of the cell the instrument was built for, and the design disagreement survives, the instrument moving *up* while the outcome-gated ladder moves down on all three seeds. That arm’s own disclosures are why it settles less than it appears to. The winning adaptive adversary beats the plain backdoor it replaces by  $+0.035$  at one seed, a margin rather than a rout, and it adapts to  $d_2$  only: it does not anticipate the upstream transform  $d_1$ . The reading we hold is therefore that the disagreement is not an artifact of a non-adaptive attacker, and that an adversary adapting to the *composition* is untested.

**8. If admission is the level that matters, is it a predictor? Could a practitioner measure admission and forecast suppression?** No, and that is reported as a refutation of our own earlier reading rather than left open. Two pre-registered tests bear on it and both return negative. The dose–response ladder’s primary pooled test asked whether the rise in ASR tracks the measured disturbance of the downstream statistic and returned  $\rho_s = +0.351$  at one-sided  $p = 0.091$  over 16 cells against a frozen criterion of  $p < 0.05$ , so **the primary prediction is reported as refuted** (§D.1). The prospective four-cell pattern we registered for *which* cells reverse was refuted on its held-out cell and, per its own frozen clause, withdrawn, the pattern demoted to a description (§D.3). Pooled, the two-channel model is unconfirmed over 26 cells ( $R^2 = 0.239$ ) with its cross-arm ordering refuted outright (§G, (L6)). What the paper claims is weaker and directional: admission is *more causally proximal* to suppression than the statistic is, in the sense that a preservation check can pass while admission and suppression both move, and no ordering or magnitude follows from that. Admission is the quantity our instrument holds fixed, not one we can hand a practitioner as a forecast.

### S13 SENSITIVITY OF THE INTERVENTION CONCLUSIONS TO THEIR TWO FROZEN THRESHOLDS

Two thresholds are frozen in this paper and a reader is entitled to ask how much rests on them. Both are swept here by `experiments/analyze_margin_sensitivity.py`, which reads `results/comparability_six_cells.json` and computes no ASR; as a provenance check it reproduces that table’s published verdict column exactly. **Two things about that path are worth stating rather than leaving a reader to notice:** the name is frozen from when the table held six cells and we do not rename it, because two pre-registrations refer to that exact path; and its hash *did* move when the seventh cell was added, so the clause in `pre_registration_randomized_composition.md` that four artifact hashes do not move is a constraint on *that* document’s analysis, which never writes here, and not a global freeze. What the hash was there to protect was checked directly instead, and the check is reported at the precision it holds at rather than rounded up to “unchanged”: **no value inside the six published cells moved**, and every field the earlier file carried is still there carrying the same number, but the records are *not* byte-identical, because two provenance fields (`missing_frozen_seeds` and `rung_coverage`) were added to each of them. The seventh cell is added alongside the six under Amendment 4 of the comparability pre-registration. **The hash has since moved a second time, and that move *did* change a published cell**, which is why the sentence above is dated rather than standing: the pre-registered endpoint top-up of `coord_median/pixel` (seeds 47–61, both designs, `experiments/pre_registration_reversal_seed_topup.md`) takes that cell to  $n=20$  on both legs, so its two means and both intervals move by design, and with them the single  $m^*$  this section derives from that row ( $0.178 \rightarrow 0.155$ , an arm that carries no equivalence reading either before or after). The md5 is `6e25ef3a`, from `2e8cced3`. No other cell moved, and the binding equivalence arm named below is unchanged. **One thing this sweep does not vary, and it should be said here rather than inferred:** the two `krum/scaling` rows below are the ones that carry an equivalence reading, CIFAR-10’s and EMNIST-byclass’s, and their identity-rung mean ASR is  $0.044$  and  $0.024$ , so on both of them  $\Delta > -m$  is satisfied by arithmetic at every rung of the ladder and what the sweep varies is the upper leg alone. Section S18 gives the one-sided bound for each,  $+0.012$  and  $+0.009$ . No  $m^*$  printed here changes and no verdict below is revised; what changes is only which of the two legs a reader should credit the pass to.

**The  $\pm 0.15$  practical-equivalence margin is not load-bearing.** A reading of no evidence of a practically meaningful change requires the paired 95% interval to lie inside  $[-m, +m]$ , so the smallest margin under which each frozen reading survives is  $m^* = \max(|lo|, |hi|)$ . For the two arms that carry an equivalence reading this is  $m^* = 0.032$  (Krum, CIFAR-10, at that arm's  $n=20$ ) and  $m^* = 0.034$  (Krum, EMNIST-byclass, at that arm's pooled  $n=5$ ), each read off Table A10's *Mode-S* interval for that arm (the equivalence reading is a statement about the controlled estimate, so  $m^*$  is not computed from the confounded column), against the frozen 0.15: headroom of  $+0.118$  and  $+0.116$ . **The CIFAR-100 arm is deliberately not among them, and the reason is worth stating because arithmetic alone would include it.** Its controlled interval  $[+0.031, +0.110]$  lies inside  $\pm 0.15$  and so passes an equivalence test at the frozen margin while *also* excluding zero, which is a real and familiar joint outcome; but that arm is reported as a detected rise inside a sign reversal, and no equivalence reading is made about it, so scoring it here would make this paragraph's binding number rest on a claim the paper does not make. The margin could have been fixed anywhere in  $(0.034, 0.5)$  without changing either reading, so the pre-registered value is a conservative choice rather than a tuned one. **That range is this table's and not the paper's, and one arm outside the table is why.** The `resnet18` architecture-alone arm of §D.2 is not scored here, because this ladder is built over the comparability design's cells and that arm has only the controlled leg, no outcome-gated twin to compare it with. Its own  $m^*$  is 0.183 at  $n=3$  (paired 95% interval  $[-0.183, +0.101]$ ), *wider* than the frozen margin, so for that arm the frozen 0.15 **is** load-bearing: it clears TOST at  $\pm 0.15$  ( $p=0.040$ , the 90% interval  $\alpha=0.05$  reads) and clears nothing tighter than 0.137. We state it here rather than let the range above be read as a claim about every equivalence reading in the paper, which is what the body sentence used to say.

**The  $\pm 0.05$  near-zero tolerance in the AGREE definition carries no verdict at all.** Sweeping it over  $[0, 1]$ , both endpoints included, all seven cells of Table A10 keep the verdict they are published with at every value, the tolerance-free limit  $\epsilon=0$  among them. Every cell is decided by its two intervals, and the tolerance is a tie-break that never fires on this table.

**The central claim turns on neither.** SIGN REVERSAL is defined as opposite signs with both intervals excluding zero, which reads the intervals alone: `coord.median/pixel` is  $-0.273$   $[-0.334, -0.213]$  against  $+0.125$   $[+0.095, +0.155]$  at  $n=20$ . No margin and no tolerance enters that determination. **The  $\delta_2$  disturbance threshold of §C is not swept here, and the reason is worse for us than a sweep would be:** no label in the 42-pair table is computed from it. `c2.holds` decides by invariance *class*, returning True for the exactly invariant statistics and False for the non-invariant ones, and for the three conditional classes it defers to the stored per-pair call, so there is no numeric  $\delta_2$  anywhere in that column whose value could be varied. Its provenance is censused in §S22 (3 of the 14 admissions rest on algebra and 11 on that call), and we state that rather than report a stability interval the code does not support.

## S14 SCREENING FOR A FURTHER DOWNSTREAM DEFENSE IN THE COMPARABILITY TABLE, AND WHY WE DO NOT ADD ONE

**Which axis this section is about, since the table can grow along two.** A cell is a (downstream defense, attack, dataset) triple, so a new cell can add a *defense* the table does not yet contain or move an existing defense to a new *dataset*. This section is about the first axis only, and it declines. The second axis is pre-registered separately, in Amendment 4 of `experiments/pre-registration/comparability.md`, which fixes a cell holding `coord.median` and the pixel backdoor unchanged and varying the dataset; that cell is the CIFAR-100 row of main §D.3, it adds no defense, and nothing below bears on it. Read the argument here as about defenses, never as a claim that the table is closed at seven cells.

Table A10 rests on four downstream defenses, and a reader may ask why not more. The contrast is interpretable only where the downstream defense suppresses the attack on its own, since otherwise there is no suppression whose attribution could move, so a further cell needs a defense meeting C1's own precondition: standalone ASR below 0.5 at clean accuracy at least 0.35. One axis had that unmeasured, and this section measures it.

EMNIST-byclass under model scaling is that axis. `results/femnist/payoff_results.json` already carries a standalone scaling cell for seven of the twelve aggregation rules in this study, and `krum` is the only powered one, which is cell 6: `trimmed_mean` reaches 0.9999, `rfa` 0.9990, `norm_clip` 0.9986, `coord.median` 0.6567, and `fedavg` and `multi_krum` both 0.3333 at 0.041 clean accuracy. The remaining five had no cell at all: we measured four of them at  $n=3$  under the identical protocol and runner (`experiments/measure_femnist_cl_inputs_scaling.py`, `results/femnist_cl_inputs_scaling/`) and exclude `fltrust` by the argument at the end of this section, so all twelve are accounted for.

| $d_2$ alone, EMNIST-byclass / scaling | mean ASR | mean clean acc. | precondition             |
|---------------------------------------|----------|-----------------|--------------------------|
| reputation                            | 0.3465   | 0.7039          | met, and declined below  |
| cos_krum                              | 0.6667   | 0.0494          | fails the accuracy floor |
| foolsgold                             | 0.6664   | 0.0459          | fails the accuracy floor |
| cos_reputation                        | 0.9993   | 0.1002          | fails the accuracy floor |

**No defense outside the table’s four is admissible.** `foolsgold` and `cos_reputation` are the only two screened candidates that would contribute a downstream defense the comparability axis does not already have, and both fail on the *accuracy* floor rather than the ASR threshold: a  $10\times$  update rescaling destroys the model under them, to 0.046 and 0.100 clean accuracy on a 62-class task. `foolsgold`’s third seed is the clean illustration of why the precondition gates on accuracy at all, ASR 0.0000 at 0.044 accuracy: a model that has learned nothing suppresses every attack, and reading that as suppression is the error the floor exists to prevent.

**reputation meets the precondition and we still decline the cell.** Its mean is 0.3465 at 0.704 accuracy, inside the rule as written. Its three seeds are 0.9947, 0.0152 and 0.0296: one total failure beside two near-total suppressions, and the mean resembles neither. Because `reputation` is already one of the table’s four defenses, such a cell would be a second-dataset replication rather than a new defense, and we are not willing to rest a replication on an input whose spread is this wide at  $n=3$ . We report the screen and leave the cell unrun, and note that what would settle it is more seeds on the C1 *input*, not a ladder built on this one.

**The screen is gated on reproducing a frozen cell.** These runs measure the scaling arm under the same runner as the frozen pixel-arm cells the main text reports for this dataset, and the two arms share their data poisoning, differing only in the update-level rescaling. So the screen first re-measures `krum/scaling` and refuses to run any target unless it reproduces the frozen value: it returns 0.0239 at 0.765 accuracy against the frozen 0.0270, where the corresponding pixel cell is 0.660. That gate distinguishes the arm under test from its pixel twin, and it makes these four cells protocol-comparable with the frozen matrix rather than merely indicative.

**These are C1 inputs, not predictions.** C1 is defined on standalone suppression, so the numbers above are the criterion’s inputs and no prediction of any kind is tested here; the convention that such inputs are measured before any per-pair prediction is frozen is `experiments/pre-registration-prospective.md`. No composition is evaluated, and no pre-registration is spent. `fltrust` is excluded by argument rather than measurement: the main text records it as bit-identical per seed under positive per-client rescaling, which is exactly the Mode S dose, so a ladder on it would observe an algebraic identity.

## S15 THE ORACLE-FREE BOUNDARY INSTRUMENT: THE FREEZE WE SUPERSEDED, AND THE CORRECTED INSTRUMENT’S MACHINERY

This section holds the provenance of the boundary-blend instrument of main §D.8: the measurements behind our first freeze of it failing its own share gate, the correction that freeze’s own sentence needed, and the machinery of the corrected instrument. The gate in question is `SHARE_TOL` =  $10^{-6}$ , imported from the admission measurement and the tolerance Mode S itself meets; the *flip* gate is that the shipped selections differ across the arm pair. The verdicts of main §D.8 rest on the corrected instrument alone, and nothing here revises any of them.

**What the first freeze measured, and the sentence in it that needs correcting.** It bisected the crossing of the float64 Gram statistic; the shipped `krum_selection` runs in float32 and crosses at a slightly different  $t$ , so the converged bracket is tight around the wrong predicate. Measured on live stacks before any arm was run, that instrument’s supremum share gap exceeds `SHARE_TOL` in 13 of 13 flip rounds, up to  $1.059\times 10^{-4}$ , and its realized gap exceeds it in 11 of the 12 rounds that carry an adversary; it also fails its own flip gate, the shipped selections differing across the pair in only 10 of 13. One sentence in that document needs correcting rather than defending: the  $5.109\times 10^{-7}$  it offered as an existence proof of a within-tolerance pair is the *minimum* over those same 12 rounds and the only one inside tolerance, and the supremum on that same round is  $8.368\times 10^{-5}$ . The repair its own linearity argument implies, shrink the bracket and bisect the same float64 predicate to convergence, is measured too: it is share-neutral to  $3.3\times 10^{-16}$  and the shipped selections differ in 0 of 13, so it is perfectly neutral and informative on nothing.

**The cost bound, and the cost.** The step cap of 40 is a *cost* bound, so exhausting it is a gate failure and never a relaxed tolerance. One shipped evaluation costs 66.0 ms at  $D = 1,117,354$ , and the corrected bisection needs 2 to 26 shipped Krum evaluations per round. The realized cost, re-derived from the finished artifact rather than carried over as the arithmetic we froze, is 40 runs in 8.7 h, of which the search is 29,925 shipped Krum evaluations, or 0.55 h and 6% of the total; we had projected 0.95 h of search on an 11.5 h ladder from main §D.7’s measured per-run cost, so this arm came in under its own estimate rather than over it.

**How the supersession is enforced, and the four harness checks that ran before the ladder.** The first freeze, `experiments/pre-registration-oracle-free-decomposition.md` at

commit bed6562, is left byte-untouched and superseded by `experiments/pre_registration_oracle_free_decomposition_refined.md` at 947c000, which cites it, records the failure with the numbers above, pins the premise artifact by md5, and corrects that sentence; the runner refuses to start unless the superseded document is present, last committed at its recorded hash, and clean, so the append-only claim is enforced mechanically rather than promised. Four harness checks ran before the ladder and their verdict dict is persisted into the artifact rather than left in a terminal. The one keyword this arm adds to shared code is bit-neutral on a frozen Mode S cell at  $\Delta = (0, 0)$ ; the blend at  $t=0$  reproduces main §D.7's identity arm at a shared seed, also at  $\Delta = (0, 0)$ , and reports that the hook was invoked 50 times over 50 rounds, which is the assertion that matters, because the standing failure mode in this codebase is a manipulation hook that is never called while the run passes silently; the Gram proposal equals the shipped selection at both endpoints on live stacks; and both gates pass on the ladder's own pair in all 12 flip rounds that carry an adversary, re-measured live in the same process rather than read back from a file.

**The same instrument on a second dataset: pre-registered, gated on three of its six gates, and its contrast is not reported here.** `experiments/pre_registration_oracle_free_femnist.md`, committed alone at 9cbe359, applies this instrument unchanged to EMNIST-byclass with `simple_cnn`,  $d_2$  Krum and the committed scaling attack, importing `make_hook` and `identity_hook` rather than copying them, so the share gate and the flip gate are the same code at the same tolerance. As elsewhere in this paper the prose reads EMNIST-byclass while the artifact path, the pre-registration filename and the JSON dataset key retain the `femnist` spelling, for the provenance reason main §D.2 gives. Its harness check passed all five of its checks, with the verdict dict persisted into `results/oracle_free_femnist/summary.json` rather than left in a terminal, and those five cover three of that freeze's six numbered gates. Gate 1, bit-identity, recomputes the published EMNIST-byclass identity rung at seed 42 two independent ways, through the ladder's call shape and through the hook path, and lands on accuracy 0.7645951359576352 and ASR 0.023899769324709396 with a difference of (0, 0) both times, the hook path also reporting 50 invocations over 50 rounds. Gates 2 and 3, the bisection invariant (this codebase's flip gate) and the share gate, are measured live on EMNIST-byclass update stacks, where all 3 of 3 crossing rounds differ in the shipped selection and the mask-free supremum share gap over every nonempty proper subset of the  $K=5$  participants reaches  $8.727 \times 10^{-8}$  against the same `SHARE_TOL` of  $10^{-6}$ . The self check that witnesses the instrument is the frozen one is `cifar10-only` by construction, its dataset being hardcoded in the module, so it says nothing about this dataset and is not counted here as though it did.

**What that freeze fixes, and why no contrast from it appears anywhere in this paper.** It sets  $n = 20$  per arm with both arms new, no imported leg and no mixed  $n$ , and it contains no stop-early provision, so there is no pre-specified form in which a partial version of it could be scored. The ladder delivered 11 of arm A's 20 runs and 0 of arm B's before the compute for this submission ran out, at a mean of 23.1 flip rounds per run of 50, between 16 and 27, and a supremum share gap of at most  $9.997 \times 10^{-8}$  across all 11: the dose is delivered oracle-free on a second dataset, which is the one thing this arm does establish. The artifact's `Delta_decision` accordingly records  $n = 0$  and the literal `NOT COMPUTED` for each of its four verdict labels, and gates 5 and 6, the accuracy gate and the ceiling/floor gate, are gates on the scored contrast and carry that same literal. **We compute no  $\Delta$  at any  $n$  from this arm and report no single leg of it as a finding,** because a one-legged report is the mixed- $n$  comparison the freeze exists to prevent. What exists is an instrument with a passing harness check and an unrun contrast, and main §D.7's scope sentence, which licenses no general statement in either direction about what an oracle buys, is unchanged by an arm that produced no verdict.

## S16 THE PRE-REGISTERED COMPOSITION AUDIT: RUBRIC, FRAME AND THE 59 CODED PAPERS

§F reports one rate out of this audit and withdraws one claim because of it. This subsection is the artifact behind both. The rubric is `experiments/pre_registration_literature_audit.md`, the coder's records are one JSON file per paper under `results/literature_audit/coding/`, and every count in this subsection is emitted by `experiments/analyze_literature_audit.py` into `results/literature_audit/summary.json`. **The body of Tables S6 and S7 is a file that script writes,** `results/literature_audit/table_rows.tex`, so no code and no count in them is transcribed by hand.

**Not all of the rubric predates the coding, and that is the first thing to report rather than the last.** Table S5 is the chronology. Pre-registered without qualification, and unchanged from the freeze to now: the primary quantity, the five criteria's text, the four ambiguity rules, the frame and its screening procedure, the ten-paper floor, and the three admissible outcomes. Amendment 1 re-pointed the retrieval endpoint before any paper had been retrieved, and changed no query string, no criterion and no threshold. Amendments 2 through 5 are adjudication rules for questions the frozen criteria underdetermine, and they were written with rows already in hand. **Exactly one of them moves a reported number.** Under Amendment 4a's mechanism-distinctness rule the primary is 6/59; without it, 5/59. The single row that turns on it is 2601.06466, whose three-way

threshold decision over a Mahalanobis score and a trajectory score is three components under 4a and one under the coarser reading, and both rates are reported everywhere either is.

**The frame is a procedure, not a reading list.** Three routes in: every `references.bib` entry that proposes or evaluates an FL poisoning or backdoor defense; five frozen query strings issued against the arXiv search interface, paginated to exhaustion, over 2019–2026; and the complete title listings of USENIX Security, NDSS, IEEE S&P and PMLR screened by the frozen regex `federated AND (backdoor OR poison OR byzantine OR robust)`. A one-hop snowball through the defense-comparison tables of Nguyen et al. (2022), Rieger et al. (2022) and Fenaux et al. (2025) closes it, and no second hop is taken. **Adding a query after seeing results is forbidden by the rubric and none was added.** The bias this realizes is stated in the same breath as the rate: coding requires quotable full text, so the frame is *arXiv-reachable* FL poisoning-defense work, ACM CCS is a coverage gap rather than a sample because its proceedings pages return HTTP 403, **and CCS is not the only gap:** OpenReview is unscreened, so ICLR and NeurIPS proceedings enter only by the `references.bib` and snowball routes, as do IEEE venues other than S&P on IEEE Xplore. A paper whose title carries none of the regex’s tokens enters only by those same two routes. **We do not close these gaps by hand:** the rubric forbids adding a retrieval route after seeing results, and a hand-added paper would void the pre-registration it is supposed to be evidence under, so the gaps are named and left open.

**The funnel.** 622 records retrieved, 593 after deduplication, all 593 screened on title and abstract, 59 full texts sought and 59 obtained, 59 included, 534 excluded. Of the exclusions, 362 are single-mechanism or non-pipeline on their own title and abstract, 71 are a retrieval artifact in which a bibliography line was captured as a record, 38 were adjudicated individually on other stated grounds, 24 name two mechanism families with no joint application, 18 are retrieval-stage losses with neither an abstract nor a resolvable identifier, 15 are surveys, reviews or position papers, and 6 are outside federated learning. Of the 593 screening decisions, 495 are mechanical and every one of those is an exclusion; the remaining 98 were adjudicated by hand, the 59 inclusions and 39 further exclusions, because the mechanical rule can only exclude and *no* paper enters the audit without a hand decision. Every exclusion carries its reason string in `results/literature.audit/screening.json`, and the analyzer refuses to print any rate if one does not.

**The criteria, and why the primary is a conjunction with a negated term.** C-a: two or more mechanisms applied jointly to the same round. C-b: the pairing is selected or motivated by the constituents’ standalone effectiveness or their complementary threat coverage, and by nothing else. C-c: the composed system’s own adversarial outcome, its ASR, backdoor accuracy or robust accuracy, is what validates it. C-d: an identifying contrast, meaning an experiment or derivation that varies the *upstream* stage with the downstream mechanism and the attack held fixed. C-e: a mechanism-attribution claim is actually made. **The primary is C-a^C-b^C-c^C-e^¬C-d**, and the C-e term is what stops it from measuring “nobody runs the contrast we invented”, which would be near total and worth nothing. Any UNCLEAR criterion sends the paper’s verdict to UNCLEAR, and UNCLEAR papers leave both the numerator and the denominator, so the measured prevalence can only be smaller than a guessing coder’s. No row resolved to UNCLEAR.

**The result, with the secondaries it must be read against.** Primary  $6/59 = 10.2\%$ , or  $5/59$  without Amendment 4a. The outcome-gated *design*,  $C-a^C-b^C-c$ , holds for  $33/59 = 55.9\%$ . An identifying contrast is *present* in  $46/59 = 78.0\%$ , and among the 33 design papers, 27 run one. **So the six are papers that had the design and did not run the contrast, not papers for which it was unavailable**, and any sentence implying the contrast is rare in this literature is false against our own audit. The pre-registered secondary that would count against the sharp reading, a design paper making no attributional claim at all, is  $0/33$ ; the one C-e = NO row in the frame, 1909.05125, has no design either, and a paper with neither is evidence in neither direction. The six are 2307.00356, 2310.13403, 2502.07011, 2507.16134, 2601.01053 and 2601.06466.

**What the 21 C-a = NO rows are made of, in three kinds.** First, Amendment 3’s purpose test excises constituents whose own stated purpose is privacy, communication efficiency or server verifiability rather than resisting corrupted updates, which is what leaves 2607.06612 and 2609.03420 with exactly one defense-purposed mechanism per round. Second, several papers decompose a *single* mechanism into stages that exist to serve it: 2505.17226’s median filter repairs an estimate the Krum-derived rule needs, and 2311.15894’s four steps are four steps of one rule. Third, 2302.07173 evaluates its schemes as alternatives to one another, which C-a’s own exclusion list names as a menu rather than a composition. The per-row basis is in each record’s C-a locator, and the reader who disagrees with a call can see the sentence it rests on.

**C-b fails on five of the 38 C-a = YES rows, for four distinct reasons, and the reasons matter more than the count.** 2508.12978 composes for *guarantee preservation*, proving that adding its compression does not destroy what the aggregator already provides, which is a reason about the composition rather than about the constituents’ effectiveness. 2601.04930 and 2602.22269 give a real constituent-property reason of the residual-coverage kind, but the surface being closed is one an Amendment-3-excluded constituent creates, so the pairing reason does not connect two counted mechanisms. 2503.18284 names the deficiencies of *prior* schemes as motivation for its framework and states the integration of its own two stages as a fact and an optimization convenience. 1909.05125’s only stated reason is computational and communication efficiency. **We record these rather than fold them in:** three of the four are pairing reasons of kinds the rubric could

Table S5: What was frozen when. The rubric is part pre-registration and part post-hoc adjudication record, and this is the only honest way to report that. Rows in hand is the number of papers already coded at the moment the text was written.

| what                                                        | state                 | coded rows in hand |
|-------------------------------------------------------------|-----------------------|--------------------|
| Base rubric: five criteria, four ambiguity rules, the floor | committed 9b8a395     | 0                  |
| Amendment 1: retrieval endpoint re-pointed                  | committed 3c61301     | 0                  |
| Amendment 2: blind-coding disclosure widened by one paper   | written during coding | 5                  |
| Amendment 3: purpose test for what counts as a defense      | written during coding | 5                  |
| Amendment 4: distinctness, distributive test, ordering      | written after coding  | 59                 |
| Amendment 5: <code>c_d_route</code> routes and precedence   | written after coding  | 59                 |

plausibly have admitted, and it does not, so C-b is stricter than it might have been in a direction that shrinks the primary.

**The C-d route field stratifies the 46 contrasts and decides nothing.** Of them, 21 are hyperparameter sweeps of an upstream defense parameter with an adversarial outcome read at each setting, 21 are component contrasts in which an upstream stage is removed, substituted or cumulatively built up, and 4 are analytic. Precedence when a row satisfies more than one is sweep, then contrast, then analytic, fixed in advance of the backward pass rather than deferred to which finding a paper calls central. **Amendment 5's backward pass was run across all 46 rows and relabelled four**, 2409.01435, 2502.00587, 2505.10297 and 2604.03862, moving the stratification from 25/17/4 to 21/21/4; no rate moved, because the field enters none. **It is a floor and not a census**, since it records only routes a record actually quotes. LASA is the proof: its appendix sweep of the upstream sparsification level, with accuracy under three attacks at every rung and the downstream filter fixed, was in the paper and merely pointed at in the record until the backward pass quoted it.

**One contrast in the frame is confounded by its own design, and it is coded YES anyway.** 2609.03420 varies its upstream privacy budget as a graded dose with the downstream median and the attacks fixed, and states the reason: "This selection isolates privacy-fairness interactions from aggregation-induced variance." But the client learning rate is a declared function of that budget, so two things move together along the ladder, and the paper demonstrates that this matters by reporting collapse rather than a graded decline under a static rate. The rubric's fourth ambiguity rule gives C-d the benefit of the doubt, and the code follows the rule rather than our preference: a rule that resolves toward finding the contrast present is a rule that shrinks our own primary.

**Quotation integrity, and the substitutions this table cannot avoid.** Every quotation in every record is asserted verbatim against the paper's own extracted full text by `experiments/verify_quotes_literature_audit.py`, which exits non-zero on any failure: 59 records, 295 quotations, 334 spans, all verbatim, with four spans under 40 characters and two pairs out of source order reported rather than failed, the locator saying in each case that the spans are not one passage. **The first run of that check over the whole set failed 28 quotations in 15 records, and the repair log is committed** at `results/literature_audit/quote-repair_log.md` with the pre-repair records beside it. **No code changed in any repair**, and two repairs produced *stronger* evidence for the code already recorded, which is the uncomfortable part: the defective quotations were faithful paraphrases, so a reader checking the argument would not have caught them and only the machine check did. One consequence lands in this subsection. Verbatim spans contain a true minus, an en dash inside a citation range, typographic apostrophes, `ff` and `fi` ligatures, Greek and math-italic letters and a set-membership sign, none of which this paper's source may carry, so quoted spans are rendered with a LaTeX equivalent, for instance  $\text{CSFT}'s$  for the typographic apostrophe and  $\delta$  for math-italic delta. **The JSON records, not this section, are the authority for what each paper says.**

**Extraction hazards, named because they are what a re-adjudication will hit first.** `pdftotext` interleaves the columns of a two-column layout, which spliced a word from the adjacent column into a sentence in 2503.18284 and produced adjacent-but-unrelated spans in 2602.16480 and 2607.06612; three papers, 2606.31066, 2608.08574 and 2608.21172, emit table bodies in column-major order, so a number's row label is not the label above it in the stream; and displayed fractions and page numbers appear mid-sentence. Every affected locator says so. Separately, 2609.03064 carries a journal line dated 2024 against a 2609 arXiv identifier; we code it under the arXiv year and record the inconsistency rather than choose silently.

**What this audit does not establish, recorded before the numbers existed and unchanged by them.** It is one coder with no inter-rater statistic, and the mitigation is the quotation rather than a claim of reliability. Four rows were known to us before the freeze and are marked in the tables: coding them is not blind. The rate is about this frame, which is arXiv-reachable, preprint-heavy and 2026-heavy, and it is not an estimate of "the FL literature". And the rate carries no claim that the six papers' conclusions are wrong: the audit measures whether an evaluation design can identify the mechanism it attributes to, which is a question about what the evidence licenses and not about whether the defense works.

Table S6: The pre-registered composition audit, rows 1–30 of 59, ordered by arXiv identifier. ✓ is YES and × is NO; the C-d route cell carries an absent-value dash exactly when C-d is NO, since a paper that runs no contrast has no route. *primary* is the outcome-gated verdict  $C-a \wedge C-b \wedge C-c \wedge C-e \wedge \neg C-d$ . Every cell is emitted by `experiments/analyze_literature_audit.py` from the per-paper records, and each YES or NO in those records carries a machine-checked verbatim quotation and a locator. *short name* is that script’s handle, an acronym where the paper has one and otherwise the head of the title before any colon, kept whole up to 24 characters and cut at 21 with an ellipsis beyond that, which is why four rows here read as a stem: 2207.01982 is “Defending against the Label-flipping Attack in Federated Learning”, 2302.07173 is “An Experimental Study of Byzantine-Robust Aggregation Schemes in Federated Learning”, 2409.17754 is “Byzantine-Robust Aggregation for Securing Decentralized Federated Learning” and 2409.19302 is “Leveraging MTD to Mitigate Poisoning Attacks in Decentralized FL with Non-IID Data”. We extend them here rather than in the table, whose body is machine-written; every full title is in `results/literature_audit/screen_pool.json` under the identifier in the first column.

| arXiv id                | short name                | yr   | C-a | C-b | C-c | C-d | C-e | C-d route | primary |
|-------------------------|---------------------------|------|-----|-----|-----|-----|-----|-----------|---------|
| 1909.05125              | Adaptive FedAvg           | 2019 | ✓   | ×   | ✓   | ×   | ×   | —         | no      |
| 2012.13995 <sup>†</sup> | FLTrust                   | 2021 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2101.02281 <sup>†</sup> | FLAME                     | 2022 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2104.06685              | BROADCAST                 | 2021 | ×   | ×   | ✓   | ✓   | ✓   | analytic  | no      |
| 2110.02940              | SHARE                     | 2021 | ×   | ×   | ✓   | ✓   | ✓   | sweep     | no      |
| 2201.00763 <sup>†</sup> | DeepSight                 | 2022 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2206.12100              | zPROBE                    | 2022 | ×   | ×   | ✓   | ✓   | ✓   | sweep     | no      |
| 2207.01982              | Defending against the ... | 2022 | ×   | ×   | ✓   | ×   | ✓   | —         | no      |
| 2208.10161              | MUDGUARD                  | 2022 | ×   | ×   | ✓   | ✓   | ✓   | contrast  | no      |
| 2210.01437              | Robust-FL                 | 2022 | ×   | ×   | ✓   | ✓   | ✓   | sweep     | no      |
| 2210.07714              | CrowdGuard                | 2024 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2302.07173              | An Experimental Study...  | 2023 | ×   | ×   | ×   | ×   | ✓   | —         | no      |
| 2306.04984              | G <sup>2</sup> uardFL     | 2023 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2306.12608              | DP-BREM                   | 2023 | ×   | ×   | ✓   | ✓   | ✓   | sweep     | no      |
| 2307.00356              | Fedward                   | 2023 | ✓   | ✓   | ✓   | ×   | ✓   | —         | yes     |
| 2309.10607              | SPFL                      | 2023 | ×   | ×   | ✓   | ✓   | ✓   | contrast  | no      |
| 2310.13403              | BRFL                      | 2023 | ✓   | ✓   | ✓   | ×   | ✓   | —         | yes     |
| 2311.15894              | refined-Krum              | 2023 | ×   | ×   | ✓   | ×   | ✓   | —         | no      |
| 2402.04409              | FRECA                     | 2024 | ×   | ×   | ✓   | ×   | ✓   | —         | no      |
| 2404.04139              | FedZZ                     | 2024 | ×   | ×   | ✓   | ✓   | ✓   | sweep     | no      |
| 2409.01435              | LASA                      | 2024 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2409.17754              | Byzantine-Robust Aggr...  | 2024 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2409.19302              | Leveraging MTD to Mit...  | 2024 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2411.01040              | MASA                      | 2024 | ×   | ×   | ✓   | ✓   | ✓   | sweep     | no      |
| 2502.00587              | RKD                       | 2025 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2502.05547              | DDFed                     | 2025 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2502.07011              | DROP                      | 2025 | ✓   | ✓   | ✓   | ×   | ✓   | —         | yes     |
| 2503.18284              | FedSAC                    | 2025 | ✓   | ×   | ✓   | ✓   | ✓   | contrast  | no      |
| 2505.01454              | SafeSparse                | 2025 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2505.10297              | FeRA                      | 2025 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |

<sup>†</sup> known to us before the rubric was frozen; coding is not blind.

Table S7: The pre-registered composition audit, rows 31–59 of 59. Columns and conventions as in Table S6. Rows are identified by arXiv identifier and short name rather than by citation: citing all 59 would add 49 entries to the bibliography, since ten are already cited, and most of the rest are preprints no resolver carries, and the identifier is what a re-adjudication needs.

| arXiv id                | short name        | yr   | C-a | C-b | C-c | C-d | C-e | C-d route | primary |
|-------------------------|-------------------|------|-----|-----|-----|-----|-----|-----------|---------|
| 2505.12851              | FLTG              | 2025 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2505.17226              | ArKrum            | 2025 | ×   | ×   | ✓   | ×   | ✓   | —         | no      |
| 2507.14588              | FORTA             | 2025 | ×   | ×   | ✓   | ✓   | ✓   | contrast  | no      |
| 2507.16134              | DP2Guard          | 2025 | ✓   | ✓   | ✓   | ×   | ✓   | —         | yes     |
| 2508.10315              | CLIP-Fed          | 2025 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2508.12978              | Fed-DPRoC         | 2025 | ✓   | ×   | ✓   | ✓   | ✓   | sweep     | no      |
| 2509.08089 <sup>†</sup> | Hammer and Anvil  | 2025 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2509.08449              | DSFL              | 2025 | ×   | ×   | ✓   | ×   | ✓   | —         | no      |
| 2509.18044              | HRA               | 2025 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2511.09294              | GuardFed          | 2025 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2512.11760              | SpectralKrum      | 2025 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2601.01053              | PQ-BRFL           | 2026 | ✓   | ✓   | ✓   | ×   | ✓   | —         | yes     |
| 2601.04930              | PPFA-BA           | 2026 | ✓   | ×   | ✓   | ✓   | ✓   | contrast  | no      |
| 2601.06466              | SecureDyn-FL      | 2026 | ✓   | ✓   | ✓   | ×   | ✓   | —         | yes     |
| 2602.16480              | SRFed             | 2026 | ×   | ×   | ✓   | ✓   | ✓   | analytic  | no      |
| 2602.22269              | CQSA              | 2026 | ✓   | ×   | ×   | ✓   | ✓   | analytic  | no      |
| 2603.04422              | FedEMA-Distill    | 2026 | ×   | ×   | ✓   | ✓   | ✓   | contrast  | no      |
| 2603.23472              | Byz-Clip21-SGD2M  | 2026 | ×   | ×   | ✓   | ✓   | ✓   | sweep     | no      |
| 2603.28652              | FedBBA            | 2026 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2604.03862              | SecureAFL         | 2026 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2605.11122              | FedSurrogate      | 2026 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2605.22506              | EnCAgg            | 2026 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2606.31066              | Secure-CHG        | 2026 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2607.06612              | PRoVeFL           | 2026 | ×   | ×   | ✓   | ✓   | ✓   | analytic  | no      |
| 2608.08574              | R2CFL             | 2026 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2608.14861              | STAR-FL           | 2026 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2608.21172              | Thermo-FL / TERRA | 2026 | ✓   | ✓   | ✓   | ✓   | ✓   | contrast  | no      |
| 2609.03064              | DP-BR-FedAvg      | 2026 | ✓   | ✓   | ✓   | ✓   | ✓   | sweep     | no      |
| 2609.03420              | GeoIndist-FL-NIDS | 2026 | ×   | ×   | ✓   | ✓   | ✓   | sweep     | no      |

<sup>†</sup> known to us before the rubric was frozen; coding is not blind.

## S17 IS RANDOMIZATION ITSELF COSTLY? A COMPUTE-MATCHED, PRE-REGISTERED TEST

**What this is, and what it is not.** This is a new experiment, pre-registered in `experiments/pre_registration_randomized_composition.md` and committed at `e3fb038` before the first run. It is *not* a reinstatement of the randomized-menu comparison §F withdraws: that artifact stays withdrawn, because it is unpaired and compute-confounded, and nothing below is scored against it. The question here is narrower and answerable: holding defense computation fixed, does randomizing *which* composition is applied each round cost suppression relative to committing to the best member of the same set? The direction was fixed in advance as  $H_1 : \Delta > 0$ , randomization *hurts*, because that is the claim the withdrawn comparison made and the claim this test is built to substantiate.

**Three defects in the predecessor, and what each fix costs.** (a) *The arms were not seed-matched.* The old runner tested `randomize_order` and `rng.random() < 0.5`, which short-circuits, so the fixed arm never drew that number and the randomized arm consumed one extra draw per round from the same generator that selects participants. The arms therefore ran on divergent participant streams and no paired test was licensed. Fixed with a separate `policy_rng`, leaving `rng` for participants only, which is why every arm here was re-run rather than partly reused. (b) *A defense-count confound.* Every CM→FG round reduces to FoolsGold alone, so the old gap conflated randomization with dropping a stage. Every member of every randomized set here keeps FoolsGold as a genuine per-client upstream, so no round degenerates. (c) *A mixed- $n$  trap.* Every leg of every contrast below is recomputed per seed from `results/randomized_composition/summary.json` at seeds 42–46; the analyzer prints both legs’  $n$  on every row and marks any mismatch.

| arm            | condition         | def./rd | scaling ASR            | pixel ASR              | acc. (sc./px.)     |
|----------------|-------------------|---------|------------------------|------------------------|--------------------|
| 1              | fixed_fg_cm       | 2       | 0.1068 ± 0.0306        | 0.0988 ± 0.0201        | 0.682/0.664        |
| 1              | fixed_fg_rfa      | 2       | 0.0510 ± 0.0096        | 0.0452 ± 0.0118        | 0.499/0.495        |
| 1              | fixed_fg_tm       | 2       | 0.3843 ± 0.2594        | 0.2495 ± 0.1026        | 0.694/0.705        |
| 2              | rand_comp_strong  | 2       | <b>0.1522 ± 0.1091</b> | <b>0.1115 ± 0.0440</b> | 0.607/0.598        |
| 2              | rand_comp_all3    | 2       | 0.2317 ± 0.2780        | 0.0879 ± 0.0301        | 0.623/0.635        |
| 3 <sup>†</sup> | temporal_mix      | 1       | 0.5712 ± 0.3737        | 0.6600 ± 0.1543        | <b>0.307/0.765</b> |
| 4 <sup>†</sup> | pure_coord_median | 1       | 0.5186 ± 0.1413        | 0.4425 ± 0.1290        | 0.764/0.767        |

All seven conditions,  $n = 5$  seeds 42–46, population sd. <sup>†</sup> **Compute matching is claimed for arms 1–2 only:** arms 3 and 4 apply one defense per round and use roughly half the defense computation, so they are reference points and not comparators. Arm 2’s primary set is the *smaller* one, {FG→CM, FG→RFA}, whose members each suppress both committed attacks alone; FG→TM is a criterion-FAIL pair whose canonical scaling arm is  $0.508 \pm 0.269$ , above threshold, so a set containing it could degrade merely because a non-suppressing member was drawn. That 0.508 is the frozen  $n=3$  reading (seeds 42–44) and is *not* the table’s 0.3843, which is the same arm at  $n=5$ ; the two agree exactly on the three shared seeds by control 3, and the difference is seeds 45–46 alone. Both sets were fixed by rule in the pre-registration, not by any outcome measured here.

**Four positive controls, because this project has twice shipped a hook that silently never fired.** (1) *The hook fired:* every member of every randomized set appears in the stored per-round policy log across all 15 streams. (2) *The arms differ:* neither randomized arm is bit-identical to any fixed arm at any seed, the closest approach being  $\max \text{per-seed } |d| = 0.2207$  (`rand_comp_strong/scaling` against `fixed_fg_cm`) and  $0.0489$  (`rand_comp_all3/pixel`). (3) *The harness did not change the run:* all eight per-seed identity checks of the fixed arms against the canonical artifacts return  $\max |d| = 0.000e+00$  on their shared seeds, against `all_compositions` (seeds 42–44) and `fg_cm_survivor`’s composed (42–46). (4) *The end-to-end path reproduces a published standalone:* `pure_coord_median` returns 0.5186 and 0.4425 against `wave2_held_out`’s `fedavg.then_coord_median` at  $\max |d| = 0.000e+00$ , the same  $n = 5$  seeds. Controls 3 and 4 are what license comparing any arm here to a number already in print.

**Control 1 fails as literally pre-registered, and we report it rather than restate it.** The frozen assertion was a per-stream two-sided 99% binomial interval with no family-wise correction. Three of the 15 streams carry an excursion (four member-level flags): `rand_comp_strong` seed 46 draws 36/14 against [16, 34] (tail  $P = 0.00130$ ), and `rand_comp_all3` draws 26 at seed 45 and 27 at seed 46 against [8, 25] ( $P = 0.00492$ ,  $P = 0.00208$ ). Family-wise,  $P(\text{at least one excursion}) = 1 - 0.99^{15} = 0.140$ , so excursions at this rate are expected. We do not treat this as invalidating, and say why: the failure mode the control targets is a *dead* hook, which yields a member with count 0 or a 50/0 log, and every stream here draws every member. The pre-registered assertion is nonetheless the stricter one and it did not hold on all streams; that is the honest statement. One further correction of the pre-registration’s own arithmetic: it wrote `rand_comp_all3`’s interval as approximately 8–26, the exact interval is [8, 25], and the exact one is used above.

**The primary contrast, paired per seed.** `rand_comp_strong` against the best member of its own set, where “best” is fixed by rule as the lower full-artifact mean on that same attack, which selects `fixed_fg_rfa` on both. On *model scaling* the per-seed gaps are +0.0352, +0.2899, +0.1564, +0.0102, +0.0143, giving  $\bar{\Delta} = +0.1012$ , 5/5 concordant signs, one-sided Wilcoxon  $p = 0.0312$  and paired  $t = 1.866$  ( $p = 0.0677$ ). On the *pixel backdoor* they are +0.0816, +0.0428, +0.1581, +0.0133, +0.0356, giving  $\bar{\Delta} = +0.0663$ , 5/5 signs, Wilcoxon  $p = 0.0312$  and  $t = 2.603$  ( $p = 0.0299$ ). At  $n = 5$  the minimum attainable Wilcoxon  $p$  is  $1/32 = 0.031$ , stated in advance, so only 5/5 signs can reach  $p < 0.05$ . Post hoc: the randomized arm exceeds both members, not just the better one, in 3 of 5 seeds on each attack, so the gap is not interpolation between the two fixed arms. The secondary contrast, `rand_comp_all3` against the best of all three arm-1 conditions, runs the same way:  $\bar{\Delta} = +0.1806$  with 4/5 signs on scaling ( $p = 0.0625$ ) and +0.0427 with 4/5 on pixel ( $p = 0.0625$ ).

**The pre-registered outcome: all three admissible outcomes obtained, on three different objects.** Materiality required both  $\bar{\Delta} \geq 0.10$  and at least 4/5 positive signs. (i) *Material, in the pre-registered direction, on model scaling:*  $\bar{\Delta} = +0.1012$  with 5/5 signs, so we state, scoped to this menu and this attack, that randomizing over non-degenerate compositions costs suppression at matched defense computation. (ii) *Not material on the pixel backdoor:*  $\bar{\Delta} = +0.0663$  clears the sign leg but not the magnitude leg, and the honest phrasing at  $n = 5$  is that no material difference was detected and the smallest effect this design resolves is  $\approx 0.10$ . (iii) *The accuracy floor is breached by one arm:* `temporal_mix` under model scaling means 0.307 clean accuracy against a frozen floor of 0.35, so that cell is reported as uninterpretable for ASR rather than compared, and every contrast involving it inherits the exclusion. This is why arm 3 is a reference point in the table and carries no verdict.

**Three caveats, and what survives them.** First,  $\bar{\Delta}$  clears the 0.10 threshold by 0.0012 and is carried by two of five seeds (+0.2899 and +0.1564); on the three others the gap is  $\leq 0.036$ . A threshold met by 1.2% of itself is met, and it is also the weakest way a threshold can be met, so we do not describe the scaling result as large. Second, the two tests disagree in opposite directions across the two attacks: Wilcoxon reaches  $p = 0.0312$  on both while the  $t$ -test reaches  $p = 0.0299$  on pixel, where materiality *fails*, and only  $p = 0.0677$  on scaling, where it holds. The stable part of the result is the sign structure, 10/10 across both attacks, and not either magnitude. Third, control 1 fails as literally pre-registered, as recorded above. What survives all three is the weakest claim consistent with them and the one the design was built to license: at matched defense computation and with the degeneracy confound removed, randomizing over a set of compositions did not *help* on either attack and was consistently worse on both, materially so on one. Nothing here licenses a claim about randomized menus in general, about randomized *internals* such as noise or subsampling (L3), or about any menu but this one.

**And the attack is committed in both arms, which is the scope condition that bites hardest on what this test can be read as answering.** Randomization’s theoretical case is against an adversary that *adapts* to the composition: a menu the adversary cannot predict is a menu it cannot optimize against. Both attacks here are fixed in advance, identical in every arm, and blind to the draw. So this arm bounds the *cost* of randomizing under a committed threat and measures none of the benefit randomization is proposed for, and a favourable outcome would not have shown that benefit either. No adaptive-adversary test of a randomized menu is run here or anywhere else in this paper: the criterion-aware adversary of §E adapts to a composition that is *fixed* and known to it, which is the opposite of the setting randomization is meant to deny an adversary, and building an adversary that adapts to a distribution over menus is a separate experiment we do not run.

## S18 EQUIVALENCE TESTING AT THE EXISTING SEED COUNTS: WHERE THE FROZEN MARGIN PASSES AND WHERE IT ONLY APPEARS TO

**Why this is here and what it is not.** The pre-registered rule scores a *point* estimate against a margin:  $|\text{mean ASR}(\kappa=2) - \text{mean ASR}(\kappa=0)| < 0.15$  and every rung  $< 0.5$ . A point estimate inside a margin is consistent with a negligible effect *and* with an effect the sample cannot locate, and those are different claims. Table S9 scores the same frozen margin as a two-one-sided- $t$  (TOST) equivalence test on the *paired* per-seed differences, declaring equivalence only when the 90% interval lies inside  $(-0.15, +0.15)$ . It is **post-hoc**, reads only frozen artifacts, **adds no runs and revises no published verdict**; the emitter is `experiments/analyze_tost_existing.py`. Pairing is not a modelling choice: every rung of an arm runs at the same seeds, so each difference holds one data partition fixed and the seed-to-seed spread that dominates this suite drops out of the contrast.

**Which leg is a test: on most of these arms ASR’s own floor satisfies the lower one before any run.** Equivalence at a symmetric margin is two claims at once, that the effect does not rise past +0.15 and that it does not fall past -0.15, and on most of these arms the second one cannot fail. ASR is bounded below by 0, so for *any* dosed outcome whatsoever, a perfect defense included, the paired mean obeys  $\text{mean}_i(\Delta_i) \geq -\text{mean}_i \text{ASR}_i^{\text{id}}$ : the largest fall arithmetically available on a cell is its identity rung’s own mean ASR. **Where**

Table S8: **Is the lower TOST leg a test?** For each arm of Table S9: the identity rung’s mean ASR, which is the largest fall arithmetically available because ASR cannot go below 0; whether that exceeds the frozen 0.15 and so makes the lower leg a test; the 95% upper bound, which is what the one-sided reading is reported with; and the two legs’ own  $p$  values, whose *maximum* is the  $p$  Table S9 prints. A “no” in the third column is not a criticism of the arm’s result; it says that one of the two claims the margin makes was never at risk on that cell, and the row’s verdict is therefore reported as a non-increase with its upper bound. Read-only over frozen artifacts, no runs added, no published interval or verdict revised. Emitted by `experiments/measure_margin_reachability.py`.

| Arm                              | $n$ | identity mean | lower leg a test? | 95% upper | $p_{\text{lower}}$ | $p_{\text{upper}}$ |
|----------------------------------|-----|---------------|-------------------|-----------|--------------------|--------------------|
| Krum / scaling (flagship)        | 5   | 0.062         | no                | +0.026    | 0.0014             | 0.0004             |
| Krum / scaling (score-only)      | 5   | 0.062         | no                | +0.061    | 0.0071             | 0.0024             |
| Krum / scaling (EMNIST-byclass)  | 3   | 0.027         | no                | −0.001    | 0.0004             | 0.0003             |
| the same arm, pooled             | 5   | 0.024         | no                | +0.009    | 0.0000             | 0.0000             |
| Krum / scaling (ResNet18)        | 3   | 0.071         | no                | +0.101    | 0.0404             | 0.0143             |
| Reputation / scaling             | 5   | 0.017         | no                | +0.469    | 0.0175             | 0.6002             |
| cos_krum / pixel                 | 8   | 0.584         | yes               | −0.118    | 0.9639             | 0.0015             |
| coord_median / pixel             | 5   | 0.443         | yes               | +0.178    | 0.0005             | 0.0734             |
| coord_median / pixel (CIFAR-100) | 5   | 0.703         | yes               | +0.110    | 0.0001             | 0.0025             |

that mean is below the margin, the lower leg is satisfied before a single run happens. It is not a test, and the honest form of the claim is a pre-registered *non-increase* reported with its 95% upper bound. Where the identity mean exceeds the margin both legs are tests, the two-sided reading is earned, and those arms keep it. Table S8 scores every arm of Table S9 on this, and §D.2 already states the same arithmetic for one arm’s ladder branch, in the same words (*arithmetically unreachable*, at an identity rung of 0.062); what is added here is that it holds on nearly every arm this paper reads an equivalence off, which we had not said. **The one Mode S arm on which the lower leg is a test reports a fall that leaves the margin** (§D.9, identity mean 0.3232,  $\Delta = -0.286$ ): the two-sided reading fails there in the direction of more suppression while the one-sided bound on a rise holds at  $-0.087$ , which is why the distinction drawn here is substantive rather than a change of wording.

**Which leg sets the published  $p$ , and the answer is the wrong one.** TOST reports  $\max(p_{\text{lower}}, p_{\text{upper}})$ , so exactly one leg sets the number each row prints, and on 5 of the 6 arms this table marks equivalent that maximum is attained by the leg ASR’s floor has already settled. The flagship’s printed  $p=0.0014$  is its *lower* leg’s; the  $p$  for the claim actually being made, that suppression does not meaningfully *rise*, is 0.0004. The same holds for the score-only control (0.0071 printed, 0.0024 one-sided), the EMNIST-byclass row (0.0004, 0.0003) and ResNet18 (0.0404, 0.0143). **So the relabelling reports a stronger number than the one already published on each of those arms, not a weaker one**, and no published  $p$  is revised anywhere:  $p_{\text{tost}}$  stays exactly as printed and the upper-leg  $p$  is reported beside it as the separate quantity it is. The one equivalent arm whose  $p$  is set by the leg that *is* a test is the CIFAR-100 row, at an identity mean of 0.703.

**Beyond this table, and the count that may be quoted.** Three further arms carry an equivalence reading and are scored by the same emitter. The body’s  $n=20$  top-up of row 1 has identity mean 0.044 and upper bound +0.012; Mode M, the coordinate-masking class of §D.2.1, has 0.053 and +0.019; the 200-round ResNet18 replication has 0.045 and +0.138. All three have an arithmetic lower leg. Counted over *distinct arms* rather than rows, because the flagship and its EMNIST replication are each scored in more than one place at more than one seed block, the paper makes equivalence readings about 5 arms and 4 of the 5 **have an arithmetically satisfied lower leg**. The fifth, `coord_median / scaling` under the outcome-gated design (identity mean 0.557, 95% CI [+0.023, +0.102],  $n=20$ ), keeps a genuinely two-sided reading and is reported as one; after its seed top-up that interval is inside the margin *and* excludes zero, which separates the margin reading from a reading of no effect on this arm. So *every* equivalence claim is not the right quantifier and is not used. **The frozen point-estimate rule has the same arithmetic on its lower side:**  $\Delta > -0.15$  is free whenever the identity mean is below 0.15, so a sentence reading  $|\Delta| < 0.15$  off one of these arms inherits the same one-sidedness. **Nothing here re-margins, re-tests or withdraws a published interval;** the emitter is `experiments/measure_margin_reachability.py`, which reads frozen artifacts only, recomputes every interval with the estimator its own table used, and asserts each one equal to the published value before it reports anything about that arm’s legs (`results/margin_reachability.json`). **The fifth arm’s  $n=20$  figures above are the one exception to that provenance, and not to the claim:** that artifact predates this arm’s top-up and still carries its identity mean at 0.519 on  $n=5$ , so the 0.557 and the interval quoted beside it are read from `results/coord_median.scaling.topup.summary.json` instead, and no frozen value is rewritten to agree with them.

Table S9: **TOST against the frozen  $\pm 0.15$  margin at the seed counts already on disk**, on the paired per-seed difference (top rung minus identity). Equivalence requires the 90% interval inside the margin, which at  $\alpha=0.05$  is  $\max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$ . <sup>†</sup> **equivalent and detected**: the 95% interval also excludes zero, so the verdict means “no effect larger than 0.15”, **never** “no effect”. Those two readings are not exclusive and the marker exists so a checkmark cannot be read as the stronger one; on the CIFAR-100 row the paper reports a *measured rise* (§D.3) and this row does not retract it. **One row’s 95% interval is itself wider than the margin**: ResNet18’s  $[-0.183, +0.101]$ . TOST at  $\alpha=0.05$  reads the 90% interval, so that row passes the test defined here while failing the 95%-interval criterion §S13’s margin ladder uses; both readings are disclosed at §D.2, and the two conventions are not interchangeable. Post-hoc; no runs added. Emitted by `experiments/analyze_tost_existing.py`.

| Arm                                           | $n$ | paired $\Delta$ | 95% CI             | 90% CI (TOST)      | verdict                 |
|-----------------------------------------------|-----|-----------------|--------------------|--------------------|-------------------------|
| Krum / scaling (flagship)                     | 5   | -0.026          | $[-0.078, +0.026]$ | $[-0.066, +0.014]$ | equivalent              |
| Krum / scaling (score-only)                   | 5   | -0.023          | $[-0.108, +0.061]$ | $[-0.088, +0.042]$ | equivalent              |
| Krum / scaling (EMNIST-byclass)               | 3   | -0.017          | $[-0.033, -0.001]$ | $[-0.028, -0.006]$ | equivalent <sup>†</sup> |
| the same arm, pooled                          | 5   | -0.013          | $[-0.034, +0.009]$ | $[-0.029, +0.004]$ | equivalent              |
| Krum / scaling (ResNet18)                     | 3   | -0.041          | $[-0.183, +0.101]$ | $[-0.137, +0.055]$ | equivalent              |
| Reputation / scaling                          | 5   | +0.178          | $[-0.112, +0.469]$ | $[-0.045, +0.402]$ | not est.                |
| <code>cos_krum</code> / pixel                 | 8   | -0.425          | $[-0.732, -0.118]$ | $[-0.671, -0.179]$ | not est.                |
| <code>coord_median</code> / pixel             | 5   | +0.098          | $[+0.018, +0.178]$ | $[+0.036, +0.160]$ | not est.                |
| <code>coord_median</code> / pixel (CIFAR-100) | 5   | +0.071          | $[+0.031, +0.110]$ | $[+0.040, +0.101]$ | equivalent <sup>†</sup> |

**What it shows, including against us. The flagship passes**, and passes at  $n=5$ :  $p=0.0014$ , the pairing holding sd to 0.042, so *the equivalence claim on the adjudicating arm does not depend on a larger seed count*. On this arm that  $p$  is set by the margin’s lower leg, which ASR’s own floor satisfies before any run: the one-sided  $p$  for the claim actually made, that suppression does not weaken, is the smaller 0.0004. Table S8 prints both legs for every row of this table and revises no published  $p$ . So do the score-only control ( $p=0.0071$ ) and the EMNIST-byclass replication ( $p=0.0004$ ,  $n=3$ ,  $sd=0.007$ , and  $p < 0.0001$  at the pooled  $n=5$  of §D.2, where the interval widens across zero but stays well inside the margin). **The ResNet18 replication passes as well, and it is the closest call in the table**:  $p=0.040$  at  $n=3$  on two endpoint rungs, its 90% interval reaching  $-0.137$  against the  $-0.15$  boundary. **Its 95% interval,  $[-0.183, +0.101]$ , is wider than the margin**, so that arm’s equivalence is established at the frozen  $\pm 0.15$  and at no margin below 0.137; §D.2 discloses this beside its pre-registered point-estimate verdict, and §S13’s ladder statement is scoped to the comparability arms for exactly this reason. The two arms whose suppression genuinely moves correctly **fail** it, which is the positive control working: reputation (+0.178) and `cos_krum` ( $-0.425$ ). **And one arm passes the frozen rule while failing the test**: `coord_median`’s  $+0.098$  ( $n=5$ ) sits inside  $\pm 0.15$ , but its 90% interval reaches  $+0.160$  and TOST gives  $p=0.073$ . We therefore report that arm’s equivalence as **not established** rather than as inherited from the point estimate. **This is the one place the paper’s frozen scoring rule was more permissive than the evidence warrants, and we say so rather than leave the width implicit**. `cos_krum` additionally sits above the 0.5 ceiling the frozen rule conjoins with the margin (0.584 at the identity), which the body already states as its never suppressing.

**Two arms are equivalent and detected at once, and the table marks them**. The EMNIST-byclass row at the frozen  $n=3$  and the CIFAR-100 row of §D.3 both have 90% intervals inside the margin and 95% intervals excluding zero: a real effect, smaller than the margin. Nothing is inconsistent here, but a bare “equivalent” on such a row would be read as “no effect”, so both carry <sup>†</sup>. The two rows differ in one further way, which Table S8 scores: the EMNIST-byclass row’s identity-rung mean ASR is 0.027, so its lower leg is satisfied by arithmetic and its only test is the upper one, which bounds a rise at  $-0.001$ ; the CIFAR-100 row’s identity mean is 0.703, so both of its legs are genuine tests and it is the one row *marked equivalent* here whose two-sided reading is earned. **The CIFAR-100 row is the sharper case**: §D.3 reports it as a *rise* of  $+0.071$  against the confounded design’s fall of  $-0.213$ , and that reading stands. What this row adds is its size, under half the frozen margin, which is why the sign reversal there is reported as a reversal of *direction* and never as a large controlled effect. **This arm is scored here and excluded from the binding-arm search of §S13**, on the separate ground that a margin-sensitivity analysis may only rest on arms about which the paper makes an equivalence claim, and it makes none about this one.

**Where  $\pm 0.15$  came from, reconstructed and not pre-registered**. The margin was frozen at 5130cec with *no written rationale*, which we state plainly rather than supply a derivation after the fact. What the pre-freeze record does support: the effect the design was built against was the pre-registered ladder’s own  $+0.762$  rise at  $\rho=54.6$ , five times the margin, and the margin never stands alone, the frozen rule conjoining it with *every rung*  $< 0.5$ . So the margin is deliberately generous and **the conjunction is what stops a generous margin**

**from certifying a high-ASR arm;** on the flagship cell every rung’s *mean* in fact sits below 0.07, the quantity the frozen rule scores. **And on five of the six rows above that this table marks equivalent, one side of that margin was never at risk:** the lower leg is satisfied by ASR’s own floor before any run, so the verdict column on those rows is read as a non-increase with the 95% upper bound that Table S8 prints beside it. The exception is the `coord.median / pixel` (CIFAR-100) row, whose identity-rung mean ASR is 0.703 and whose two-sided verdict is earned.

Table S10: **The frozen  $\pm 0.15$  is not load-bearing; every equivalence reading in the comparability table survives the whole ladder.**  $\checkmark$  if the arm’s *controlled* 95% interval lies inside  $[-m, +m]$ ;  $m^*$  is the smallest margin at which it does. An arm is scored only where its controlled interval contains zero, and the others are marked absent (—) because their verdict is a detected effect and reads no margin at all. The one row marked  $\times$  at every rung, reputation / scaling, is scored because its interval contains zero, but that interval is too wide ( $[-0.112, +0.469]$ ) for any margin here and the paper makes no equivalence claim about that arm, so the headline above is not contradicted by it. **These are the controlled intervals of the comparability design** (§D.3), *not* Table S9’s paired 90% TOST intervals: the two differ by design and by seed count, so Krum / scaling is  $n=20$  here against  $n=5$  there, and `coord.median / pixel` is  $n=20$  here (both legs, after the pre-registered endpoint top-up, seeds 47–61) against the frozen  $n=5$  there. **coord.median / scaling is the one row this artifact still reads at  $n=5$  while §D.3 now reports it at  $n=20$ :** `results/margin_sensitivity.json` predates that arm’s top-up and is left byte-untouched, so the  $m^*$  printed for it is the  $n=5$  bound 0.280 and the  $n=20$  bound is 0.247, given here so the frozen figure cannot be read as the current one. Two absent arms have  $m^*$  inside the ladder’s own span (0.155 and 0.110), named here so the absence cannot be mistaken for a hidden flip. **One arm of the paper is not in this artifact at all:** the `resnet18` replication (§D.2) has only the controlled leg and so no comparability cell, and its own  $m^*$  is 0.183, *wider* than the frozen margin. The emitter derives and prints that number beside this table, so the ladder cannot be read as covering it. **On both rows this ladder marks  $\checkmark$ , and on that `resnet18` arm,  $m^*$  is `|lo|` and not `|hi|`:** identity-rung mean ASR is 0.044, 0.024 and 0.071 there, so a fall of 0.15 is arithmetically unavailable and the lower leg is satisfied before any seed runs, which means those three  $m^*$  values measure the side of the margin that was never at risk. Read on the leg that is a test, the 95% bound on a rise is  $+0.012$  for krum / scaling,  $+0.009$  for its EMNIST replication and  $+0.101$  for `resnet18`, all three inside the frozen margin (Table S8). Emitted by `experiments/analyze_margin_sensitivity.py` part (D); the fuller sweep, including the  $\pm 0.05$  AGREE tolerance over all of  $[0, 1]$ , is supplement §S13.

| Arm                                           | $m=0.05$     | 0.10         | 0.15         | 0.20         | $m^*$ |
|-----------------------------------------------|--------------|--------------|--------------|--------------|-------|
| krum / scaling                                | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ | 0.032 |
| <code>cos.krum / pixel</code>                 | —            | —            | —            | —            | 0.732 |
| reputation / scaling                          | $\times$     | $\times$     | $\times$     | $\times$     | 0.469 |
| <code>coord.median / pixel</code>             | —            | —            | —            | —            | 0.155 |
| <code>coord.median / scaling</code>           | —            | —            | —            | —            | 0.280 |
| krum / scaling, EMNIST                        | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ | 0.034 |
| <code>coord.median / pixel</code> , CIFAR-100 | —            | —            | —            | —            | 0.110 |

**What the ladder shows, and the one arm it marks  $\times$ .** Three arms are scored, the three whose controlled interval contains zero, and none of their verdicts changes anywhere on the ladder, so the reading the body rests on would be identical at  $\pm 0.05$ . **What the ladder does not show is that the lower side of any of those margins was ever at risk:** on the two krum rows the identity-rung mean ASR is 0.044 and 0.024, so  $\Delta > -m$  holds by arithmetic at every rung of the ladder and what varies down the column is the upper leg alone (Table S8). The exception is reputation / scaling, which fails at every rung: its controlled interval contains zero but is wide,  $[-0.112, +0.469]$ , so no margin in this range licenses an equivalence claim there, and the paper makes none. *The sign reversal turns on neither threshold:* it is defined by opposite signs with both intervals excluding zero.

## S19 COMPOSITION AND ADAPTIVE-ATTACK DETAILS

### S19.1 COMPOSITION BASELINE DETAILS

Simultaneous composition (NormClip  $\tau=5$  then reputation every round),  $N=10$ ,  $K=5$ ,  $f=0.2$ , 50 rounds, `cifar.cnn`, 5 seeds. Per-seed results:

| Seed           | model_scaling     |                                     | backdoor_pixel    |                                     |
|----------------|-------------------|-------------------------------------|-------------------|-------------------------------------|
|                | Accuracy          | ASR                                 | Accuracy          | ASR                                 |
| 42             | 0.783             | 0.766                               | 0.774             | 0.837                               |
| 43             | 0.788             | 0.783                               | 0.774             | 0.792                               |
| 44             | 0.794             | 0.835                               | 0.784             | 0.828                               |
| 45             | 0.762             | 0.889                               | 0.771             | 0.931                               |
| 46             | 0.780             | 0.763                               | 0.766             | 0.823                               |
| Mean $\pm$ std | $0.781 \pm 0.011$ | <b><math>0.807 \pm 0.048</math></b> | $0.774 \pm 0.006$ | <b><math>0.842 \pm 0.047</math></b> |

Composition partially suppresses scaling (ASR 0.807 vs. pure NormClip  $\sim 0.95$ , and 0.961 under the NE3 temporal-mixing policy defined in §S19.2) but does not approach pure reputation (0.017). Norm clipping preserves the attack direction while reducing magnitude; the clipped update then passes reputation’s consensus-distance filter. Pixel attacks are similarly persistent under composition (ASR 0.842). Both attack families remain effective despite simultaneous application of both defenses.

### S19.2 DBA AT $f=0.4$ : PER-SEED DETAILS

**Three run labels are carried over from an earlier framing and used here only to identify runs**, not to make any equilibrium claim: *NE3* is the mixed server policy that framing computed, which randomizes over defenses across rounds; *reputation eq* is the pure-reputation policy; and *BR* is the adversary’s best response to the policy it faces. NE3 BR on CIFAR-10 with DBA in place of model scaling at  $f=0.4$  (raising adversarial co-participation to  $K \cdot f = 2.0$  from the  $f=0.2$  value of 1.0).  $N=10$ ,  $K=5$ , 5 seeds 42–46, 50 rounds. Per-seed realized ASRs:

| seed           | DBA realized ASR  | backdoor_pixel realized ASR |
|----------------|-------------------|-----------------------------|
| 42             | 0.222             | 0.972                       |
| 43             | 0.103             | 0.945                       |
| 44             | 0.176             | 0.913                       |
| 45             | 0.603             | 0.993                       |
| 46             | 0.077             | 0.934                       |
| mean $\pm$ std | $0.236 \pm 0.190$ | $0.952 \pm 0.028$           |

The DBA per-seed spread is  $7.8\times$  between min and max; only seed 45 clearly persists, three (0.08, 0.10, 0.18) are below the threshold ( $\geq 0.20$ ) we would call active persistence, and one (0.22) is borderline. Pixel persists cleanly at  $f=0.4$  ( $0.952 \pm 0.028$ ), so the persistence mechanism survives an adversarial-fraction shift on the original attack. DBA’s seed-variability reflects per-round co-participation alignment (whether  $\geq 2$  of the 4 adversarial clients land in the same round to coordinate the distributed trigger) rather than the structural argmax-collapse mechanism we describe for scaling. The cleanest evidence for the headline persistence claim therefore remains model scaling; DBA at  $f=0.4$  is suggestive but not a robust second-attack demonstration.

### S19.3 DBA AT $f=0.6$ : PER-SEED DETAILS AND BIMODALITY REPRODUCTION

Follow-up to Appendix S19.2, designed to test whether the DBA bimodality at  $f=0.4$  is purely a per-round co-participation effect or whether it persists as a per-seed admissibility latent at higher  $f$ . At  $f=0.6$ , expected adversarial co-participation per round is  $K \cdot f = 3.0$ , making  $\geq 2$  adversarial co-participation per round near-certain. NE3-style BR with committed adversary playing DBA; 5 seeds (42–46), 50 rounds,  $N=10$ ,  $K=5$ , *cifar\_cnn*. Two deployed policies: *NE3 mix* (FedAvg 26% + NormClip 74%) and the *reputation equilibrium* (NormClip 83% + reputation 17%).

| seed           | DBA ASR (NE3 mix) | DBA ASR (reputation eq) |
|----------------|-------------------|-------------------------|
| 42             | 0.360             | 0.339                   |
| 43             | 0.295             | 0.278                   |
| 44             | 0.386             | 0.415                   |
| 45             | <b>0.734</b>      | <b>0.705</b>            |
| 46             | 0.154             | 0.130                   |
| mean $\pm$ std | $0.386 \pm 0.192$ | $0.373 \pm 0.190$       |

**Observations.** (i) Mean DBA ASR improves over  $f=0.4$  (0.236) to  $\sim 0.38$  on both policies, but does not reach the  $\geq 0.85$  persistence band scaling achieves. (ii) Per-seed bimodality persists at  $f=0.6$ : seed 45 alone reaches  $> 0.7$  in both policies; the remaining four seeds cluster in  $[0.13, 0.42]$ . (iii) Crucially, *the same seed*

is the strong-persistence seed under both policies (NE3 mix: 0.734; reputation eq: 0.705), and the same four seeds are the weak-persistence seeds under both: the per-seed cross-policy correlation rules out per-round co-sampling alignment as the dominant source of bimodality. The bimodality is a per-seed training-initialization latent in DBA itself: whether the distributed trigger embeds at all in the early rounds. This is the DBA analogue of the model-scaling binding-cell Bernoulli, but at a much lower mean (admissibility prevalence  $\approx 1/5$  for DBA at  $f=0.6$  vs.  $\approx 4/5$  for scaling at  $f=0.2$ ).

**Pre-specified verdict: INTERMEDIATE.** The four-way rule that emits it, STRONG then PARTIAL then NEGATIVE then INTERMEDIATE with its thresholds, is in `experiments/run_dba_f06.py`, so the label is decided by the runner rather than chosen after reading the table. Overall mean 0.380 falls between PARTIAL ( $> 0.5$ ) and NEGATIVE ( $< 0.3$ ). Reading: DBA at  $f=0.6$  does *not* replicate the persistence collapse with the cleanness scaling achieves; the trigger-alignment mechanism has its own admissibility constraint we do not control by raising  $f$ . The cleanest headline-persistence evidence remains model scaling under norm-based menus, as acknowledged in L3.

## S20 SCALE, ARCHITECTURE, AND EXTERNAL VALIDITY

### S20.1 SCALE VALIDATION ( $N = 100$ ) AND DEFENSE-INFERENCING CHECK

**Scale check at  $N = 100$  ( $4 \times 5$  game, 5 seeds). Residual material from an earlier framing, recorded rather than dropped.** Nothing in the screen, the identification result or the dose-response evidence depends on it. Write VoPD (value of policy diversity) for the gain the server’s equilibrium strategy achieves over its best *pure* defense, so  $\text{VoPD} = 0$  exactly when the equilibrium is pure. **This is not the quantity an earlier draft called “realized VoPD”, and that draft described its own quantity with the bound direction inverted, which is worth recording rather than quietly dropping.** There the reported difference was oracle-minus-committed, and it was said to *lower*-bound the gap an adversary with no information about the server’s draw could realize. It does the opposite: the committed policies are a *subset* of that adversary’s available policies, so its best policy is at least as good as the best committed one, and subtracting the larger quantity makes oracle-minus-committed an *upper* bound on the gap. That quantity is withdrawn with its framing, nothing in this paper uses it, and the VoPD defined above is a different object, non-negative by construction and sharing only the name. The equilibrium is pure on seeds 43 to 46 (model scaling vs. RFA,  $\text{VoPD} = 0$ ) and mixed on seed 42 (pixel + scaling vs. FedAvg + RFA,  $\text{VoPD} = 0.012$ ); per-seed values in Table S11.

Table S11: Per-seed NE, CIFAR-10  $N = 100$  ( $4 \times 5$  game). Seeds 43–46 identical.

| Seed  | Type  | Adversary       | Server       | VoPD  |
|-------|-------|-----------------|--------------|-------|
| 42    | Mixed | Pixel + Scaling | FedAvg + RFA | 0.012 |
| 43–46 | Pure  | Model scaling   | RFA          | 0.000 |

### S20.2 RESNET18 ARCHITECTURE-REPLICATION MINI-SWEEP: PER-SEED DETAILS

Per-seed details for the ResNet18 architecture-replication mini-sweep behind the architecture-conditional collapse boundary in §G. `resnet18` with BatchNorm replaced by GroupNorm (8 groups), conv1 adapted to  $32 \times 32$  (kernel 3, stride 1, maxpool  $\rightarrow$  Identity); 11.2M parameters. Identical FL parameters to `cifar_cnn`:  $N=10$ ,  $K=5$ ,  $f=0.2$ , 50 rounds,  $\tau = 5$ . Committed-scaling BR, 3 mix points  $\times$  5 seeds = 15 runs at  $\sim 100$  min/run on MPS. **Reading the table.** *mix (NC/rep)* is the server’s per-round mixture over the two upstream defenses, `norm_clip` (NC) and `reputation` (rep), so the 90/10 row draws `norm_clip` in 90% of rounds and `reputation` in the other 10%; every cell is trigger ASR under the committed scaling attack; the two *mean* columns are over the same five seeds, the second on `cifar_cnn` from `results/cifar10_mix_ratio_sweep/summary.json`, whose sweep carries five mix points of which the three run here are shown. Both spreads are population standard deviations over the five seeds, recomputed here from the per-seed values and not the Student- $t$  intervals the rest of the paper reports.

| mix (NC/rep) | s42   | s43   | s44   | s45   | s46   | ResNet18 mean                       | cifar_cnn mean                      |
|--------------|-------|-------|-------|-------|-------|-------------------------------------|-------------------------------------|
| 90/10        | 0.926 | 0.929 | 0.965 | 0.974 | 0.962 | <b><math>0.951 \pm 0.020</math></b> | $0.945 \pm 0.021$                   |
| 50/50        | 0.924 | 0.971 | 0.813 | 0.920 | 0.903 | <b><math>0.906 \pm 0.052</math></b> | $0.815 \pm 0.104$                   |
| 10/90        | 0.917 | 0.733 | 0.783 | 0.975 | 0.915 | <b><math>0.865 \pm 0.091</math></b> | <b><math>0.097 \pm 0.090</math></b> |

NC-dominant endpoint replicates within 0.01; mid-mix is  $\sim 9\%$  higher on ResNet18; the reputation-dominant endpoint shows an order-of-magnitude gap (no CI overlap). Seed 45’s accuracy collapses to 0.153 at NC10/rep90 (the persistent attack effectively breaks the model) while trigger ASR remains 0.975. **Verdict:**

**MEDIUM** (monotonic ordering preserved, endpoint criterion fails), on the scale stated in `experiments/run_mix_ratio_sweep_resnet18.py`; that label name is reused with different meanings by other runners in this repository, which is why its criterion is restated here rather than referenced. The `cifar_cnn` collapse boundary at  $\sim 70\text{--}90\%$  reputation is therefore a property of the smaller-architecture regime. This sweep bounds the architecture-conditional claim of §G and establishes no boundary of its own; we draw no conclusion here about which mixture of the two defenses survives, since mixture policies are outside this paper’s scope (§C).

## S21 AN APPLICATION: SCREENING AS EVALUATION PRIORITIZATION

Table S12: **What this design can and cannot conclude.** The middle column is about *identifiability*, not statistical power: a “no” marked *structural* cannot be repaired by more seeds, defenses or attacks (Proposition 7). Every entry is scoped to the measured setting of §D.2’s interventions and §S24’s menus.

| question                                                         | identifiable?         | answer, and its evidence                                             |
|------------------------------------------------------------------|-----------------------|----------------------------------------------------------------------|
| Does disturbing $d_2$ ’s statistic change its <i>decision</i> ?  | measured              | <b>Yes:</b> Krum’s selection flips in up to 0.80 of rounds           |
| Does it change the <i>admitted adversarial mass</i> ?            | by construction       | <b>No:</b> unchanged in every one of the 60 measured rounds          |
| Does it change <i>suppression</i> ?                              | instrumented          | <b>No:</b> $ \Delta =0.026$ , inside the frozen margin (Prop. 12(a)) |
| Does <i>adversary-correlated</i> reweighting change suppression? | instrumented          | <b>Yes,</b> non-monotonically: single-peaked (Mode A)                |
| Does admitted mass <i>order</i> outcomes across defenses?        | pre-registered        | <b>No, refuted:</b> $+0.098$ ( $n=5$ ) against a frozen $+0.150$     |
| Is (P4) <i>sufficient</i> for (P5)?                              | control arm           | <b>No:</b> <code>cos_krum</code> admits 0.000 while ASR falls        |
| Can (P4) be screened <i>without</i> adversary identity?          | <b>no, open</b>       | Both instruments are oracles: the stated open problem                |
| Does C1 identify C2’s contribution?                              | <b>no, structural</b> | Empty cell by the gate, for every menu (Prop. 7)                     |
| Does any of this certify <i>deployed</i> security?               | <b>no, scope</b>      | Committed attack only; one adaptive test breaks it                   |

**What follows is a consequence of the identification boundary, not a claim the paper argues for.** Proposition 7 says a criterion gated on the outcome it predicts cannot be validated by cross-defense contrast; what such a criterion can still do is *order an evaluation budget*, and that is the only use we claim for it. Proposition 8 bounds what it can find. Read the numbers below as a measurement of a prioritization heuristic’s cost, not as validation of the framework.

**Does every condition earn its place? C3 does not: reported as a finding.** Scoring all 42 pairs on C1, C2 and C3 *independently* rather than recording only the first failure: against a 12% base rate of low-ASR pairs, C1 alone selects 4 at 50% precision, C2 alone 14 at 36%, and  $C1 \wedge C2$  selects 2 at 100%. **The surviving C2 failures are exactly the non-invariant signal class:** after the Proposition 11 reclassification every one of them is a  $\rightarrow$ reputation pair ( $nc \rightarrow rep$  0.819,  $rfa \rightarrow rep$  0.882,  $fg \rightarrow rep$  0.798), consensus distance being the signal a positive rescaling does not preserve, while all three non-degenerate  $\rightarrow$ FoolsGold pairs are C1 failures instead. **C3 has no demonstrated incremental predictive value in this study:** alone it selects 35 pairs at 9% precision, below the base rate, and it never binds on a pair  $C1 \wedge C2$  already admits. **So we demote C3 from a gate to a margin diagnostic and recommend  $C1 \wedge C2$  as the criterion,** retaining C3 on theoretical grounds only: Theorem 9 shows margin compression is a real failure mode, and the  $\alpha=0.1$  regime below is precisely a margin collapse. **The demotion changes no number,**  $C1 \wedge C2$  selects the same two pairs at the same precision (recomputed by `analyze_screening_cost.py`), which is why we report it as a finding about the criterion rather than a revision of results. *The pre-registered PASS/FAIL labels and the 16/18 and 22/24 tallies were frozen with C3 in the gate and are not relabelled;* this ablation is post-hoc and labelled as such, and the recommendation is for use going forward.

**If the goal is to find emergent compositions, drop C1: a recommendation we price but do not validate.** Proposition 8 says an inherited-suppression gate cannot certify a composition whose suppression neither constituent possesses, so an evaluator whose object is discovery rather than prioritization should not run C1 at all, and this is a consequence of the proposition rather than a finding of ours. Two designs follow, and §S22 prices both on our own menu. **Sample the ungated menu** when the budget allows: 42 pairs at 12% precision reaches every low-ASR pair by construction, and it is the only design with no structural blind spot.

**Otherwise gate on the mechanism condition alone**, C2 without C1: it admits 14 of the 42 at 36% precision, a third of the ungated cost, and it reaches FoolsGold→RFA where every C1-bearing variant misses it. **The caveat is ours and it is not small**: C2-alone’s recall on this menu is 100%, but 11 of those 14 admissions rest on a per-pair call made in sample rather than on algebra, and all 5 low-ASR pairs are among the 11 (§S22), so that recall is a development-set number and we offer the design as a recommendation rather than as a validated procedure. Finding emergence efficiently is a search problem, and nothing here solves it: what the propositions establish is which designs cannot solve it.

## S22 THE SCREENING FRONTIER: PRECISION, RECALL AND EVALUATION COST

*What does the screen cost, and what does it cost to find out?* 40% **recall is an operating point, not a verdict on the conditions**. Scoring C1, C2 and C3 independently over the 42 pairs traces a frontier, and the criterion the paper reports is its high-precision end. Emitted by `experiments/build_screening_frontier.py`, which joins the two existing generators and introduces no number of its own.

**Neither screen is noise, and neither is a predictor**. Both C2 alone and  $C1 \wedge C2$  beat a size-matched random draw from the same 42 pairs ( $P \leq 0.005$  at  $10^5$  resamples against the menu’s 12% base rate, seeded `rng(42)`; emitted by `experiments/blind_selection_analysis.py`). Beating a random draw is the whole of what that establishes: it separates the screens from chance and not from each other, and it says nothing about either one’s ability to predict suppression.

| Variant                                   | selected | precision | recall | mean ASR | runs | reaches FG→RFA |
|-------------------------------------------|----------|-----------|--------|----------|------|----------------|
| no screen (all pairs)                     | 42       | 12%       | 100%   | 0.743    | 420  | ✓              |
| C3 alone                                  | 35       | 9%        | 60%    | 0.760    | 350  | ✓              |
| C2 alone                                  | 14       | 36%       | 100%   | 0.588    | 140  | ✓              |
| C2 algebraic only <sup>†</sup>            | 3        | 0%        | 0%     | 0.869    | 30   | —              |
| $C2 \wedge C3$                            | 8        | 38%       | 60%    | 0.562    | 80   | ✓              |
| C1 alone <sup>‡</sup>                     | 4        | 50%       | 40%    | 0.514    | 40   | —              |
| $C1 \wedge C3^{\ddagger}$                 | 4        | 50%       | 40%    | 0.514    | 40   | —              |
| $C1 \wedge C2^{\ddagger}$ (the criterion) | 2        | 100%      | 40%    | 0.242    | 20   | —              |
| $C1 \wedge C2 \wedge C3^{\ddagger}$       | 2        | 100%      | 40%    | 0.242    | 20   | —              |

<sup>†</sup> additionally requires 60 standalone runs before the condition can be evaluated at all, so this row’s evaluation budget is  $20+60 = 80$  runs, not the composition runs alone.

<sup>‡</sup> the same screen restricted to the pairs whose C2 verdict Prop. 11 settles algebraically.

percentages are over small denominators: 5 of the 42 pairs are low-ASR, so the criterion’s precision is  $2/2$ , C1 alone’s and  $C1 \wedge C3$ ’s are  $2/4$ , and every C1-bearing recall is  $2/5$ .

*mean ASR* averages max-committed ASR over the admitted pairs;

a lower mean is not a better screen, since the C1 end reaches it by discarding the menu’s lowest-ASR pair.

*mean ASR* scores every pair at the menu’s uniform  $n=3$ , not at the marked

seed counts of Table S13, where the two admitted pairs read 0.107 and 0.463.

This table is the only home of these rates.

**The blind spot is attributable to exactly one condition, and the theory named it before the ablation was run**. Every C1-bearing variant misses FoolsGold→RFA and every C1-free variant reaches it, the algebraically restricted <sup>‡</sup> row aside, which admits no low-ASR pair at all: FoolsGold→RFA’s row is C1 *false*, C2 *true*, C3 *true*, max-committed ASR 0.045. C1 is the inherited-suppression gate, and Prop. 8 proves no gate of that form can certify an emergent composition, so the miss is a *predicted* property of one condition rather than an unexplained failure, and it is tradeable: C2 alone recovers every low-ASR pair in the menu at 36% precision. **We report the high-precision end because false positives deploy an insecure stack**, not because the conditions cannot be operated at higher recall.

**The C2 column is not all one kind of evidence, and the high-recall claim depends on the weaker kind**. We describe C2 as read off the two defenses’ definitions, and for two of its four branches that is literally what happens: a  $d_2$  reading a cosine is exactly invariant under any positive per-client rescaling (Prop. 11(a)) and a  $d_2$  reading a consensus distance, a residual magnitude or a norm is not (Prop. 11(c)). The remaining branch is *conditional*: `coord.median` and `trimmed.mean` preserve a coordinate ordering only under a separation condition, and Thm. 9(2) gives `rfa` an analogous one, so `analyze_condition_ablation.py` marks all three `None` and `c2.holds` defers to a *per-pair call*, the stored category. Nothing in the C2 column distinguishes a verdict reached by algebra from one reached by that call, so we census it (`c2.provenance`, same generator): of the 14 pairs C2 admits, 3 **rest on algebra and 11 on the per-pair call**, the 11 being every `coord.median`, `trimmed.mean` and `rfa` row. **And all 5 low-ASR pairs are in the 11**: 0 of 5 are algebraically settled. Hence the <sup>‡</sup> row above. **Worse for us, all 11 are development-set pairs**, so those

calls were made *in sample*, while the criterion was being written; they are not pre-registered, and the generator derives the split from which results artifact holds each pair rather than taking our word for it. **Two consequences we accept rather than repair.** First, C2 alone as we score it *is not a pure statistic-preservation screen* for those three  $d_2$  classes, so the body’s contrast between it and C1/C2 is a contrast between two screens that both borrow from the same in-sample call, not between algebra and measurement. Second, *within this menu* the region where our own invariance result decides the question contains no low-ASR pair at all, so the algebraically settled sub-region cannot discriminate any strategy from any other *here*, and the 100% recall of C2 alone is carried entirely by in-sample judgement. **The second consequence is the one we then removed**, by enlarging the algebraic region until it closed rather than by relabelling this one: the 3 pairs of the  $\ddagger$  row are all three upstream defenses against the single downstream defense `foolsgold`, which is one column of a  $3 \times 3$  census over  $d_1 \times d_2$ , and completing that census prospectively puts a low-ASR pair inside the algebraic region and separates the two strategies more sharply there than anywhere in this table (§S23). The *first* consequence stands and needs an invariance theorem for the conditional classes, which we do not have; what neither needs is a relabelling, and no number in this appendix or the body changes. This is (L5) in concrete form: the conditions are not always readable off a  $(d_1, d_2)$  pair.

**What the screen saves, in runs rather than in pairs.** `experiments/analyze_screening_cost.py`: unscreened, the 42 pairs under 2 committed attacks at 5 seeds are 420 runs; the reported criterion evaluates 16 cells, 80 runs, of which 60 are the standalone measurements C1 needs. That is 340 runs avoided, 81% of runs. We give the run-level figure rather than the pair-level 95.2%, which overstates the saving by ignoring C1’s own standalone cost. **The unit is runs, and the hours behind them are now measured rather than assumed:** the runners print per-run seconds into their logs, and the six the script reads ship with the code, so the figure is checkable; over the 228 logged CIFAR-10 pair-evaluation runs the median is 11.1 min (mean 13.1, IQR 10.7–12.5), so the 340 avoided runs are 63 GPU-hours on the one machine, or 74 at the mean. *Correcting our own caveat:* through the previous version this paragraph said no per-run timing existed anywhere in the results and the artifact assumed 11.0 min/run. The runner logs did carry it; the assumption happens to have been accurate to 0.14 min, and we report the measurement in its place because a figure that can be measured should not be assumed. EMNIST-byclass arms run 52 min/run and are excluded, so this accounting is CIFAR-10 only.

## S23 A CLOSED CENSUS OF THE ALGEBRAICALLY SETTLED REGION, PRE-REGISTERED

*What happens where the algebra settles preservation exactly, with no cells unspent? The  $\ddagger$  row above is the `foolsgold` column of a  $3 \times 3$  box over  $d_1$  and  $d_2$ , and the two cells of that box which had never been run are what this wave measured.* The region where mechanism preservation is settled on *both* sides is small and enumerable:  $d_1$  emitting strictly positive per-client coefficients  $T(u_i) = c_i u_i$ , so Prop. 11(a) applies, against a  $d_2$  reading an exactly scale-invariant statistic. Over this project’s defenses and the metric-swap mirrors that region is `{norm_clip, rfa, reputation} × {foolsgold, cos_krum, cos_reputation}`, exactly 9 ordered pairs, of which 7 were already measured at the identical protocol and 2 were unspent. Enumeration is what turned a planned 20-pair sample into a census: 118 **ordered pairs of this project are already spent**, and once the admissibility constraints (a non-degenerate  $d_1$ , and C0) are imposed only 6 candidates remain anywhere. **A census needs no sampling argument, and an all-negative census is as informative as a positive one**, which is why this design was worth 24 runs and 4.87 h. Pre-registered at 96f8856 with per-pair C0–C3 verdicts, a frozen scoring rule and a two-sided resolution table (`experiments/pre_registration_wave3_emergent.md`); scored by `experiments/analyze_wave3_emergent.py`, which refuses to run against an uncommitted prereg.

| $d_1 \downarrow d_2 \rightarrow$ | <code>foolsgold</code> | <code>cos_krum</code>   | <code>cos_reputation</code> |
|----------------------------------|------------------------|-------------------------|-----------------------------|
| <code>norm_clip</code>           | 0.887 HIGH             | 0.508 HIGH <sup>†</sup> | 0.935 HIGH <sup>†</sup>     |
| <code>rfa</code>                 | 0.914 HIGH             | 0.566 HIGH              | 0.861 HIGH                  |
| <code>reputation</code>          | 0.807 HIGH             | <b>0.263 LOW*</b>       | 0.787 HIGH*                 |

Max-committed mean ASR over seeds 42, 43, 44; LOW iff  $< 0.5$ .

\* the two unspent cells, labels frozen at 96f8856 before the output directory existed.

<sup>†</sup> `norm_clip`’s clip is inactive on the pixel arm (max update norm  $3.043 < \tau=5$ ), so those *pixel* cells are bit-identical to  $d_2$  alone; both verdicts are carried by the scaling arm, where it is not.

**`foolsgold` is excluded from the  $d_1$  column by measurement, and that measurement changed the design before any prediction was frozen.** It was going to supply two more C1-fail  $\wedge$  C2-true pairs. But `foolsgold` normalizes with `weights - weights.min()`, forcing at least one coefficient to *exactly* zero every round, which puts it outside Prop. 11(a)’s strict  $c_i > 0$  hypothesis and under Lemma 2 instead. Measured through the shipped transform over 3 seeds  $\times$  3 rounds: 2/5 exact zeros every round, and both downstream cosine statistics moved in 9 of 9 rounds. Synthetically at 1000 trials the effect is entirely due to the zero: strictly positive coefficients change `cos_krum`’s selection 0/1000 times, one exact zero changes it 471/1000. `reputation`

by contrast emits 0/5 zeros every round at a positive-part ratio up to 18839.8, and under that same extreme rescaling in the same rounds the two cosine statistics moved 0/9 while non-invariant krum's selection moved 8/9 and reputation's own ordering 9/9: **a contrast between statistics on identical inputs, not between conditions.** So the two foolsgold  $\rightarrow$  cos\_\* pairs were entered as *measured-C2-fail controls* rather than as members of the census. **Had C2 been assumed from the definitions instead of measured, this wave would have reported two mislabelled pairs** (experiments/verify\_wave3\_invariance.py, results/wave3\_invariance\_check.json).

| # | pair                                    | cell                       | pred. | measured                        |
|---|-----------------------------------------|----------------------------|-------|---------------------------------|
| 1 | reputation $\rightarrow$ cos_krum       | certified (C1 $\wedge$ C2) | LOW   | <b>0.263 LOW</b> ✓              |
| 2 | reputation $\rightarrow$ cos_reputation | C1-fail $\wedge$ C2-true   | HIGH  | 0.787 HIGH ✓                    |
| 3 | foolsgold $\rightarrow$ cos_krum        | measured-C2-fail control   | HIGH  | HIGH <sup>§</sup> (raw 0.086) ✓ |
| 4 | foolsgold $\rightarrow$ cos_reputation  | measured-C2-fail control   | HIGH  | 0.824 HIGH ✓                    |

§ scored HIGH *only* by the frozen DEGENERATE-ACC rule: mean clean accuracy  $0.255 < 0.35$  on the arm supplying the max, so its raw 0.086 is a collapsed model and not a suppressed one.

4/4 as registered, including the pair we flagged uncertain, and the near-identity check came back clean on every arm (0 of 3 seeds identical to  $d_2$  alone on all 8 arms; max per-seed differences 0.073 to 1.000), so these are compositions and not  $d_2$  in disguise. Pair 1 was the live risk and the prereg said so: its LOW rests on two *different* attacks being suppressed by two *different* constituents, reputation holding model-scaling (0.017 at accuracy 0.777) and cos\_krum the pixel backdoor (0.300 at 0.598), and on the scaling arm the certification and the two nearest measured analogues (norm\_clip  $\rightarrow$  cos\_krum 0.508, rfa  $\rightarrow$  cos\_krum 0.566) pointed *opposite* ways. It came out LOW on both arms (0.064 scaling, 0.263 pixel). **So the one low-ASR pair in the closed census is a pair neither constituent suppresses alone**, and it is the pair C1  $\wedge$  C2 selects.

**Both strategies priced over the same closed census, which costs no further runs because C1 is defined on standalone numbers that already exist.**

| strategy over the 9-cell census      | selected | precision   | recall      | runs      |
|--------------------------------------|----------|-------------|-------------|-----------|
| A: statistic preservation (C2 alone) | 9        | 11%         | 100%        | 90        |
| B: the criterion (C1 $\wedge$ C2)    | 1        | <b>100%</b> | <b>100%</b> | <b>10</b> |

In this region C2 alone is not a screen at all: it admits everything, because everything here satisfies it by construction. **That inverts the body's trade**, where B bought precision by giving up recall; here, at the frozen seed count, B dominates A on precision, recall and cost at once, and disclosure ( $v$ ) below is the qualification that phrase needs. It is also the answer to the objection of §S22 that the algebraically settled sub-region cannot discriminate strategies: it could not, at 3 pairs; at 9 it discriminates them completely.

**Five disclosures, none of which the numbers above carry on their face.** (i) *B's 1/1 is a gated reading.* C1 requires suppression at mean clean accuracy  $\geq 0.35$ , the gate applied throughout §S24.2, and it flips exactly two census cells: cos\_krum alone reaches ASR 0.078 on model-scaling but at accuracy 0.146, barely above CIFAR-10's 0.10 chance, so it is a destroyed model rather than a defended one. Read at the bare 0.5 threshold, C1 additionally admits norm\_clip  $\rightarrow$  cos\_krum and rfa  $\rightarrow$  cos\_krum, both HIGH, and B selects 3 for 33% precision. The gate is the paper's pre-existing convention rather than a knob turned for this table, and analyze\_wave3\_emergent.py emits both readings. (ii) *The one positive is  $n=3$  and wide:* pair 1's pixel arm is 0.000/0.107/0.681 per seed, so one of three seeds sits above the threshold and the LOW verdict is the frozen mean rule applied to a spread that a fourth seed could move. We do not add seeds after seeing this. (iii) *The controls are weaker evidence than intended.* Every foolsgold-as- $d_1$  scaling arm sits at chance accuracy (0.100), and pair 3 is HIGH only through the accuracy gate, its raw ASR being 0.086, and its scaling arm tripped the frozen "ASR identically 0.0 across all 3 seeds" flag, which on a model at 0.100 accuracy means the attack had nothing left to succeed against; so pairs 3 and 4 establish that the C2 probe is not inert, which is their stated role, but they are not independent confirmations of a HIGH prediction. (iv) *7 of the 9 cells are retrospective.* Only the two reputation cells were blind; the other seven were read from frozen artifacts, with each transcription in the prereg checked against its source by the analyzer. So the census's closure is prospective at the margin, not throughout, and pair 2's scaling arm tripped the second of the wave's two frozen sanity flags (clean accuracy 0.790 against the FedAvg reference 0.667). Both flags are reported, neither adjusted. (v) *The C1 input that certifies pair 1 is the lowest-seed reading of a high-variance cell.* C1 holds for pair 1 on the pixel arm because cos\_krum alone measures 0.300 there, and that is the mean over the three frozen seeds 42–44 (0.003/0.269/0.630). The identical cell, same protocol and identity upstream, is measured twice more in this paper at larger seed counts, and the cell's mean climbs monotonically with the count: 0.490 at  $n=5$  (Table A6, arm 3's  $\kappa=0$  rung, which adds 0.864/0.686) and 0.584 at  $n=8$  (Table A7, which adds 0.578/0.845/0.795), the last being the reading §D.2 states as cos\_krum never suppressing. Six

of the eight seeds are above 0.5 and the frozen three are the two lowest plus the fourth-lowest. So B’s precision 1/1 and recall 1/1 here are contingent on the frozen seed count as well as on the accuracy gate of (i), and we state the contingency at the resolution the numbers support rather than rounding it against ourselves: 0.490 is still *below* the threshold, by 0.010, so C1 survives at  $n=5$ ; at  $n=8$  C1 fails for pair 1, B certifies nothing in this census, and its one low-ASR pair becomes another false negative of the criterion rather than its one hit. We report this instead of rescoring, because the pre-registration fixes the seed set at three and forbids adding seeds afterwards (`pre_registration_wave3_emergent.md`, §5). What the disclosure costs is the claim that B *dominates* A on this region, which holds at the frozen seeds and is not established beyond them.

## S24 A SUBORDINATE BOUNDARY STUDY: WHAT THE SCREEN BUYS, AND WHAT IT DOES NOT

Criterion 4 predicts for each two-defense pair whether composition achieves low max-committed ASR. All experiments in this section and §E use one deployment regime throughout (CIFAR-10,  $N=10$  clients,  $K=5$  per round, adversarial fraction  $f=0.2$ , 50 rounds, 4-layer CNN), with committed attacks `model_scaling` and `backdoor_pixel`. **Convention:** unless stated otherwise, ASR means *max-committed ASR* (the maximum over the two committed attacks of the per-attack mean across seeds) thresholded at 0.5: the level at which the trigger fires on the majority of triggered inputs, so that “suppresses” means the backdoor is not the model’s dominant response. **That 0.5 is an operational classification threshold, not a theoretically privileged security boundary:** nothing in the propositions depends on it, and a deployment with a lower tolerance should read the per-cell ASR rather than the label. adaptive-attack, order-sensitivity and per-seed figures are labelled explicitly where they differ. A second gate is needed because a destroyed model has a low ASR for the wrong reason: every cell in this section, §E and §D–§D.2 must also hold mean clean accuracy  $\geq 0.35$  (our **accuracy floor**), below which we call the cell *uninterpretable* rather than suppressed, in either direction. **That scope is wider than an earlier version of this sentence stated:** it named §D–§D.2 alone, while the paragraph it sits in scopes itself to this section and §E, so the floor was announced as a second gate and then withheld from most of the cells it governs. Widening it to what this paragraph already claims reclassifies exactly one cell, `fg→rfa` at  $\alpha=0.1$  (§S24.4(E)), and that reclassification *weakens* a limitation we state against ourselves, which is why both readings are printed there rather than only the gate-applied one. **Sample sizes are small and were fixed before the runs:**  $n=5$  seeds per cell in the interventional suites ( $n=8$  for `cos_krum`,  $n=3$  frozen for the second-dataset replication, which also reads at a pooled, post-hoc  $n=5$ , §D.2),  $n=30$  for the FoolsGold→RFA characterization, so intervals are wide, and no null result is read as flatness unless it clears a pre-registered practical-equivalence margin. **“Refuted” means one thing throughout:** a pre-registered *directional* hypothesis was rejected against the rule frozen for it, never that a mathematical claim was disproved. Adversaries are the fixed client identities  $\{0, \dots, fN - 1\}$  and each round samples  $K$  participants uniformly without replacement from all  $N$ , with no stratification, so the per-round adversary count is *hypergeometric* rather than fixed: for  $N=10$ ,  $f=0.2$ ,  $K=5$ ,  $P(n_a=0) = P(n_a=2) = 56/252 = 0.222$  and  $P(n_a=1) = 140/252 = 0.556$  (measured over 100 rounds: 0.19/0.54/0.27). Two consequences the data bear out: CoordMedian’s count-majority condition  $n_a < K/2 = 2.5$  holds for *every* attainable  $n_a \in \{0, 1, 2\}$ , i.e. in 100% of rounds, while TrimmedMean’s budget  $\lfloor \beta K \rfloor = \lfloor 0.2 \cdot 5 \rfloor = 1$  is exceeded whenever  $n_a=2$ : 22.2% of rounds,  $\approx 11$  of 50. That is a second, count-based reason FoolsGold→TrimmedMean fails (0.508) while FoolsGold→CoordMedian succeeds (0.107) under an identical upstream transform; we retain the pre-registered C3-FAIL label, the budget violation being an additional mechanism rather than a relabelling. We validate in two waves. The **development set** (18 targeted pairs, Table S13) spans all criterion prediction categories and was used to refine the criterion (adding C3). The **held-out set** (24 pairs, Table S14) was *pre-registered*: all 24 predictions were committed to version control (commit `7e4d9b1`) *before* any held-out experiment was run, eliminating post-hoc classification concerns. The criterion predicts LOW ( $<0.5$ ) or HIGH ( $\geq 0.5$ ) max-committed ASR. The verdict categories are:

**PASS precision:** 2 of the 2 selected pairs are low-ASR. That is 100% as a descriptive statistic, and the positive class is two cases, so it resolves the direction and not the rate. Under the tightened per-attack C1 (§C) the certified in-distribution set is 2 pairs, `fg→cm` (0.107) and `rep→cm` (0.463 at  $n=30$ ), both correct; `fg→rfa` (0.093) is no longer certified and is reported below as a false negative of our own criterion. This is the practitioner-relevant result: the criterion’s positive predictions are reliable, at the cost of 40% recall. Overall development-set accuracy: **16/18 = 89%**. Two misses, both in the HIGH direction:

**One certified pair is measured to the threshold, and a pre-registered top-up did not resolve it.** `Rep→CM`’s LOW label rests on its pixel arm, whose  $n=5$  interval  $[0.231, 0.616]$  crossed 0.5 with one seed at 0.665, so we pre-registered seeds 47–71 (`de77a7f`: seed list, unchanged 0.5 rule and the three admissible outcomes all fixed before any run) and took that cell to  $n=30$ , the same power as the pair the criterion rejects. It is **still unresolved**: mean 0.463, 95% interval  $[0.412, 0.514]$ , with 10 of the 30 seeds above the threshold. Six times the seeds narrowed the half-width  $3.8\times$  ( $0.192 \rightarrow 0.051$ ) and moved the mean *up* from 0.423, toward the threshold rather than away, so the cell is near-threshold rather than merely noisy. The LOW label holds by the frozen mean rule, the pre-registration forbids adding seeds once the outcome is seen, and we therefore stop

Table S13: Composability criterion predictions, scored (max across committed\_scaling and committed\_pixel;  $\pm$  is per-seed std where reported). Threshold: LOW  $< 0.5$ , HIGH  $\geq 0.5$ . **Seed counts differ by row and are marked:** unmarked rows are  $n=3$ , \* is  $n=5$ ,  $\dagger$  is  $n=30$ ,  $\S$  is split across arms and noted in the table.

| Pair                                                                                                                                                                                       | Failure mode | Predicted | Max ASR           | Correct? |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------|-------------------|----------|
| <i>PASS: <math>C1 \wedge C2 \wedge C3</math> all hold (predicted LOW)</i>                                                                                                                  |              |           |                   |          |
| foolsgold $\rightarrow$ rfa $^\dagger$                                                                                                                                                     | —            | LOW       | $0.093 \pm 0.082$ | ✓        |
| foolsgold $\rightarrow$ coord_median*                                                                                                                                                      | —            | LOW       | $0.107 \pm 0.034$ | ✓        |
| reputation $\rightarrow$ coord_median $^\S$                                                                                                                                                | —            | LOW       | $0.463 \pm 0.136$ | ✓        |
| <i>C1-FAIL: neither defense individually suppresses all attacks (predicted HIGH)</i>                                                                                                       |              |           |                   |          |
| reputation $\rightarrow$ trimmed_mean $^\dagger$                                                                                                                                           | C1 (pixel)   | HIGH      | 0.679             | ✓        |
| reputation $\rightarrow$ rfa                                                                                                                                                               | C1 (pixel)   | HIGH      | 0.825             | ✓        |
| reputation $\rightarrow$ foolsgold                                                                                                                                                         | C1 (pixel)   | HIGH      | 0.807             | ✓        |
| reputation $\rightarrow$ norm_clip                                                                                                                                                         | C1 (pixel)   | HIGH      | 0.813             | ✓        |
| norm_clip $\rightarrow$ foolsgold $^\ddagger$                                                                                                                                              | C1           | HIGH      | 0.887             | ✓        |
| <i>C2-FAIL: <math>d_1</math> destroys <math>d_2</math>'s signal direction (predicted HIGH)</i>                                                                                             |              |           |                   |          |
| norm_clip $\rightarrow$ reputation                                                                                                                                                         | C2           | HIGH      | 0.819             | ✓        |
| rfa $\rightarrow$ reputation                                                                                                                                                               | C2           | HIGH      | 0.882             | ✓        |
| <i>C3-FAIL: <math>d_1</math> compresses boundary below <math>d_2</math>'s threshold (predicted HIGH)</i>                                                                                   |              |           |                   |          |
| norm_clip $\rightarrow$ trimmed_mean                                                                                                                                                       | C3           | HIGH      | 0.780             | ✓        |
| norm_clip $\rightarrow$ rfa                                                                                                                                                                | C3           | HIGH      | 0.853             | ✓        |
| rfa $\rightarrow$ trimmed_mean                                                                                                                                                             | C3           | HIGH      | 0.678             | ✓        |
| rfa $\rightarrow$ coord_median                                                                                                                                                             | C3           | HIGH      | 0.485             | ×        |
| foolsgold $\rightarrow$ trimmed_mean                                                                                                                                                       | C3           | HIGH      | 0.508             | ✓        |
| norm_clip $\rightarrow$ coord_median                                                                                                                                                       | C3           | HIGH      | 0.433             | ×        |
| <i>DEGEN: <math>d_1</math> has no per-client transform (predicted HIGH)</i>                                                                                                                |              |           |                   |          |
| fedavg $\rightarrow$ norm_clip*                                                                                                                                                            | DEGEN        | HIGH      | 0.935             | ✓        |
| fedavg $\rightarrow$ trimmed_mean*                                                                                                                                                         | DEGEN        | HIGH      | 0.846             | ✓        |
| Overall: 16/18 = 89%. HIGH direction: 13/15 correct (87%). Under the tightened C1 the certified PASS set is 2/2 correct; fg $\rightarrow$ rfa is reclassified as a synergy false negative. |              |           |                   |          |
| $^\ddagger$ Reclassified from C2-FAIL: by Proposition 11(a) clipping preserves FoolsGold's similarities exactly, so C2 holds and the pair fails C1. Prediction and outcome unchanged.      |              |           |                   |          |
| $^\S$ Pixel arm $n=30$ (seeds 47–71 pre-registered at de77a7f), scaling arm $n=5$ ; the pixel arm sets the max, and its interval [0.412, 0.514] still crosses 0.5 at $n=30$ .              |              |           |                   |          |

here and report it unresolved: the certified in-distribution set is 2/2 correct on the mean, and only fg $\rightarrow$ cm is certified and resolved below threshold. The frozen seed reproduced bit-for-bit before the top-up ran (0.43178 at seed 42), so the 25 new seeds extend the original five rather than replacing them; the scaling arm stays at  $n=5$  (0.020), where it cannot set the max. Clean accuracy is 0.755 over the 30 runs, and 0.50, 0.68, 0.76 on the three cells, all well clear of the collapse regime.

**Two boundary conditions, stated alongside the primary result.** Both are developed in full later, but neither should be met late. (i) **Extreme heterogeneity breaks the PASS pairs: two of three on the raw reading, one of three once our own accuracy floor is applied.** At  $\alpha=0.1$ , fg $\rightarrow$ rfa reaches 0.569 and rep $\rightarrow$ cm 0.548; only fg $\rightarrow$ cm survives, at 0.433. fg $\rightarrow$ rfa's cell sits at clean accuracy 0.20, below our 0.35 floor, so it is *uninterpretable* rather than a crossing and the gated count is 1/3; both counts are carried, because the floor softens a limitation we state against ourselves and rep $\rightarrow$ cm's crossing (0.548 at 0.48) survives it either way. What makes this more than a negative datapoint is that the failure appears in the theorem's own quantity: RFA's separation margin  $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ , measured on the transformed points, collapses from an all-round mean of 3.24 to 1.01 (exactly the  $m \rightarrow 1$  boundary at which Corollary 4's bound degrades to 40% adversarial mass at  $n_a=2$ ), so a practitioner running the 5–10 round pilot would measure  $m \approx 1$  and decline to deploy. We emphasise that this is a post-hoc measurement of a known failure, not a pre-registered prediction of it; full sweep and caveats (bimodality at  $n=3$ , accuracy collapse to 0.20–0.48) in §S24.4(E). (ii) **Mechanism preservation is not adaptive-attack robustness.** A criterion-aware adversary solving FoolsGold's and RFA's constraints jointly (decorrelated coordinate blocks to evade the similarity test,  $\epsilon$ -ball projection to stay inside the geometric median's radius) raises fg $\rightarrow$ rfa from 0.064 (seed-matched at  $n=30$ , and on the pixel arm the attack is built from) to 0.157 within our evaluated composition-aware attack family (§E). Theorem 9 bounds

what the preserved mechanism admits, not what an adversary reshaping its update to satisfy both constraints at once can do, and the  $2.4\times$  increase is the size of that gap on our flagship pair.

**Held-out validation (pre-registered): 22/24 = 92%** (*Screening*: an evaluation ordering, and by Proposition 7 not an identification of the mechanism’s contribution). Table S14 reports the 24 pre-registered pairs. Combined across both waves: **38/42 = 90.5%**. **Read against its base rate that figure is nearly uninformative, which is why it appears here, where it is computed, and in no summarizing position.** Only 5 of the 42 pairs are low-ASR (the 12% base rate of §S21), so the constant predictor that calls every pair HIGH already scores  $37/42 = 88.1\%$ ; 90.5% is one pair better than that, and the tally is carried by correct negatives. The informative quantities are precision 2/2 and recall 40% (§S21). The two held-out misses are both LOW-predicted pairs that were flagged as *speculative* in the pre-registration (neither is a clean  $C1 \wedge C2 \wedge C3$  PASS):

Table S14: Held-out pre-registered validation (24 pairs,  $n=5$  seeds; max across committed\_scaling and committed\_pixel). All predictions committed to version control (commit 7e4d9b1) before experiments. Threshold:  $LOW < 0.5$ ,  $HIGH \geq 0.5$ . Only the two LOW predictions are shown individually; the 22 HIGH predictions (all correct) are summarized.

| Pair                                                                                                                                                                                           | Category       | Predicted | Max ASR | Correct? |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|---------|----------|
| <i>Speculative LOW predictions (not clean <math>C1 \wedge C2 \wedge C3</math> PASS)</i>                                                                                                        |                |           |         |          |
| fedavg $\rightarrow$ coord_median                                                                                                                                                              | DEGEN          | LOW       | 0.519   | ×        |
| foolsgold $\rightarrow$ norm_clip                                                                                                                                                              | C1 (uncertain) | LOW       | 0.927   | ×        |
| <i>HIGH predictions (22 pairs: 6 CM-as-<math>d_1</math>, 6 TM-as-<math>d_1</math>, 4 fedavg-as-<math>d_1</math>, 3 weighting-<math>d_1</math>, 3 RFA-as-<math>d_1</math>; range 0.70–0.94)</i> |                |           |         | 22/22 ✓  |
| Held-out: 22/24 = 92%. Combined with development set: 38/42 = 90.5%.                                                                                                                           |                |           |         |          |
| Constant-HIGH baseline on the same 42 pairs: 37/42 = 88.1%.                                                                                                                                    |                |           |         |          |

**Rep $\rightarrow$ tm: composition dominates despite C1-FAIL.** Reputation $\rightarrow$ TrimmedMean is correctly classified as C1-FAIL (pixel ASR 0.679 > 0.5), yet it is the paper’s main composition result. The resolution: the criterion predicts *max-committed* ASR, while the deployment-relevant metric is *oracle* ASR (0.316). The oracle cannot exploit the pixel gap because scaling (which reputation *does* suppress) dominates its utility. The criterion thus correctly identifies that rep $\rightarrow$ tm does not suppress every committed attack, while the game-theoretic analysis shows it is strategically dominant.

**Tightening C1 costs us our own flagship, and reveals a synergy case.** Under the per-attack quantifier in C1, **fg $\rightarrow$ rfa no longer certifies**: neither FoolsGold (0.732) nor RFA (0.845) suppresses the pixel backdoor individually, and FoolsGold’s 0.200 against model-scaling is hollow because it comes with 10% clean accuracy. The composition nonetheless reaches 0.093. This is genuine *synergy* (suppression neither constituent possesses) and Proposition 8 proves that *no* inherited-suppression screen can certify it, so this is a scope boundary rather than a tuning failure. We therefore report our lowest-ASR composition as a false negative of our own criterion. The certified in-distribution set is fg $\rightarrow$ cm (0.107) and rep $\rightarrow$ cm (0.463), both correct: precision 2/2, recall  $2/5 = 40\%$ .

**A directional C1 would empty that certified set, and we report the price rather than pay it.** The natural next tightening is to replace C1’s “some constituent suppresses  $a$ ” with the directional “ $d_2$  suppresses  $a$ ”, on the ground that C0 forces  $d_2$  anyway. **C0 does not force it.** C0 is existential over attacks while C1 is universal over them, so Lemma 1 pins  $d_2$  only on the witnessing attack  $a^*$ ; off  $a^*$  the constituent holding that arm may be  $d_1$ , and the two definitions accordingly disagree on 4 of the 42 pairs. On all four the downstream defense is coord\_median, whose standalone model-scaling ASR is 0.519 and so above threshold, while the upstream foolsgold (0.200) or reputation (0.017) is what holds that arm; every C1 input here is the  $n=5$  single-defense baseline on seeds 42–46. **Both certified pairs are of exactly that kind**, so a directional C1 selects nothing in distribution and leaves precision *undefined* instead of 2/2. **What the substitution correctly points at is real, and it is a quantifier and not a direction:** on the model-scaling arm of these two pairs the suppression is inherited from an *upstream* defense, which nothing downstream can undo, so the signal-preservation argument contributes nothing to that arm even though the arm counts toward the max-committed label. The criterion and Theorem 9 are therefore aligned *per attack* and not per pair. We keep C1 and disclose the asymmetry; the census is emitted by experiments/analyze\_condition\_ablation.py.

#### S24.1 A PROSPECTIVE TEST ON A NEVER-RUN DEFENSE MENU

*Does the criterion predict on a menu it has never seen?* **Result: 13/20, and the shortfall is in our experimental design.** At the same 0.5 threshold: TP = 8, FP = 0, TN = 5, FN = 7 (Table S15). **The suite carried the sanity gate our out-of-sample battery lacked**, pre-registered with the labels: a predicted-FAIL

pair counts only if the attack clears ASR 0.5 against FedAvg, which both committed attacks do (0.667 scaling, 0.768 pixel), so a pair cannot be scored correct because the attack failed on its own. Every predicted-LOW pair came out LOW, 8/8, and that figure is far weaker than it looks, because **FLTrust suppresses both committed attacks on its own** (max-committed ASR 0.048 at 0.575 clean accuracy), the first defense in our menu that does. All 13 pairs containing FLTrust land in  $[0.041, 0.078]$ , within 0.030 of FLTrust alone, in either slot. The two slots fail for different reasons, and the distinction matters:

**The two slots fail differently, and the difference is the finding.** (i) *FLTrust as  $d_1$  (7 pairs, 2/7)*.  $d_1$  alone already suppresses, so the composition inherits that suppression whatever  $d_2$  does; all five predicted-HIGH misses have real C2/C3 failures, but destroying  $d_2$ 's mechanism is costless when  $d_2$  was never load-bearing. This is C0, invisible on the original 42 pairs because no defense there suppresses alone (the minimum single-defense max-committed ASR is CoordMedian's 0.519, and the lowest ASR among all 39 non-PASS pairs is 0.433), so we count these 7 pairs out of scope *retrospectively*: a weaker claim than a pre-registered exclusion, and C0's own provenance is Remark 6. (ii) *FLTrust as  $d_2$  (6 pairs, 6/6)*, but the prediction is nearly an identity. FLTrust gates on the scale-invariant  $\cos(u_i, g_{\text{srv}})$  and then renormalizes every update to  $\|g_{\text{srv}}\|$ , discarding magnitude entirely, so a positive per-client rescaling is *exactly* cancelled: five of the six pairs are bit-identical to FLTrust alone or differ only within seed noise. The single genuine deviation is FG→FLTrust (0.064), the one  $d_1$  producing exact *zeros* rather than positive rescalings and hence a Lemma 2 interaction. This arm is a consistency check between theory and implementation, **not** a falsifiable test: every  $d_1$  in our menu is a positive rescaling and Proposition 11(a) permits no other outcome. (iii) *Neither defense is FLTrust (7 pairs, 5/7)*: the only arm that could have surprised us. Five predicted-HIGH pairs are correct (0.870–0.943), though only 2 of the arm's 7 pairs compose two real mechanisms once the pass-through and MultiKrum degeneracies are accounted for (below); both misses are diagnosable and neither is a mechanism error: fg→krum measures 0.078 at 9.9% clean accuracy (the collapsed-model artifact, not suppression), and krum→cm measures 0.450 against a 0.5 threshold as a degenerate pair reducing to CoordMedian alone, whose own baseline is 0.519. The arm contains *zero* predicted-LOW pairs, so it bears on specificity only. Per-pair numbers for all three arms are in §S1.

**What the suite does and does not establish.** Counting only in-scope pairs gives 11/13; the arm that carried real risk of falsification gives 5/7 with both misses explained. But it is not the prospective positive the framework needs, because **it contains none**: by (i) and (ii) the 8 predicted-LOW pairs are 6 near-identities plus 2 out-of-scope inheritances.

**We withdraw the three matched contrasts we previously drew here**, which we had read,  $d_1$  fixed while  $d_2$  moved from FLTrust to a Euclidean rule, as a 19–20× ASR gap turning on exactly the property Proposition 11 identifies. Two ran against MultiKrum, which at  $K=5$  selects *all five* participants and is bit-identical to FedAvg ( $1.9 \times 10^{-6}$  per aggregation; a genuine selector again at  $K=10$ ), so they carried no downstream defense at all. Substituting Krum does not repair them, and that is the finding: all three Krum arms take their max-committed ASR from the pixel backdoor, which Krum *fails to suppress standalone* (0.583), so the gap measures C1 failure and not disturbed invariance, and nc→krum is bit-identical to Krum alone on all three seeds because under that attack no client exceeds  $\tau=5$  and  $d_1$  is the exact identity. Counting the pass-through  $d_1$  rows too, 8 of the 20 rows reduce to a single defense rather than the 5 labelled DEGEN at freeze, and arm (iii) composes two real mechanisms in 2 pairs, not 7 (App. S2). The frozen labels and the 13/20 score stand; what we retract is our reading of them. The confound is accordingly worse than we reported: in this menu the invariant-signal  $d_2$  is also the only  $d_2$  that suppresses alone, so C1 and C2 co-vary. Lemma 1 explains why no other choice of defense would have helped: in scope, C0∧C1 *force*  $d_2$  to suppress  $a^*$  alone, so the confounding is structural. Breaking it requires holding the suppression principle and standalone effectiveness fixed while varying only the invariance class, which is the design of §S24.2; that section's  $\Delta$  analysis is the only contrast in either paper holding C1 exactly fixed, and it is where our C2 evidence now rests.

## S24.2 BREAKING THE CONFOUND: A MATCHED METRIC-SWAP ABLATION

*Can the confound be broken by swapping the metric instead of the defense?* Lemma 1 says mechanism–effectiveness confounding cannot be escaped by picking a different defense off a menu: in scope, C0∧C1 *force* the invariant-signal  $d_2$  to be one that suppresses  $a^*$  alone. So we break the confound by construction instead, holding the suppression *principle* fixed and varying only the invariance class of the statistic  $d_2$  reads.

**The design's premise was refuted by its own baselines.** Matching failed in both arms, for two independent reasons, and because the C1 inputs are measured before the freeze we learned this before the suite ran: the only reason it can be reported as a finding rather than a post-hoc excuse. `cos_reputation` does not merely weaken relative to its Euclidean original, it collapses, from Reputation's 0.017 on model-scaling to **0.982**. `cos_krum` reaches 0.078 on model-scaling, which passes C1's numeric threshold, but at **0.146** clean accuracy, the hollow-suppression artifact already documented for FoolsGold (0.200 at 0.10) and fg→krum (0.078 at 0.099), so C1 fails under the accuracy gate we apply everywhere else. The mechanism is visible in the baselines rather than asserted: undefended FedAvg under model-scaling is itself at 0.100 accuracy, so Krum's 0.577 is *restoring* utility the attack destroyed and `cos_krum`'s 0.146 fails to. **Magnitude information is what lets this**

Table S15: Prospective suite: 20 pairs, labels frozen at commit 262cf35 before any run. Max-committed ASR over 2 attacks  $\times$  3 seeds; threshold 0.5. Categories are verbatim from the pre-registration, including its own uncertainty hedges. **The two scope headers are not:** C0 was formulated after this suite scored and applied to it retrospectively (Rem. 6), so the in-scope/out-of-scope split is ours and post hoc, and the 11/13 it produces is a weaker number than the frozen 13/20.

| Pair                                                                                     | Frozen category     | Pred. | Max ASR | Acc   | Hit |
|------------------------------------------------------------------------------------------|---------------------|-------|---------|-------|-----|
| <i>In scope (C0 holds): <math>d_2</math> load-bearing</i>                                |                     |       |         |       |     |
| Rep→FLTrust                                                                              | certified           | LOW   | 0.046   | 0.577 | ✓   |
| NC→FLTrust                                                                               | certified           | LOW   | 0.046   | 0.576 | ✓   |
| RFA→FLTrust                                                                              | certified           | LOW   | 0.047   | 0.574 | ✓   |
| Krum→FLTrust                                                                             | DEGEN               | LOW   | 0.048   | 0.574 | ✓   |
| MKrum→FLTrust                                                                            | DEGEN               | LOW   | 0.048   | 0.574 | ✓   |
| FG→FLTrust                                                                               | certified           | LOW   | 0.064   | 0.547 | ✓   |
| FG→Krum                                                                                  | C1-fail             | HIGH  | 0.078   | 0.099 | ×   |
| Krum→CM                                                                                  | DEGEN               | HIGH  | 0.450   | 0.770 | ×   |
| Krum→RFA                                                                                 | DEGEN               | HIGH  | 0.870   | 0.780 | ✓   |
| MKrum→RFA                                                                                | DEGEN               | HIGH  | 0.870   | 0.780 | ✓   |
| RFA→MKrum <sup>†</sup>                                                                   | C1-fail             | HIGH  | 0.875   | 0.782 | ✓   |
| Rep→Krum                                                                                 | C1-fail             | HIGH  | 0.880   | 0.530 | ✓   |
| NC→MKrum <sup>†</sup>                                                                    | C1-fail             | HIGH  | 0.943   | 0.761 | ✓   |
| <i>Out of scope (C0 violated): FLTrust as <math>d_1</math> suppresses alone at 0.048</i> |                     |       |         |       |     |
| FLTrust→CM                                                                               | certified-uncertain | LOW   | 0.041   | 0.505 | ✓   |
| FLTrust→MKrum <sup>†</sup>                                                               | C2-fail             | HIGH  | 0.045   | 0.576 | ×   |
| FLTrust→TM                                                                               | C3-fail-uncertain   | HIGH  | 0.046   | 0.529 | ×   |
| FLTrust→NC                                                                               | C2-fail             | HIGH  | 0.047   | 0.576 | ×   |
| FLTrust→RFA                                                                              | certified           | LOW   | 0.048   | 0.512 | ✓   |
| FLTrust→Rep                                                                              | C2-fail             | HIGH  | 0.048   | 0.518 | ×   |
| FLTrust→Krum                                                                             | C2-fail             | HIGH  | 0.078   | 0.361 | ×   |

Full suite 13/20 (TP 8, FP 0, TN 5, FN 7); in scope 11/13; non-FLTrust arm 5/7.

<sup>†</sup>MultiKrum at  $K=5$  keeps all five participants, so it is bit-identical to FedAvg and these rows carry no downstream defense; MKrum→FLTrust and MKrum→RFA repeat their Krum rows exactly. 8 of 20 rows reduce to one defense; frozen labels unchanged (App. S2).

**defense family handle a magnitude attack:** a direction-only mirror of a Euclidean scorer cannot see a scaling attack at all, so removing the magnitude dependence removes the suppression with it: exactly the coupling the pre-registered contingency anticipated, stated there as the possibility that “both mirrors are magnitude-blind and the Euclidean originals may be catching model-scaling via magnitude”. On the magnitude-neutral pixel backdoor the mirror does survive (0.300 at 0.598), but there its control Krum fails C1 (0.583), so those arms are unmatched in the opposite direction and the pixel cells cannot carry the test either.

The consequence is that **the confound re-materialized inside an experiment built specifically to break it**, which we report as this ablation’s methodological finding rather than as a failed run. Lemma 1 establishes that mechanism–effectiveness confounding is structural in scope; this is what trying to escape it looks like when the mechanism being varied is the mechanism the defense relies on. Two responses were available and we took neither. We did not switch to a pixel-only protocol, which would have made `cos_krum` pass C1 and manufactured two predicted-LOW cells: choosing the attack after seeing the C1 inputs is attack-shopping regardless of the resulting labels. We did not adjust the accuracy gate to admit 0.146, which would have produced a predicted-LOW pair by weakening the very criterion that catches hollow suppression elsewhere. The pre-registration’s second non-negotiable (“no switching of committed attack, threshold, seed count, or accuracy gate”) forecloses both, and either would have bought a positive test at the cost of the property that makes the test worth running.

**Outcome: 8/8 correct, and zero positive tests.** Because C1 binds for all eight pairs (on model-scaling through the two baseline failures above, on the pixel backdoor by design) every frozen label was HIGH, so the suite bears on specificity only, as a consequence of inputs measured before the freeze rather than a choice made afterwards. We ran all 8 pairs  $\times$  2 attacks  $\times$  3 seeds anyway, because a suite whose labels are all one class is still falsifiable in the direction that matters: a single LOW cell would have been a false positive against a criterion claiming 100% precision. All eight came out HIGH (8/8 correct, TP = 0, FP = 0, TN = 8, FN = 0, max-committed ASR spanning 0.508–0.935) with labels frozen in `pre_registration.metric.swap.md` at commit 356897b, timestamped before `results/metric.swap/` existed and reproduced verbatim in the runner’s PAIRS list so scoring cannot drift. We state the limitation in the same breath as the result: the suite

contains **no falsifiable prospective positive test**, and 8/8 is not validation of the framework’s predictive claim: eight correct negatives on pairs our own conditions could only have called HIGH is weak evidence, and it supports a claim about specificity, not prediction.

One prediction in the suite was genuinely two-sided, and it is the only prospective positive result either screening suite produced. `cos_krum`’s C1 verdict on model-scaling turns entirely on whether the accuracy gate is applied (0.078 passes the numeric threshold; 0.146 accuracy fails the gate), so rather than choose a reading afterwards we froze *both*, with the resolution rule fixed in advance: accuracy-gated  $\Rightarrow$  HIGH, numeric-only  $\Rightarrow$  LOW, and a LOW outcome at accuracy  $\geq 0.35$  would have recorded our two primary labels as misses. Both pairs came out **HIGH** (NormClip  $\rightarrow$  `cos_krum` 0.508, RFA  $\rightarrow$  `cos_krum` 0.566), refuting the secondary reading and vindicating the accuracy gate against a pre-registered alternative specification of itself on data neither of us had seen. The weaker half is flagged in §S3: the NormClip arm sits barely above threshold and it is the RFA arm that carries the conclusion.

**A post-hoc contrast that is not confounded.** The cross-pair scoring above is confounded in exactly the way Lemma 1 predicts, but the frozen suite happens to contain a comparison that is not, and it is the strongest C2 evidence in either paper. Compare a composition against *the same*  $d_2$ ’s own standalone ASR on the same attack, under two different upstream transforms. C1 is then identical by construction (same downstream defense, same attack, same measured standalone effectiveness on both sides), so the residual  $\Delta = \text{ASR}(d_1 \rightarrow d_2, a) - \text{ASR}(d_2, a)$  cannot be attributed to  $d_2$  being a stronger or weaker defense. What does vary is how much the upstream transform actually disturbs  $d_2$ ’s statistic, and that quantity was measured on the mechanism itself (`results/cos_invariance_check.json`) rather than inferred from any ASR. The frozen label scoring is fixed at  $n=3$  and must not move, so we topped the six powered cells up to  $n=5$  in a separate directory, with the tested direction fixed in the top-up script before the new seeds ran (`results/metric_swap_topup/`). Because  $\Delta$  is a difference, the subtrahend moves with it: each topped-up cell’s standalone is recomputed at the *same* five seeds, from the dose ladder’s  $\kappa=0$  rung, which is that  $d_2$  alone under the identical configuration (§D) and agrees with the published standalone baselines bit-for-bit on every seed they share (largest deviation 0.00e+00 in ASR and accuracy across 22 seed-level comparisons), so it is the same quantity carried to five seeds and not a second measurement of it. Scoring a five-seed composition against the three-seed baseline would hide a mixed- $n$  comparison inside the subtrahend, where it is invisible, and on one cell below it would invert the conclusion. For Krum under model-scaling (standalone 0.062 at 0.565 accuracy over those five seeds; 0.061 at 0.577 on the frozen three): NormClip changes Krum’s selection in 0/9 rounds and the composition reaches 0.062 ( $\Delta = +0.001$ , per-seed 0.120/0.018/0.047/0.058/0.068); RFA changes it in 6/9 rounds and the composition reaches 0.779 ( $\Delta = +0.717$ , per-seed 0.974/0.116/0.982/0.904/0.916): a gap of +0.716 between two compositions sharing a downstream defense, an attack, and a baseline, and at  $n=5$  it is statistically resolved (Welch  $t = 4.28$ , two-sided  $p = 0.012$ ; paired  $t = 4.60$ ,  $p = 0.010$ ; Mann-Whitney one-sided  $p = 0.008$ ), which it was not at  $n=3$ .

The pattern holds across the suite once cells without power are excluded:  $\Delta$  is observable only where standalone  $d_2$  genuinely suppresses, since a baseline already above 0.5 shows  $\Delta \approx 0$  by ceiling effect whatever the transform did. Restricting to standalone ASR  $< 0.5$  at accuracy  $\geq 0.35$  leaves six cells, and at  $n=5$  measured disturbance still splits them perfectly: the three undisturbed cells (0/9) give  $\Delta \in [-0.044, +0.001]$ , the three disturbed ones ( $> 0/9$ ) give  $\Delta \in [+0.717, +0.878]$ : a separation of +0.716 with no overlap, wider than the +0.629 the frozen  $n=3$  scoring gives. That separation is the *same* +0.716 as the Krum contrast above and not independent evidence on top of it: both extremes are those two Krum cells, which share a standalone, so the width of the split is algebraically that contrast’s gap. The other four cells extend the two halves outward rather than the distance between them. Two of the three undisturbed cells are `cos_krum` on the pixel backdoor, where the mirror suppresses standalone (0.300 at 0.598 accuracy on the frozen three seeds, 0.490 at 0.607 on all five, since two of the added seeds are high-ASR) and carries that suppression through under *both* upstreams. Under NormClip it carries through *exactly*: NormClip never binds on this attack (max client norm 3.04 against  $\tau = 5$ , 0/9 rounds), so the composition is bit-identical to the baseline at all five seeds and  $\Delta = +0.000$  is an identity rather than a measurement; under RFA the composition is 0.447 against that 0.490, i.e.  $\Delta = -0.044$ . Kept against the frozen three-seed baseline these same rows would instead read +0.190 and +0.147, and the natural reading of *that*, the magnitude channel of Lemma 2 moving  $\Delta$  where the selection is exactly invariant, would credit an effect to a transform that provably did nothing: the movement is entirely the baseline’s seed set. So this contrast has no power on the magnitude channel in either direction, because its one exactly-invariant downstream sits under an upstream that never rescales; measuring that channel requires moving the dial on purpose, which is what the ladder’s arm 3 (§D) does prospectively. The ten excluded cells are listed rather than dropped silently in §S3.

The suite contains a second such contrast, on Reputation under model-scaling (standalone 0.017 at 0.777 accuracy), and the two are informative in opposite ways. NormClip disturbs Reputation’s weight ordering in 3/9 rounds and the composition reaches 0.806; RFA disturbs it in 9/9 and the composition reaches 0.895. The gap is only +0.089 (both arms are far above threshold, because Reputation’s ordering is disturbed under *both* upstreams) but within-arm variance is small, all five seeds move in the predicted direction (per-seed deltas +0.149/ +0.089/ +0.030/ +0.061/ +0.115), and it is resolved at both  $n=3$  and  $n=5$  (Welch  $t = 3.06$ ,

two-sided  $p = 0.018$ ; paired  $t = 4.31$ ,  $p = 0.013$ ). So the two contrasts trade off exactly as the disturbance measurements imply: Krum has the large effect because one of its upstreams leaves it untouched, and Reputation was resolvable even at  $n=3$  because neither upstream does.

**This is post-hoc, and we do not present it otherwise.** It was not pre-registered, it re-uses a suite frozen for a different purpose, the two extra seeds per cell were added *after* seeing the  $n=3$  result, and the six-cell split is a description of that suite rather than a test on new data. The top-up could have cut in either direction and we report what it did: it resolved the Krum gap (Welch  $p = 0.16$  at  $n=3$ , where per-seed paired deltas  $+0.854/+0.097/+0.935$  left one seed not separating, to  $p = 0.012$  at  $n=5$ ) and left the invariant control at  $\Delta \approx 0$  ( $+0.000$  under NormClip,  $-0.044$  under RFA), the added seeds raising its baseline and its compositions together. What the analysis does have, which neither the FLTrust contrasts of §S24.1 nor the cross-pair scoring has, is C1 held exactly fixed and a disturbance measurement taken on the statistic rather than read back from the outcome. We offer it as the cleanest available piece of C2 evidence and as the shape a properly prospective version of this test would take: fix a downstream defense that suppresses the attack, then vary the upstream along measured disturbance rather than along defense identity. Supplement Table S12 sets out, claim by claim, what this design can and cannot conclude.

#### S24.3 THE SCREEN ON A SECOND DATASET AT THE COMPOSITION LEVEL: IT PREDICTS LOW FOR NO PAIR, SO IT COINCIDES WITH THE CONSTANT-HIGH BASELINE CELL FOR CELL AND ITS RECALL ON THE LOW CLASS IS 0

*Does the screen's behaviour survive a change of dataset at the composition level, where every other dataset arm in this paper is a single-arm ladder?* It does not, and the way it fails is sharper than a fall in agreement: on CIFAR-100 **no single defense suppresses the committed pixel backdoor alone**, and a pair is predicted LOW only when C1 is met for *every* committed attack, so the screen returns HIGH on every scored pair and is therefore the constant-HIGH predictor on this menu cell for cell rather than merely equal to it in aggregate. **The precondition does not fail uniformly, and the part of it that held is reported rather than absorbed into the headline:** reputation does suppress committed scaling alone, at ASR 0.005, and on five scored pairs C1 and C2 both hold for that attack, so the screen reaches a LOW per-attack prediction five times and still predicts no LOW pair. The failure is attack-specific and belongs to the pixel backdoor, which is how Table S13 already records this condition failing on CIFAR-10, four of its five C1 rows reading *C1 (pixel)*, and how §G reports it for EMNIST-byclass.

**What was run, reported as coverage and never as a menu.** The 42-pair composition menu of Table S13 was re-run on CIFAR-100 with `cifar_cnn` against both committed attacks at seeds 42–46, at  $N=10$ ,  $K=5$ ,  $f=0.2$ , Dirichlet  $\alpha=0.5$ , 50 rounds and 2 local epochs, pre-registered in `experiments/pre_registration_cifar100_composition_suite.md` and committed alone at `f559b85`, with the runner refusing to start unless that document is committed at the recorded hash with a clean working tree. **This is not the full 42-pair menu and is not described as one:** the freeze capped the arm to wave 1's 18 pairs before the first run, so wave 2's 24 held-out pairs are deliberately unrun and no held-out generalization claim rests on this arm. Of wave 1's 18 pairs, 18 completed all 10 of their runs and 0 did not. What the cap keeps is the part where the question is decidable: all five of CIFAR-10's LOW pairs sit in wave 1.

**Three of the 18 complete pairs fall below the frozen 0.35 clean-accuracy floor, and they are counted rather than dropped:** `fedavg.then.trimmed.mean` at minimum mean accuracy 0.3388, `norm.clip.then.foolsgold` at 0.3481 and `foolsgold.then.rfa` at 0.2316. A collapsed model can carry a high ASR, so a below-floor cell is *uninterpretable*: it is excluded from scoring and counted in every denominator that describes coverage. The headroom is genuinely thin here and the freeze anticipated it rather than discovering it: every scored pair's minimum mean accuracy lies between 0.3519 and 0.3887, so no cell clears the floor by as much as 0.04.

**The screen agrees with the observed class on 13 of the 15 scored pairs, 86.7%, against a constant-HIGH predictor's 13 of 15 on the same pairs, 86.7%, which is also the base rate, with 3 of the 18 complete pairs excluded below the floor and 0 excluded for a missing baseline.** Those two figures are equal by construction rather than by coincidence, and that is the finding: the screen returns HIGH on all 15, so its prediction vector *is* the constant predictor's, and no agreement figure computed on this arm can separate them. Precision on the LOW class is undefined because the screen predicts LOW for no pair at all, and recall on the LOW class is 0: the two observed-LOW pairs, `foolsgold.then.coord.median` at max committed ASR 0.3561 and `foolsgold.then.trimmed.mean` at 0.4940, are both missed. **So §6's recommendation not to screen on inherited suppression at all now rests on two datasets rather than one**, and it is the stronger of the two failures, since on CIFAR-10 the screen at least made LOW predictions to be wrong about.

**The precondition fails on one attack rather than on the dataset as a whole, and that is a measurement rather than an inference from the agreement figure.** The seven aggregators the screen reads give fourteen standalone readings, and **exactly one falls below the screen's own C1 threshold of 0.3:** `reputation` against committed scaling, at 0.005. The other thirteen run from `fedavg`'s 0.658 to `rfa`'s 0.908, every one of them

above the 0.5 mark at which a composition counts as suppressing. **The committed pixel backdoor is the attack with no standalone suppressor at all:** its seven readings run from `coord_median`'s 0.673 to 0.888, so C1 has no pixel input anywhere on this menu, and because a pair is predicted LOW only when C1 is met for every committed attack, the one scaling suppressor cannot yield a LOW pair. **That is the same shape as \$G reports for EMNIST-byclass**, where the minimum single-defense pixel ASR is 0.609 and C1's precondition fails over the menu measured there. Both datasets that leave CIFAR-10 at the composition level fail C1's precondition on the same attack, so the screen's scope condition is not a CIFAR-10 artifact of one menu. This is also the hypothesis under which Corollary 2's cap reaches 0, the suppression that is observed being emergent rather than inherited from a constituent: neither observed-LOW pair contains a defense that suppresses either committed attack alone, and the measured recall here is 0; we report that as an instance of the corollary on one menu and not as a confirmation of it.

**One departure from the frozen run order, and it is the order alone.** The freeze fixes wave 1's pairs in the imported list order so that any partial completion is a pre-specified prefix. Obeying it would have stopped short of wave-1 position 18, `foolsgold.then.rfc`, whose CIFAR-10 max committed ASR of 0.0454 is the lowest in the menu and which is one of the five LOW pairs the estimand is computed over, so the remaining pairs were run as disjoint slices with the LOW slices taken first (`experiments/run_cifar100_suite_shard.py`, merged by `experiments/merge_cifar100_suite_shards.py`). **Execution was serial, one run at a time;** no rule, threshold, seed, attack, gate or menu entry was changed, and the freeze is not edited. What licenses recombining the slices was measured and not merely argued: `run_one_seeds` torch and numpy as its first two statements, so a run is a deterministic function of its pair, attack and seed, and an already-banked cell was deliberately recomputed under the new scheduler and returned accuracy and ASR *bit-identical* to the artifact's, a difference of exactly zero on both rather than a difference within a tolerance. No run appears in more than one slice, the slices being disjoint by construction, so the merge's overlap check is empty and that emptiness is a property of the design and not a second passed check. Every figure above is emitted by the frozen runner's own reporting path over the merged rows; the merge computes no fraction, base rate, precision or recall.

Per seed, for `foolsgold.then.coord_median`, the more clearly LOW of the two pairs the screen misses. This is the same form §S19.1 uses for the CIFAR-10 menu, one pair's table in the paper and the artifact for the rest; every run of this arm is in `results/cifar100_composition_suite/summary.json`, in the `per_seed` rows. The rows below are emitted from that artifact by `experiments/emit_perseed_cifar100_menu.py`, whose `--check` asserts each one appears verbatim here, so the table is checked against the artifact mechanically and not by eye.

| Seed | committed_scaling |       | committed_pixel |       |
|------|-------------------|-------|-----------------|-------|
|      | Accuracy          | ASR   | Accuracy        | ASR   |
| 42   | 0.389             | 0.381 | 0.391           | 0.399 |
| 43   | 0.399             | 0.228 | 0.404           | 0.271 |
| 44   | 0.380             | 0.444 | 0.397           | 0.481 |
| 45   | 0.381             | 0.322 | 0.382           | 0.343 |
| 46   | 0.382             | 0.254 | 0.383           | 0.286 |
| mean | 0.386             | 0.326 | 0.391           | 0.356 |

The pair's `max_committed_asr` is the larger of the two mean ASRs, 0.3561, which is below the 0.5 mark and so observed LOW; per-attack ASR standard deviations are 0.089 and 0.086 at  $n=5$ , so a single pair's class at this seed count is not a precise quantity and no interval is claimed for it. The screen predicted HIGH here, its reason being that C1 fails because neither constituent suppresses alone.

#### S24.4 OUT-OF-SAMPLE PASS VALIDATION

The development and held-out validation above tests predictions from the *same* attack families (`model_scaling`, `backdoor_pixel`) and scale ( $N=10$ , `cifar_cnn`) used to design the criterion. To confirm that PASS predictions generalize we pre-registered three out-of-sample tests (commit `ca627e5`, before any of them ran); all held, the accounting is in the summary below, and per-cell values for every test are in §S4. We state plainly what this is and is not: it is evidence that the conditions keep identifying compositions which survive a change of attack family, scale and architecture; it is *not* an out-of-sample classification result, because every prediction in it is positive and the matching FAIL controls proved uninformative. Negative-direction evidence rests entirely on the in-distribution tiers.

**(E) Heterogeneity: where the criterion breaks.** Earlier heterogeneity sweeps in this work covered *single defenses* only; no composition had been run at any  $\alpha \neq 0.5$ . We ran all three PASS pairs at  $\alpha \in \{0.1, 1.0, 10.0\}$  (3 seeds, both committed attacks, 50 rounds); the  $\alpha=0.5$  column is each pair's headline value at its own seed

count ( $n=30$  for  $\text{fg} \rightarrow \text{rfa}$  and  $\text{rep} \rightarrow \text{cm}$ ,  $n=5$  for  $\text{fg} \rightarrow \text{cm}$ ) rather than a 3-seed rerun. Reporting max-committed ASR / clean accuracy:

| Composition                        | $\alpha=0.1$                     | $\alpha=0.5$ | $\alpha=1.0$ | $\alpha=10.0$ |
|------------------------------------|----------------------------------|--------------|--------------|---------------|
| $\text{fg} \rightarrow \text{rfa}$ | <b>0.569</b> <sup>§</sup> / 0.20 | 0.093 / 0.50 | 0.051 / 0.57 | 0.063 / 0.62  |
| $\text{fg} \rightarrow \text{cm}$  | 0.433 / 0.40                     | 0.107 / 0.68 | 0.133 / 0.74 | 0.129 / 0.78  |
| $\text{rep} \rightarrow \text{cm}$ | <b>0.548</b> / 0.48              | 0.463 / 0.76 | 0.377 / 0.79 | 0.326 / 0.81  |

§ *uninterpretable* under the accuracy floor of §S24:

clean accuracy  $0.20 < 0.35$ , so the raw 0.569 is a collapsed model and is scored as neither a crossing nor a survival. Raw reading kept above.

The criterion transfers cleanly across  $\alpha \in \{0.5, 1.0, 10.0\}$ , including the near-IID regime where FoolsGold might plausibly founder on legitimately similar benign updates; it does not, and  $\alpha=10$  is indistinguishable from  $\alpha=0.5$ . **Under extreme heterogeneity ( $\alpha=0.1$ ), two of the three PASS compositions cross the threshold on the raw ASR reading** ( $\text{fg} \rightarrow \text{rfa}$  0.569,  $\text{rep} \rightarrow \text{cm}$  0.548); only  $\text{fg} \rightarrow \text{cm}$  survives, at 0.433.  $\text{Rep} \rightarrow \text{cm}$ 's  $\alpha=0.5$  baseline is itself at the threshold (0.463 at  $n=30$ ), so its crossing is the smaller move of the two. **Our own accuracy floor changes that count, and it changes it in our favour, which is why both readings stand here.**  $\text{fg} \rightarrow \text{rfa}$ 's  $\alpha=0.1$  cell sits at clean accuracy 0.20, below the 0.35 floor of §S24, so by the paper's own rule it is *uninterpretable* and cannot be scored as a crossing. The gate-applied reading is therefore one crossing at interpretable accuracy ( $\text{rep} \rightarrow \text{cm}$ , 0.548 at 0.48), one uninterpretable cell, and one survivor ( $\text{fg} \rightarrow \text{cm}$ , 0.433 at 0.40); the raw reading is 2/3. **We print the raw count as well as the gated one because applying our own rule here removes a datapoint from a limitation we report against ourselves**, and a reader should be able to see that softening rather than inheriting it. Two things the gate does *not* buy: the cell is uninterpretable in both directions, so it is no evidence that the criterion holds at  $\alpha=0.1$  either, and  $\text{rep} \rightarrow \text{cm}$ 's crossing is unaffected by the floor, so extreme heterogeneity still breaks a PASS pair at accuracy we are willing to read. Two caveats cut in opposite directions:  $\text{fg} \rightarrow \text{rfa}$  at  $\alpha=0.1$  is strongly seed-dependent, the six runs at this  $\alpha$  spanning 0.065–0.923 and the max-committed arm itself 0.245–0.866, so at  $n=3$  the direction is clear but the magnitude is not; and clean accuracy collapses to 0.20–0.48, so this regime stresses utility as much as security. Crucially Theorem 9 supplies a **diagnostic quantity that retrospectively explains this boundary** rather than predicting it: re-measuring on the transformed points at  $\alpha=0.1$ , RFA's separation margin  $m := \Delta'_{\text{sep}} / [2(R'_B + \delta')]$  collapses from an all-round mean of 3.24 (met in 68% of rounds) to 1.01 (met in 20%). Over the payload-carrying rounds alone (the ones Figure A1 plots) the same collapse reads  $4.71 \pm 1.51$  ( $n=17$ ) to  $2.06 \pm 2.35$  ( $n=12$ ), with the per-round minimum falling  $2.82 \rightarrow 0.00$ ; the all-round means are lower because they average in the Lemma 2 rounds, where FoolsGold zeroes every adversary and  $m$  is recorded as 0. Either way the collapse is measured, and the all-round figure is the one that reaches the vacuous limit. This sits exactly at the  $m \rightarrow 1$  boundary where Corollary 4's bound degrades to 40% adversarial mass: extreme heterogeneity inflates the benign radius  $R'_B$  until the benign cluster is no longer separated from the adversarial points, which is precisely the precondition the theorem requires. A practitioner running the 5–10 round pilot would measure  $m \approx 1$  and decline to deploy, so the criterion is mechanistically diagnostic at this boundary, though we emphasise this is a post-hoc measurement of a known failure, not a pre-registered prediction of it.

The weakest axis is still *dataset*, and it is now validated with composition experiments on one further dataset and no more: §S24.3 re-runs wave 1 of the 42-pair menu on CIFAR-100/cifar\_cnn at the coverage stated there, with wave 2's 24 held-out pairs deliberately unrun, so the screen's behaviour off CIFAR-10 rests on part of one menu on one dataset and not on a cross-dataset comparison. Every other composition in this paper is CIFAR-10, and the two remaining arms measured on another dataset are single-arm intervention ladders rather than composition suites, EMNIST-byclass/simple\_cnn in §D.2 and CIFAR-100/cifar\_cnn (100-class output) in §D.3. The conditions are dataset-agnostic in mechanism, but the downstream aggregator's decision boundary may shift with the number of classes, and the CIFAR-100 ladder is direct evidence that it can: the same cell that reverses sign on CIFAR-10 reverses sign there too, at a different magnitude (L1).

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